



## **Milestone 3**

# **Concept Design Report**

### **Bring ROAR to Life - Team 1**

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# Executive Summary

*Bring ROAR to Life* is a mechanical engineering senior design project. The goal of this project is to create an animatronic stegosaurus skeleton with the capability to walk. This project was largely inspired by the film night at the museum, which features a T-Rex skeleton that comes to life.

The project is being designed with a museum application in mind, to be an additional exhibit that guests can interact with. The animatronic requires a mechanical system, electrical system, and control system. All of these must coordinate with each other to generate the final result. The animatronic must display a high degree of anatomical accuracy, and its walk must be smooth.

The team will develop this animatronic through the use of engineering principles. The structure will be analyzed for stress and torque results. An optimized material will be selected to handle the stress and torque, as well as fit within budgetary constraints. The power system must be strong enough to actually all joints, well compact enough to fit within the skeletal structure. The control system must be able to operate the joint actuators. The team will work together to iterate and test each of these systems. Additionally, the team will analyze the scope of the project and identify any failure, modes and effects.

The length of the project is eight months, with five months remaining before a substantial prototype will be presented in a showcase format.

# Introduction

This project originated from Dr. Kurt Stresau, who many years ago purchased for his daughter a stuffed toy of a stegosaurus from a museum gift shop [48]. Several years later Dr. Stresau was inspired by his daughter and “Rex-y” modeled after a complete fossil specimen, AMNH 5027, from the American Museum of Natural History in New York City [47] in the film Night at the Museum, to submit a senior design project around the concept. This project will consist of creating a biologically-realistic stegosaurus skeleton, primary focuses will include a biologically accurate appearance, anatomically correct movement, and ease of control.



*Figure 01: Image of ROAR Stresau for Reference [48]*

When getting familiarized with the project, the team was permitted a high degree of flexibility in determining what direction a final product may look like. Ultimately, the team decided that the target customer would be a museum that would utilize the animatronic as an educational tool/aid within an exhibit or

showcase. The primary user will consist of an employee of said museum and will exclude guests or visitors. The secondary user is the companies purchasing the animatronic. It is important to keep them in mind in terms of cost of maintenance and functionality, as their needs will be different from the needs of the operator. The tertiary user is the audience; this is important as the design also needs to reflect and consider the experience from their perspective. The design is not intended to be used for military or industrial purposes and furthermore should not by any means be used to endanger injury or hurt any of the three forms of users.

The team has currently met with and interviewed the Lawrence Hall of Science to better understand the wants and needs of the user. The team is in active communication with them and will continue to do so throughout the progression of the project.

The primary feature that will stand out in the design will be that it is based on an anatomically accurate skeleton. This is different from what exists in the industry as animatronics to date typically use a “skin” that is stretched or placed over a larger inner structure. Here, the team is attempting to showcase biological movement with an anatomically accurate gait. Further, the skeleton makes for a difficult packaging challenge in order to effectively hide the mechanical and electronic components.

Engineering projects are driven by a problem. This problem can come in the form of a customer want, a consumer need, or a creative idea. Engineers compile data that establishes the full problem, then begin to develop solutions. This initial problem data can be called the design requirements, while the solution-development phase can be called concept generation. It is important to understand design requirements and research similar project areas early in the project. This ensures that the concept generation falls in-line with its use-case.

This report focuses both on design requirements and concept generation. Initial concepts will be presented for the subsystems of the animatronic, paired with the engineering thought process behind these concepts. Analysis opportunities will also be presented, paired with discussion on failure modes. This will provide the team with insight into how each individual component may fail, and the severity of each failure on both the surrounding environment, the robot itself, and potential injury to the final end user.

# Needs Analysis

To begin any engineering project, it is vital to determine the end users or customers, which will drive base requirements and what must be analyzed in order to move forward. For this project, the primary customer will be a science museum as an education tool that can interact with visitors and guests. Key requirements, therefore, include anatomical accuracy and life-like movement. The operating environment will be limited to indoor environmental conditions, including tile floor. The team has been in contact with the Lawrence Hall of Science at Berkeley

| Customer vs. Design Requirements |            |                     |               |                           |               |                |                    |         |                     |              |                                  |
|----------------------------------|------------|---------------------|---------------|---------------------------|---------------|----------------|--------------------|---------|---------------------|--------------|----------------------------------|
| Customer Requirements            | Importance | Design Requirements |               |                           |               |                |                    |         |                     |              |                                  |
|                                  |            | Visual Accuracy     | Walking Speed | Flexibility/ Articulation | Control Range | Turning Radius | Animation Segments | Runtime | Physical Durability | Maintainable | External/ Internal Sensor System |
| Walk Like a Dinosaur             | 9          | X                   | X             | X                         | X             |                |                    |         |                     |              |                                  |
| Anatomically accurate            | 9          | X                   | X             | X                         |               | X              |                    |         |                     |              |                                  |
| Engaging for Audience            | 9          | X                   |               | X                         | X             |                | X                  | X       |                     |              |                                  |
| Easy to operate                  | 7          |                     |               | X                         | X             |                |                    |         | X                   | X            | X                                |
| Repairable                       | 3          |                     |               |                           |               |                |                    |         | X                   | X            |                                  |
| Run-time of 30 minutes           | 3          |                     | X             |                           |               |                |                    | X       |                     |              |                                  |
| Semi-affordable                  | 1          |                     | X             |                           |               |                |                    |         |                     |              | X                                |
| Compact size/Easy to transport   | 1          |                     |               |                           |               |                |                    |         | X                   |              | X                                |

Figure 02: Design and Customer Requirement Matrix

to help develop these customer requirements.

To the left is the derived matrix of requirements for customer wants and needs, along with basic design requirements which will be discussed further. A weighted factor has also been placed on each requirement, which will serve as a guide for future design decisions.

## Individual Customer Questions and Wants

In the discussions with the Lawrence Hall of Science at Berkeley the following questions have been proposed in the event that this project becomes an addition to their museum [1]. A partnership has not been decided, however this gives the team a good base to determine customer requirements. Customer requirements have been discussed at length with the museum, along with the proposed requirements in the matrix above. Some of the items below have not been fully defined, as the end prototype will drive its use. But a likely use case will be to supplement

STEM education and to have a hands-on way to explain the design process and engineering. Below are some questions posed by the museum.

*What is the intended use for the Stegosaurus?*

In a museum setting, the ideal goal for the stegosaurus animatronic would be to provide a more visually interactive display. Due to the complexity of the electronic and computer system, it would not be ideal for museum goers to operate the animatronic themselves. However, the animatronic ought to be simple enough for employees to learn how to operate it. Additionally, museum goers ought to be able to touch the animatronic, though not handle it.

Is it designed to be a hands-off display, or would it be robust enough to be actively driven by visitors (including young children) on the science center floor?

*How durable are the controller units, and what is the typical range of the signal? How is the stegosaurus powered and what is the runtime on a full charge, and what is the recharge time?*

These are vital questions, and they were taken into consideration when clarifying the system requirements. Additionally, concepts were generated with these questions in mind.

*What maintenance do you anticipate will be required and would you be able to provide the full technical documentation?*

Through material selection and design decisions, the team would be able to assess how frequent maintenance ought to be performed on the animatronic. The benefit of the museum coordinating directly with the engineering team is that the team has access to all technical documentation with respect to the project.

*What led the group to design and build this remote-controlled Stegosaurus and are there any specific engineering, robotics, or design concepts that the Stegosaurus demonstrates?*

While the inception of this project originates from a Senior Design course, the team largely chose this project because of an interest in animatronics. As for specific engineering concepts, the animatronic will demonstrate mechanical, electrical, and programming concepts. On the mechanical aspect, the kinematics and dynamics of the stegosaurus dramatically impact its design, as well as material selection. Electrically, the team will have to create a power system, strong enough to move the animatronic, while still being safe. From the programming side, both the mechanical and electrical systems need to be controlled by a computer.

### **Locomotion & Aesthetics**

Customer requirements dictate that a successful project would be an anatomically accurate stegosaurus skeleton with a fluid gait. It is also vital for the model to be engaging for an audience of museum guests. In order to meet these, the final design will need to be based off of an existing 3D-scan of a Stegosaurus skeleton and display a high visual resemblance to that scan. Additionally, a large packaging effort must take place in order to effectively hide all main components, motors, and joints within the existing bone structure. Individual joints should not be limited to be exact copies of what was present in a dinosaur, but an effort should be made to create a biologically appearing joint and overall movement system.

### **User Operation**

Another important question brought up by the museum was who will be operating the dinosaur. The main goal of which would be to have guests be able to interact with a finished product. But in order to remain within the scope of this project and for operating a prototype, a museum employee should be the likely operator as they will receive comprehensive training on



how to effectively make this prototype move and interact with its environment and guests. In doing so, there will also need to be an operations manual which will be created.

### **Run Time**

For a successful exhibit, the end robot will need to last long enough to be operated for the duration of a demonstration along with the ability to charge quickly in order to use the end product multiple times a day. A simple base target for this will be a minimum run time of 30 min, which can be calculated both in testing, and estimating a needed battery capacity from the typical Power draws from electrical components.

### **Handling and Storage**

When outside of using the end product with guests, there needs to be consideration for both transport, storage, and repair if and when something breaks. As far as handling is concerned the ideal robot shall be able to be carried by a single user without the need for special lifting equipment or processes. To meet this, the goal will be for the robot to weigh under 50lbs as suggested by OSHA [2] in order to help prevent injury.

### **Repairability**

If a major component does break, there will need to be readily available instructions on where to source and replicate existing components. It will also be beneficial to use easily replaceable sub-assemblies where possible. This will both speed up the repair process and allow for non-technically trained users to possibly repair a finished device.

## **Audience Engagement**

One factor that was not previously considered in depth was audience engagement, as the primary end use case for this robot will to be used as a teaching tool for museum guests. This places a greater importance on non-functional movement separate to the primary drive motors and legs. Another aspect to consider is how will this robot be used to teach about STEM outside of being a dinosaur? Which can be translated into using this project in general to teach about the design process and how the flow of a larger engineering project works from conception all the way to a final prototype.

# Technology Assessment

In order to better understand where to take the final design, the team first needs to understand industry standards to inform design decisions. Further, as the model is based on a previously-living creature, care should be taken to understand how a stegosaurus lived in its environment. Research should also cover the current animatronic industry, along with more specific research into joints and control systems.

## I. Anatomy and Diet of the Stegosaurus

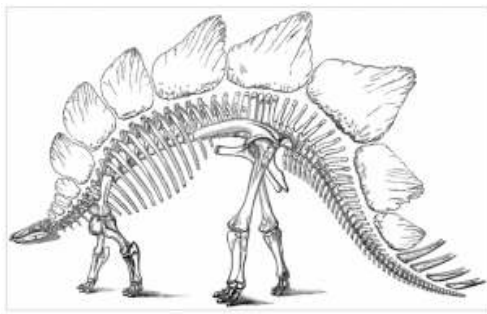


Figure 03: Marsh Othniel C. "Restoration of Stegosaurus," in: *American Journal of Science*, series 3, vol. 42 (1891), pl. 9.

The stegosaurus is a dinosaur found mostly in the mid to late Jurassic period, although specimens have been found dating from the Cretaceous period [5]. They were a member of the genus *Stegosauria*, which was a family of quadrupedal herbivores found across the globe, including China, Siberia, Morocco, and Australia. Fossil

remains of the stegosaurus, however, have been discovered primarily in North America and Europe [6]. The stegosaurus is one of the larger of the stegosaurs, with fossils reaching up to 9 meters in length.

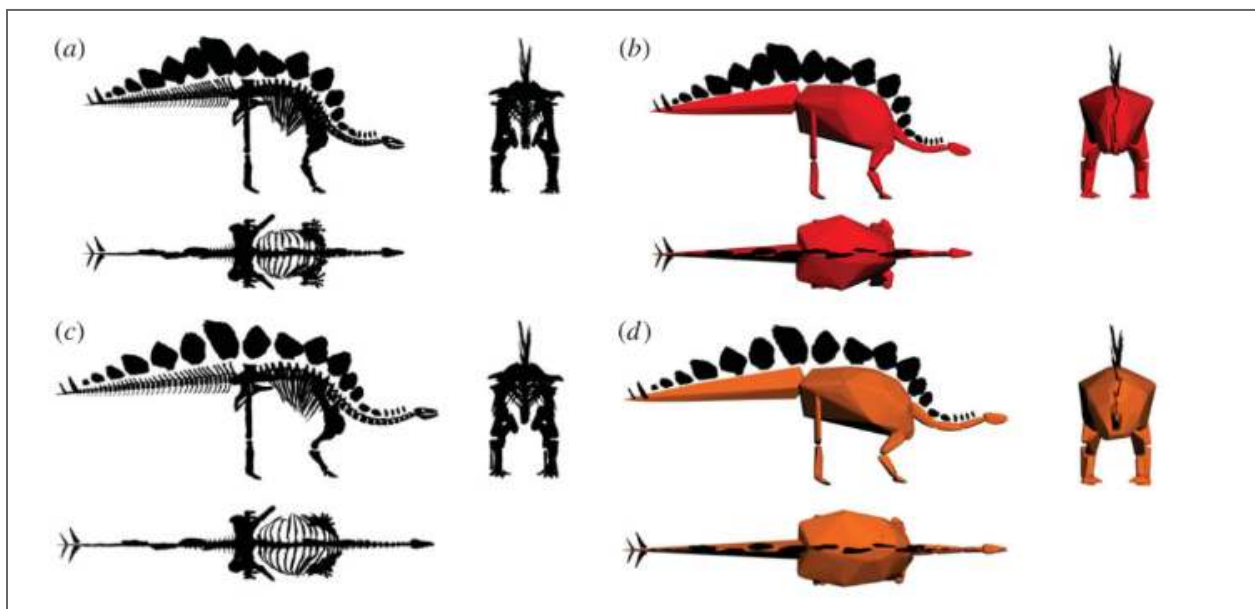
### ***Diet***

Due to the lack of soft tissue preservation, there's little evidence for what the stegosaurus ate. Many of the tooth specimens black substantial wear, making it difficult to predict the diet of the stegosaur. The common consensus is that they fed close to the ground, near the 1 m level. The one dissenting opinion suggests that the stegosaur stood on its hand limbs, using its tail to balance, similar to a kangaroo [6]. Jaw and dental analysis implies that the stegosaurus had a bite

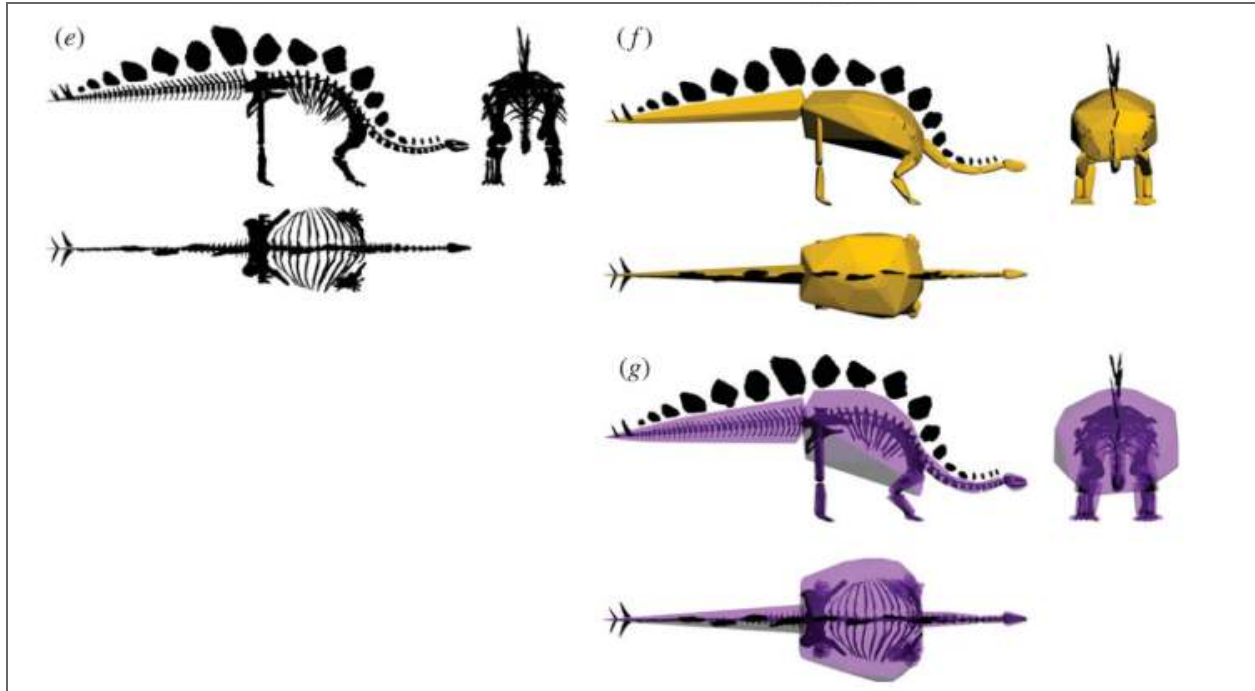
force ranging from 166 to 410 N [5]. This bite force would have allowed the dinosaur to bite through branches, though the main function of their teeth appears to be centered on grinding down leaves and other plant matter.

### *Anatomy*

Estimated masses of the stegosaurus range from 1082 kg to 2256 kg [3]. The mass of the animal depends heavily on the spread of the legs and descent of the body cavity. This information is lost to time without the presence of soft tissue or more complete skeletal specimens. Further, it is predicted that the stegosaurus grew to its full size slowly, implying it did not need to reach its final dimensions quickly for protection [4].



*Figure 04: Reconstructions of Stegosaurus stenops (NHMUK R36730) and associated convex hulls [3].*



*Figure 04.1: Reconstructions of Stegosaurus stenops (NHMUK R36730) and associated convex hulls [3], cont.*

### ***Head***

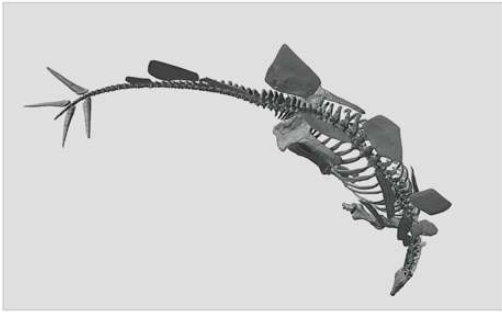
Notable features of the stegosaurus anatomy include a proportionally small head compared to the body. The top of the skull was flattened, and the cranial cavity was small. It is likely the animal had a small brain as a result of its diminished cranium [6].

### ***Body and Legs***

Fossil records demonstrate that the beginning of the rib cage was narrow, widening when approaching the pelvis. Additionally, the forelimbs were short in proportion to the hind limbs, which introduced a dramatic curve down in the spine. This results in a visual center of mass targeted over the hind legs.

The ribs of the stegosaurus are T-shaped in cross section. Females have an additional sacral rib. The neural passageway through the spine is narrow, which would limit the thickness of blood vessels and nerves able to pass through [6].

## ***Tail***



*Figure 05. 3-D Scan and Model of Stegosaurus  
Triebold Paleontology, Inc., Denver Museum of Nature and Science*

One of the most notable features of the stegosaurus were the dorsal spines running down the vertebrae. They began after the skull and extended down the tail. The plates alternated angles at 50° along the body, while falling more in line when proximal to the tail. The tail finished with four large spikes.

The function of the dorsal fins is disputed. A possible purpose could have been for defense, making the animal too large and boney for predators to easily eat. Additionally, it has been proposed that its plate could have performed thermoregulatory functions. The fins contain vertical grooves, which may indicate the presence of blood vessels. Another explanation may be that the fins distinguished each animal, allowing for the recognition of individuals. Its spikes are bony and presumed to be defensive measures [4].

## ***Summary***

The anatomy of the stegosaurus resembles many modern quadrupedal herbivores. Stegosaruses have a low center of mass. Their wide ribcage and sturdy limbs resemble animals such as elephants or rhinoceroses. In contrast to modern herbivores, stegosaurus fossils display small heads, spiked tails, and dorsal fins. Additionally, their long necks and tails result in a wider splay of the animal in comparison to the aforementioned animals.

This anatomy will create an interesting design challenge. The short forelimbs will create dramatic stability issues in the animatronic. Another potential struggle will be providing enough support and Power to animate the far-reaching head and tail. While the solutions to these

problems are not yet clear, the team will be better equipped to develop an accurate animatronic when backed by the fossil data.

## II. Animatronic Development in Industry

This project at its core is a project dealing with animatronics, specifically animatronics and robotics in entertainment. Creating a dinosaur that is lifelike in its movements and actions, as well as its interactions with the environment is an integral point of success for this project. In order to better achieve this goal, it is good to be familiar with the problems that engineers before this group encountered when attempting to take on similar projects. Looking at the specifics of their problem statements and solutions, gives the group a more detailed guide when it comes to figuring out what functionality and design specifications will likely be most important. Additionally, studying examples from past attempts in industry may illuminate potential pitfalls the group is likely to encounter during the design and prototyping process. Knowing where those pitfalls are likely to occur, and how other groups went about solving them, gives the group a roadmap to success that doesn't need to be built from the ground up, saving valuable time and resources.

### Bringing Art to Life



*Figure 06: Disney Imagineer Sonny Chapman demonstrates the features of their largest and latest animatronic, Louis, including fully electric servos and independent control of every part of its body [9].*

Companies like Disney and Universal are at the forefront in the development of robotics and animatronics in entertainment. From Walt Disney's very first bird animatronic which he found in an antique store up to today's highly

advanced models with face projection, fully electric precision servos, speakers, and expertly designed costumes.

Animatronics in this field tend to be a highly coordinated effort between interdisciplinary teams of engineers and highly skilled digital artists and animators. Robotics experts work with costume designers [9] back and forth to bring every detail of a fictional character to life from their attitudes and mannerisms to their voices. Software interfaces allow for direct and precise control of every part of the animatronic, coming together to form smoothly coordinated, natural movements. Disney's Audio-Animatronics [10] are famed precisely for their ability to unite movement and sound, bringing together songs and gestures perfectly coordinated and choreographed by teams of animators. For the purposes of the project, nailing down as small a level of precision as possible will be paramount to achieving the smoothest and most natural movement possible.

### **Disney's Droids**

Within the past few years, Disney especially has had to ramp up their development of new animatronic solutions to accommodate their new and expanding repertoire of themed parks based upon their newer intellectual properties. Most interestingly for the purposes is an animatronic robot based upon the droids from Star Wars. These new robots incorporate elements of engineering design and artistic creativity to deliver a final product that, when paired with a skilled puppeteer, behaves in an incredibly lifelike manner.



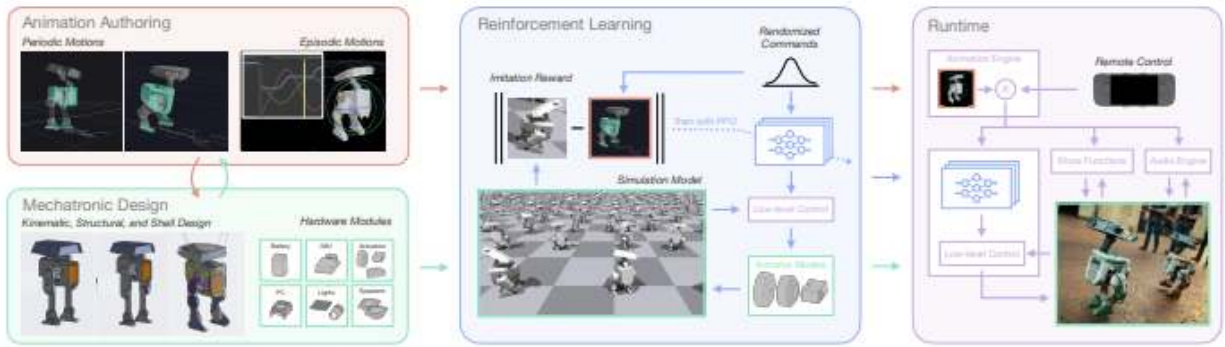


Figure 07: Summary of the Development Process [7].

Through a unification of artistic animation and computer simulation based upon the initial character design, animators are able to use kinematic simulations of the actual robot to test realistic walking patterns within the physical bounds of the robot's weight and size, as well as range and speed of motion. Using reinforcement learning [7] the robot can be trained to move within the bounds of the intended animation as closely as possible while also maintaining proper balance across a broad range of possible environmental variables. Special attention was given to the use of both 3D printing and off the shelf parts which allowed for much quicker development time and a reduction in cost from having to develop a multitude of systems from the ground up. Puppeteers use the already existing Steam Deck [7] in order to move the robot and issue commands, which also makes training puppeteers easier. The use of a video game style controller was deliberate due to widespread familiarity with the platform, making it easy to pick up on the basics of the controls. Using an existing video game platform also goes back to allowing off the shelf parts to reduce development time and cost.

This project also focuses very heavily on having creative design guide the functionality of the robot, especially in the realm of Human-Robot Interaction. Every aspect from the use of puppeteers to guide spontaneous behavior to allowing artists and animators to define pre-set animations results in a robot which can display a wide variety of emotions as well as

dynamically responding to a variety of environmental factors. Robots can appear to inspect their environment, interact with objects, respond to questions, voice commands, and touch [7] in much the same way a household pet or a very young child would.[8]



*Figure 08: (top) A robot interacts with an object in a set scene performing a routine of emotional controls, (bottom) A robot interacts with a human, showing curiosity and receiving affection [7].*

Simulation and Reinforcement Learning allow much of the heavy lifting in terms of calculation to be taken off of engineers and computer scientists, further reducing development time and allowing for more effort to be expended towards the creative goals of the project.

## **Summary**

This dinosaur project has a very broad scope that largely focuses upon a lot of creativity driven solutions rather than pure functionality. Examining this case in industry sheds light on foundational conclusions including the use of off the shelf parts, 3D printing, simulation, and machine learning as tactics to greatly reduce development time and cost. It also helps open the door to functions which the team may want to consider such as necessary range and types of motion, pre-determined animations, and Human-Robot Interaction. Considering aspects of who

and where the robot will be interacting with, gives insight as to what behaviors it may need to express.

### III. Robotic Joint Types and Movement Systems

One important aspect of creating lifelike animatronics is having joints that are able to mimic fluid movement. For this project, this will mean the use of high torque, lightweight drive systems. Another important factor will be a consideration on size and packaging. There are many different drive systems already used in robotics, the challenge of this will be inventing a way to hide these systems in order to preserve biological accuracy to the model.



Figure 09: Capstan Drive example [11].



Looking to the industry today, one drive method is a capstan drive, these systems trade out the use of gears for synthetic rope. This allows for the system to transmit high torque, while decreasing noise and slop. These

systems will typically have a reduction ratio less than 10:1 [11]. however there will need to be a careful look into packaging, as the primary drum is quite large when compared to the rest of the joint.

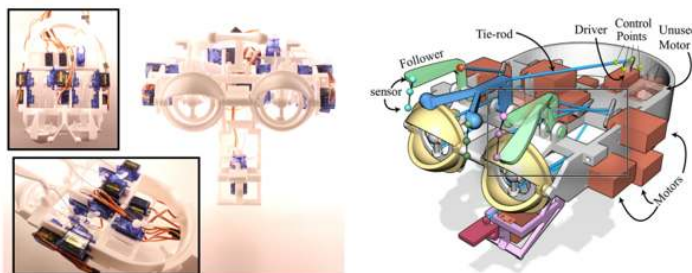


Figure 10: Head with motors and linkages for an animatronic face [12].

Another consideration could be through the use of tie rods & cables. These can allow for drive motors to be relocated away from a joint. In doing so, it allows for the combination of

electronics, drive motors, and mechanical linkages to drive separated movement. [12] This could

possibly be used to actuate finer movement, such as the head, neck, and tail. Along with relocating motors for the legs in order to preserve biological accuracy.



Figure 11: Shoulder with linkages and range of motion [12].

Along with using linkages and separated motors for facial features, another example used was for a shoulder. This uses the same basic principles however includes a larger focus on bone structure and lifelike movement. [12] Which

will become a large focus of this project.

For a more direct drive example, The team can take a look at cycloidal drives. This system uses a set of eccentric gears which results in a very space efficient gear reduction. The basic working principle of a cycloidal drive is that of a disk rolling about an outer ring of larger radius. As it rolls, the disk rotates at a velocity that is proportional to the relative radii, which provides the gearing ratio of the gearbox. [13] These are able to be low-cost, 3D-printable, and use off-the-shelf bearings and steel components. [13]

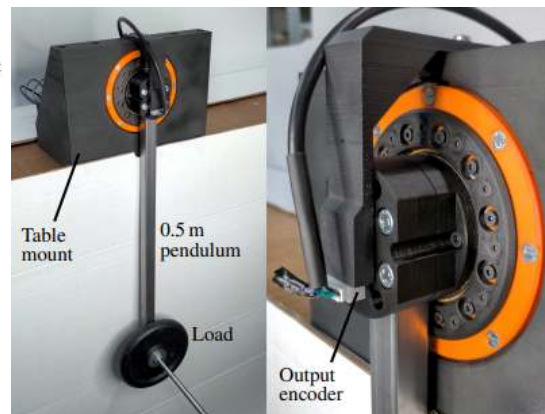
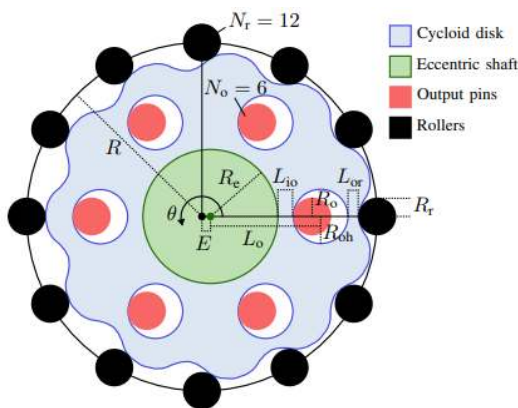


Figure 12: Cycloidal Drive Geometry [13]

When beginning any robotics project, many design considerations will need to take place. Aspects include robot weight, required torque, speed, and cost. Most of these joint types aim to increase the torque output of most electric motors which is needed on most legged robots. In order to determine these factors, The team will have to estimate the dinosaur's weight and expected forces on each leg. This will then allow the team to calculate the required mechanical advantage needed from these joints.

Another creative technique is through the use of elastic or springs along with rigid linkages. These can be used to provide a static mechanical advantage to joints, here the relationship between the applied torque and the joint torsion is identified [14]. In turn allowing for the movement mechanisms to become smaller which can aid in overall packaging. Additionally these systems can be used to help prevent backlash and reduce slop [14].

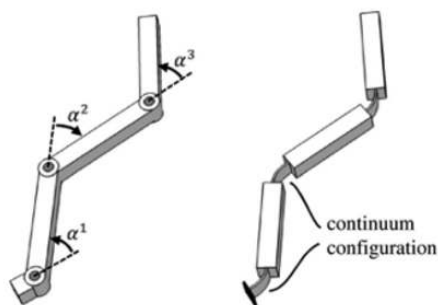


Figure 13: Comparison of Standard and Flexure Joints [15].

Moving on from joint movement, the team can take a look into types of physical joints., which will need to mimic biologically accurate joints. Here, industry standard joints may be less desirable, however required depending on specific requirements. One alternative is to use a Flexure joint, these trade out pins and bearings for a flexible material. Advantages include the lack of sliding parts, which reduce points of backlash or slop. Along with the ability to use a variety of different material types that can possibly be 3d printed at a much lower cost to traditional joints [15]. These may be worth looking at as these joint types could be hidden into a biologically accurate appearing joint, which can increase realism. One drawback to flexures as robot joints is the complex mechanical behavior that they exhibit compared with pin joints [15]. In order to determine accurate joint movement, beam

bending models may be needed, along with a higher level of research in material selection and manufacturing.

Although more research is to follow, the previous joint types allow the team to gain exposure to a few unique methods used outside of traditional robotics. Which opens up conversations about material selection, packaging, maintaining biological accuracy, and many other important aspects of animatronics. And will likely lend useful as the team progresses further into this project.

#### **IV. Simplified Linkage Types to Mimic a Realistic Gait**

Problem statement: The success of the walking dinosaur depends heavily on its ability to walk in a stable manner, while allowing movement for special effects, including standing on its 2 back legs, and using its tail for a 3<sup>rd</sup> leg. One other special effect the team agreed on implementing is having the dinosaur walk, trot, canter, gallop, etc. Unlike wheeled systems, legged locomotion requires precise coordination.

##### **Jensen Linkage**

One mechanism is the Jansen linkage. The Jansen linkage is quite simple, it is an eleven-bar mechanism, and its motion mimics a walking animals' motion. I have designed this mechanism in my internship at Siemens and have familiarity with its function. The mechanism has 13 different length numbers that can be edited. The genetic code of the numbers are called the "13 holy numbers" [16] and these numbers are a ratio, and provide the best walking motion, ensuring there are always two legs touching the ground simultaneously. Each two legs would only use one motor, unless The team has a motor for all of the legs.

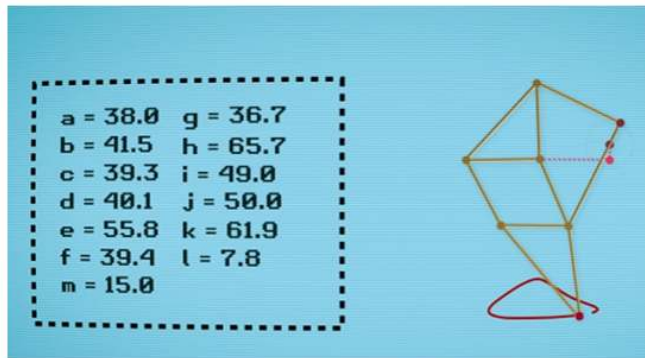


Figure 14: [16] (Veritasium 4:29)

This walking mechanism has pros and cons. The cons are that if the team has one motor for each side, it would be hard to make the robot stand up on its hind legs. One way to fix this is to have 1 motor for each leg, that way, each leg can behave independently, and the robot can stand on

its hind legs, and balance using its tail. The pros are that it is a simple mechanism to work with and only requires 2 or 4 motors. Another pro to this mechanism is that the team can add one motor on each side to not only have one degree of motion (front and back), but side to side. This would enable the robot to move its legs from side to side.

## AlienGo Robot

In this article, The AlienGo robot shows how complex locomotion can be controlled and trained. Unlike fixed linkage mechanisms, advanced robots use multiple actuators and control systems to replicate walking, trotting, and galloping with high accuracy. For the dinosaur, understanding these systems can guide the team in how to design both stability and special effects (such as standing on hind legs or using the tail for balance).

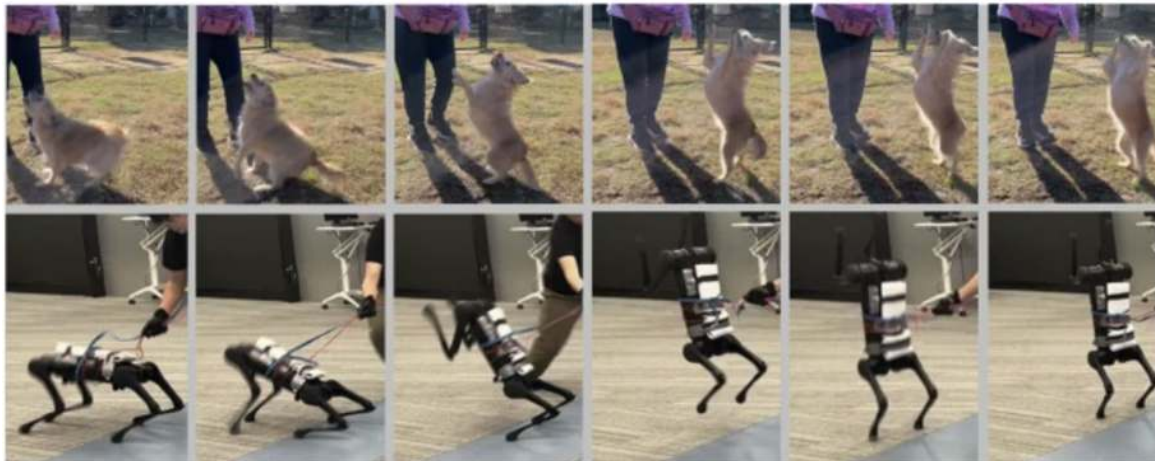
[17] AlienGo is a quadrupedal robot weighing approximately 23 kg and designed to replicate agile animal-like movement. [17] It has 12 actuated joints: 3 per leg (hip, thigh, and calf). This design allows it to mimic a wide variety of gaits, including walking, trotting, galloping, and even acrobatic maneuvers like backflips. The key advantage of AlienGo [17] is its ability to learn gait motions from videos of real animals, which makes its locomotion both



efficient and realistic.

The pros of the AlienGo approach include precise independent control of each leg, making complex behaviors like standing on hind legs or balancing with a tail much more feasible. [17] With 12 motors, each joint can contribute to stability and dynamic movement. Another strength is flexibility; new gait patterns can be programmed without redesigning the hardware. This design has the most flexibility and would be great especially if The team only made the skeleton dinosaur.

The main con is the complexity: [17] 12 motors means higher cost, more Power requirements, and greater control system complexity compared to simpler mechanisms like the Jansen linkage. For the project, this approach may be too resource-intensive for a prototype, but the principles behind AlienGo's motion studies are valuable for informing gait control in the dinosaur.



*Figure 15: [17] Liu Zhao (results section)*

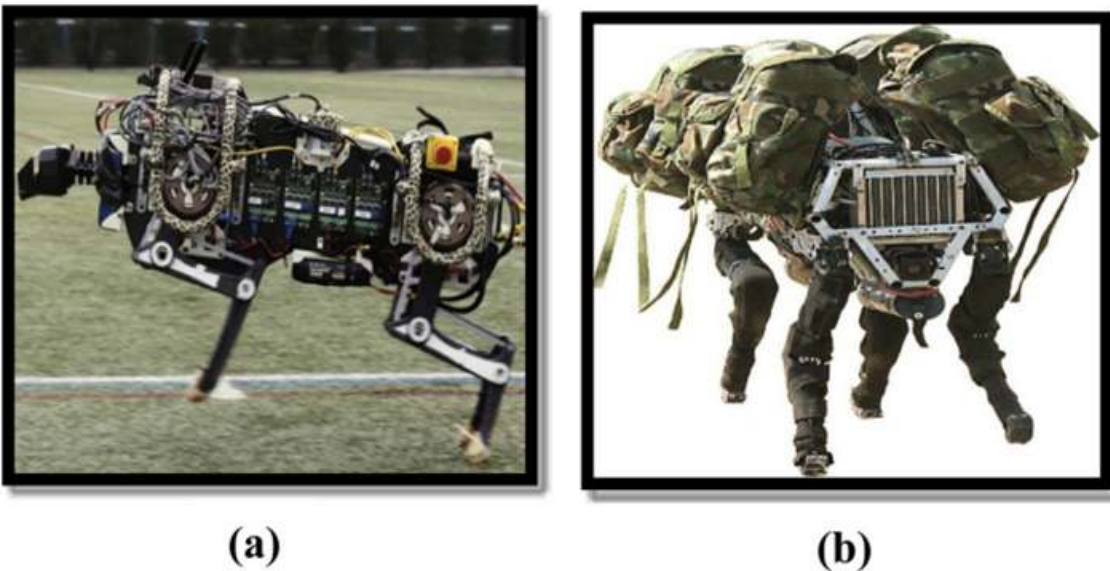
### **Development of quadruped walking robots - article**

This article [18] has many different types of quadrupedal motion. The purpose of this article is to summarize key findings from the review by Biswal & Mohanty (2020) that are particularly relevant to the walking dinosaur project. In particular, this includes how modern



quadruped robots are designed in terms of actuation (motors/joints), gait patterns, structural trade-offs. Many quadruped robots in the literature use two rotary joints per leg; each leg often has a hip and a knee that are actuated, this enables flexion/extension motions which are essential in the robot in order to perform the trot, canter, gallop motions.

Some more advanced designs have three actuated joints per leg (hip, thigh, knee / ankle), this is for dynamic gait (trotting, running) and complex terrain adaptation. While this approach includes precise independent control of each leg, it would require higher costs, and more Power requirements.



*Figure 16: [18] Complex actuated joints robot.*

## Summary

These articles collectively show the challenge of designing effective quadrupedal locomotion for the walking dinosaur: balancing stability, complexity, and realism. The Jansen linkage [16] offers a mechanically simple and motor-efficient solution, but its limited degrees of freedom restrict more advanced movements like standing on hind legs. In contrast, the AlienGo [17] quadruped demonstrates the Power of multi-motor actuation and advanced control, using 12

independent joints to achieve dynamic gaits and agile maneuvers, but the tradeoff is the cost of higher complexity and budget. Finally, the review by Biswal & Mohanty [18] situates these approaches within the broader field, showing how two to three joint legs per limb are standard in quadrupeds and highlighting tradeoffs between actuator count, gait diversity, and adaptability. Together, these insights guide the project toward a design that balances the simplicity of linkage mechanisms with the flexibility of the gate to achieve both realistic walking and special effects.

## **V. Digital and Control Systems**

The team has been tasked with constructing a free standing quadrupedal robot able to achieve a walking motion with dinosaur anatomy. To accomplish this the quadrupedal robot will require the implementation of digital and control systems that will drive its ability to walk as well as any additional functions/features that have been approved and discussed by the team such as for example opening and closing the mouth of the robot etc. The Tech memo is about the selection and a catalog of options for digital and control systems. Below the following will be discussed: a summary of control theory with a theoretical analysis as pertains to the project, discussion of the Hardware such as microcontrollers, microcomputers, and microprocessors, the programming languages and operation systems with regards to use in robotics, a discussion of references for algorithmic and mathematical methods for use in controlled and autonomous robotics systems. Achieving these objectives will allow the team to make more informed decisions when creating design requirements as well as create a reference document for future design and development.

## Control Theory and Systems

Control theory “is a mathematical framework that gives the team the tools to develop autonomous systems” [20]. A feed forward controller or open loop controller: takes in the desired output/reference and generates the control inputs without ever needing to measure the state. A feed forward controller does not rely on feedback from the state of the system which is why it is an open loop, it never loops back. Feedback control/closed loop control relies on both the reference and the current state of the system to generate the appropriate control inputs. This allows for if the state of the system begins to deviate due to errors or external disturbances the output is fed back into the feedback control so it can adjust the input accordingly. Refer to *Figure 17*. for types of feedback controllers. State Estimation is another critical part of control theory. This is because the state of the system is read using sensors, so measurement noise becomes a problem.

| Type of Controller | Key Characteristics                                                                                         |
|--------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Linear</b>      | Based on linear system assumptions; easy to analyze and design (e.g., PID, state feedback)                  |
| <b>Nonlinear</b>   | Handles nonlinear dynamics; can deal with real-world complexities                                           |
| <b>Robust</b>      | Maintains performance under model uncertainties and disturbances                                            |
| <b>Adaptive</b>    | Adjusts its parameters in real-time based on system changes                                                 |
| <b>Optimal</b>     | Minimizes a defined cost function (e.g., LQR, MPC)                                                          |
| <b>Predictive</b>  | Uses a model to predict future behavior and optimize control actions (e.g., Model Predictive Control - MPC) |
| <b>Intelligent</b> | Uses AI techniques (e.g., fuzzy logic, neural networks, reinforcement learning)                             |

*Figure 17: Types of feedback controllers*

The team must also be able to observe every state of the system even if not every state is directly measured. So the goal is to “reduce measurement noise” and “manipulate measurements

in such a way that allows the team to accurately estimate the state of the system.” [20] It is also vital to test the control system via analysis, simulation, and test to ensure it works properly and gather all data from testing for final conclusions and edits. The primary requirement of the project is to achieve a walking motion, implementing control theory will aid in the design process. In order to achieve motion control, control theory can be used to create a model to regulate the position, speed, and torque of the robot. For example a feedback controller or closed loop controller would adjust the motor input based on the desired state and the actual measured state. Another example would be for trajectory tracking, the control system can in real time adjust the input based on disturbances.

## **Hardware**

The purpose of this section is to define and create an understanding of the hardware and aims to give the reader understanding when selecting Hardware for project use. Proper selection is crucial for planning and implementation of the project. Determining the proper hardware for the project is crucial to achieve the locomotion goal, factors like processor architecture, clockspeed, RAM, Flash memory, I/O ports, Power consumption, operating voltage, and the development environment, must be taken into consideration.

The architecture of a processor is a direct determinant of the performance of the system as “microcontrollers are generally classified according to processor characteristics such as processor architecture, processor word processing length, processor clock operating frequency, and so on.” This determines how well it handles data. For example, “RISC architectural microprocessors are generally preferred in electronic applications such as control and automation, [while] CISC architectures are used in microcomputer applications such as signal processing and embedded systems” [21]. Therefore when choosing the architecture it should

account for computational Power and the complexity of its intended application. The Clock speed, measured in MHz or GHz, showcases how fast a processor can carry out instructions, thus “attention should be paid to technical features such as word processing length, working clock frequency, memory amount, bus width and internal additional hardware” [21]. If it has a higher clock frequency then the general consciousness is that there is also an increase in Power consumption and heat generation. Another key factor to take into account is memory. Random access memory or RAM “stores the information that the processor needs during its operation, while ROM, PROM, EPROM, and EEPROM memories...store information such as command set, program and program data”[21]. Enough Ram is important for dynamic operations while Flash memory is generally used to store firmware or any permanent code. Having enough Flash storage is a necessity so it can adequately hold all the required data and code. Just as important as the memory capacity, the I/O ports are equally as crucial to consider. It is recommended that the designer should list the hardware needed for different “ communication interfaces (USB, SPI, UART etc.)” and list the “port connections... for the input/output elements such as sensor, LCD display, driver and relay” [21]. How much Power the system will consume is also important to consider especially if it is battery-Powered or energy-sensitive. It is important to check the “ energy consumption, operating frequency and other variables” to assure you are making the correct choice in the micro controller [21]. When looking at operating voltage it is important to look at voltage compatibility since it can directly impact the ability and safety of a system. Finally it is important to check the “check the development environment and programming languages provided for the hardware...[and] search the programming interfaces and tools to be used” [21]. Together all these factors are crucial to understand when trying to make the proper choice for which microcontroller to be used for the project.

## **Software/Programing**

The purpose of this section is to create an understanding of the different types of software tools and programming languages. It aims to give the reader understanding when selecting software and programming language for project use. Selecting the appropriate software tools are crucial to insure performance, compatibility and development efficiency. Robotics has a diverse range of programming languages that it relies on. Each language has its own strengths in terms of control abstraction and usability. C/C++ are one of the most commonly used languages due to their low-level efficiency and hardware access in industrial and research robotics, which offered extensions for controlling sensory, actuators, and planning activities [22]. In comparison, Python is versatile for high level logic. It is popularly used for scripting and AI [22]. Robot-specific extensions for Java, like motion description languages have been developed emphasizing object-specific abstraction with portability [22]. Another programming language is MatLab, a specialized environment design for data analysis, algorithm development, and simulation [23]. ROS supports multiple languages, providing tools and libraries for robot software development making it flexible to work with [24]. Micropython is another lightweight and “efficient implementation of the Python 3 programming language that includes a small subset of the Python standard library and is optimised to run on microcontrollers” [25]. Lua is one of the lightweight scripting languages used for rapid prototyping and embedded scripting, it was “designed to be simple, small, portable, fast and easily embedded into applications” [26].

When selecting the programming language there are multiple factors to consider such as: the execution speed and real-time capabilities, hardware compatibility and low-level access, library and API support, community support, and the integration with existing robotics

platforms, as stated in FIGURE B. It is important to evaluate the programming language in relation to the intended task, performance constraints, hardware choices etc.

### **Methods for Algorithmic and Mathematical Techniques**

The purpose of this section is to identify and summarize key methods and programming techniques commonly used in robotics applications. Below is a table (*FIGURE A*) summarizing the technique, purpose, area of application and a source with the method being used. The table in Figure A can be applicable to the making of the project for reference of possible techniques to use.

### **Summary**

Further research into this subject is recommended once design requirements have been designated. The background knowledge as well as all the research provided in this document should be valuable reference for multiple of the development and design phases. I believe that the key concepts provided here should allow the team to do the following: select the hardware, software, design the control system structure and architecture, and implement the techniques mentioned or variations as needed for the needs. It is recommended that further study also looks into the cost and difficulty of implantation and use, this Technology memo fails to take these factors into account and is primarily of the purpose of familiarizing the team with the methods, models, key concepts, and systems used in digital systems and control systems in a generalized matter.

# **System Requirements**

Once primary and customer requirements have been outlined, they need to be expanded upon so individual functions, components, target values, and testing methods need to be determined. These are split up into functional and component decompositions, which are included in Appendix B and C. These will give a starting point to start generating concepts to see what potential designs will be viable for a finished project, and more importantly which ones will not be, or may lead to potential failures down the road.

## **Functional Requirements**

The functional requirements for the final robot are directly taken and expanded upon based on the customer requirements the team has received. Primary areas of focus will include Structure, Material, Power, Articulation, and Safety requirements. These are the “shall’s” that a final robot should be able to meet. Specific design choices for each system are not made at this time; however, target values are set which will guide the decisions for each sub-system as the project progresses.

The key functional requirements of this project include an animatronic that is visually anatomically accurate to a stegosaurus skeleton. The animatronic must be able to walk, and therefore balanced dynamically. Additionally, the motion of the animatronic must reflect a plausible natural motion of the living stegosaurus.

In terms of material functional requirements, the stegosaurus must be durable, and economical. Further, the material should be easily repairable.



Looking at the power requirements, the animatronic must have a reasonable runtime, and a reasonable recharge time. The control system must be able to coordinate between the mechanical and electrical components. The animatronic must be able to respond to user input.

All these requirements have key specific target data, as well as validation methods, included in Appendix B and C

### **Component Requirements**

The component decomposition outlines each individual component and subsystem. The team will need to create or select in order to create the robot. The components are categorized into Control Systems, Electrical Systems, Main Structure and Joints. Each individual component is listed in Appendix B, and the expected requirement the component must achieve to be viable. Target values for validation are set with their associated units which will guide individual design choices for each component. And should be carefully considered when choosing between various component specifications for a single integrated system.

When reviewing the component requirements, the team has determined that the main structure should not exceed 3 feet. Originally, the team determined that each leg will have a minimum of 3 degrees of freedom.

There have been a few updates to the requirements following our discussions with Matt Dhempsey. In short a stegosaurus would have had a limited range of motion compared with what was first envisioned. This boils down to the legs only needing 2 degrees of freedom, as a stegosaurus would have had an extremely limited range of motion in both stepping side to side, and the ankle would have essentially been a fixed element, similar to an elephant's foot.

Additionally, each joint will have a set limit on its range of motion reflected by the skeletal structure of the stegosaurus.

The electrical system must exert sufficient torque to move the limbs to the specified position. It must also exert sufficient torque to output the speed required by user input. exert sufficient to work to move the limbs to the specified position. It must also exert sufficient torque to output the speed required by user input. The control system will be operated from a remote control with a minimum range of 50 feet. It will contain a computer that is capable of controlling the joint actuators so that they reach the appropriate position. Further component specifications are included in Appendix C, along with their validation method.

With each functional and component requirement, multiple analyses will need to take place. In this report, concept selection will be the primary focus, however it is vital to understand why each concept or mechanism is selected over the others, which can be evaluated through comparison matrices. Another important topic to discuss will be the ways in which each of these components can fail, and the severity of damage to the user, environment, or itself that may occur with each type of failure.

## Concept Generation and Selection

In order to come to a final design for the robot, the team first needs to come up with conceptual ideas for what each component or section may look like. More importantly it is good practice to come up with ideas that will not work, as to push the team iteratively towards the final decisions.

Each sub component will need to be analyzed as seen in the component decomposition [Appendix C]. And a run through of the thought processes are below. Starting with simple paper sketches in order to get a rough idea of how the robot will come together and move. Additionally, these will be split into multiple sections, such as Mechanical joints, electrical design, and control design.

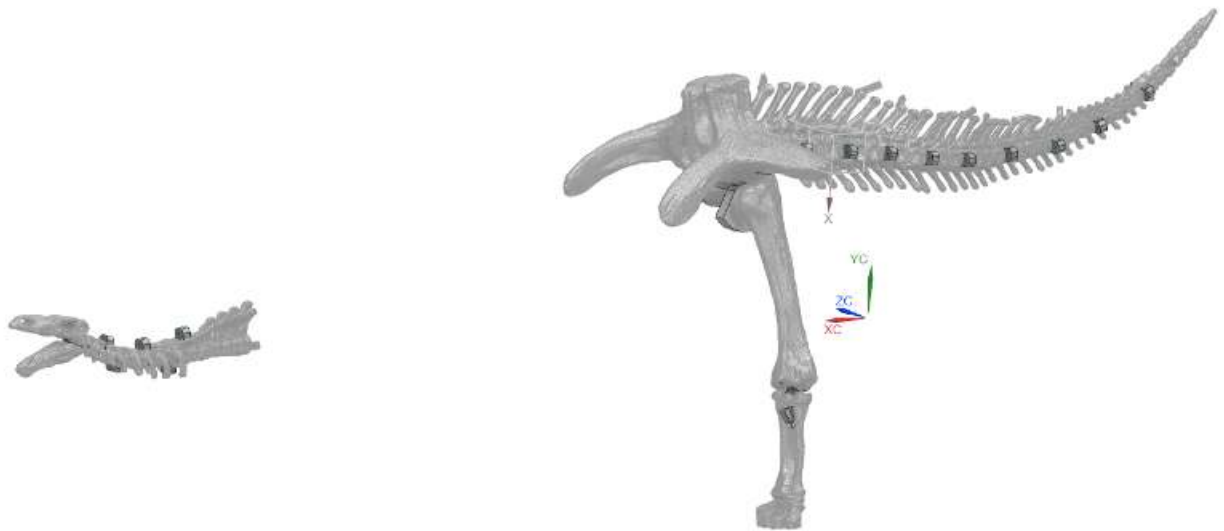
To begin generating concepts, although crude, an extremely effective method at rapid concept generation is through physical sketching. This allows ideas to be written on paper and visualized. After a few concepts are realized for each component. It can then be analyzed deeper with the use of Computer Aided Design (CAD) and Analysis. For this project, The team will be utilizing Siemens NX for physical modeling. Along with Blender and Meshmaker to fine tune the existing 3D scan.

### **Preliminary Layout:**

*Figure 18* is the current representation of our dinosaur. All joints have yet to be fully designed and will be finalized in M5. However it is good to get a rough idea of what the stegosaurus will shape up to be. A few things to note are that the drive mechanisms for the rear legs have been determined, the front legs have not, however their base joint type will be identical to the front legs and will effectively be mirrored to the rear. Additionally, the base model has a target of a 36in length from nose to tail to maximise the available room for packaging

electronics. If later on, it is determined that a size change needs to take place, each segment of the robot can be scaled up or down depending on the specific need of the project.

Our current assembly model of the stegosaurus can give a generalized idea of joint placement, many elements are missing, however almost all of the specific joint mechanism types have been determined and are outlined below.



*Figure 18: Current Assembly*

### **Legs:**

Looking at requirement C.J.1 - C.J.3 of Appendix C, outlines the needed degrees of freedom each leg must have. For the final design, each leg must have at minimum 2 degrees of freedom, and be able to move at a specified angle in order to meet the gait studies. These will be split up into Hips, Knees, and Ankles. And can be flipped or mirrored for front, rear, left, and right legs/arms respectively.

Additionally, there have also been a few recent changes to the design requirements. Recently The team has been in contact with Matthew Dempsey who has a PHD in

musculoskeletal biology and who has done extensive research on dinosaur skeletons including the stegosaurus. After the talkies with him, specifically on the movement of a Stegosaurus, The team can decrease the required degrees of freedom for the legs down to 2.

Another topic for discussion is in the linkage types for the legs, in that consolidated linkages, such as the Jensen Linkage in figure 12 shall be considered. If outright speed and simplicity were the main focus of this project, it may be worth a higher level of consideration. Along with if The team were to use a skin or stuffed dinosaur. The team may be able to better incorporate such a linkage. However the desire to keep as accurate of a skeleton as possible will likely make such a system unrealistic for the scope of this project. So each joint will effectively be an independent system.

### **Hip Joints:**

The hip/ shoulder joints of the dinosaur will be critical to achieving an acceptable range of motion, speed, and rigidity in order to meet the design requirements. In this case, a single degree of freedom will be selected. As in with the talks with Matt Dempsey, the stegosaurus had a limited flexibility range, and would not have been able to “splay” their legs outwards very much. This can also be beneficial for packaging as 4 motors will be able to be removed, which will improve overall packaging and Power requirements. This will also help to make the hips more rigid, which will make the programming and overall build process simplified.

To drive each axis of the hip joints, there are a few viable options, these include a direct drive, a gear reduction, and a capstan drive as seen in figure 8. Below is a breakdown of each drive type and their advantages/ disadvantages. For problems like this, a Pugh Matrix can be used to compare different concepts to desired attributes. Below, figure 18 outlines these 3 options, and it can be seen that a capstan drive may be favored over the others. As the team can

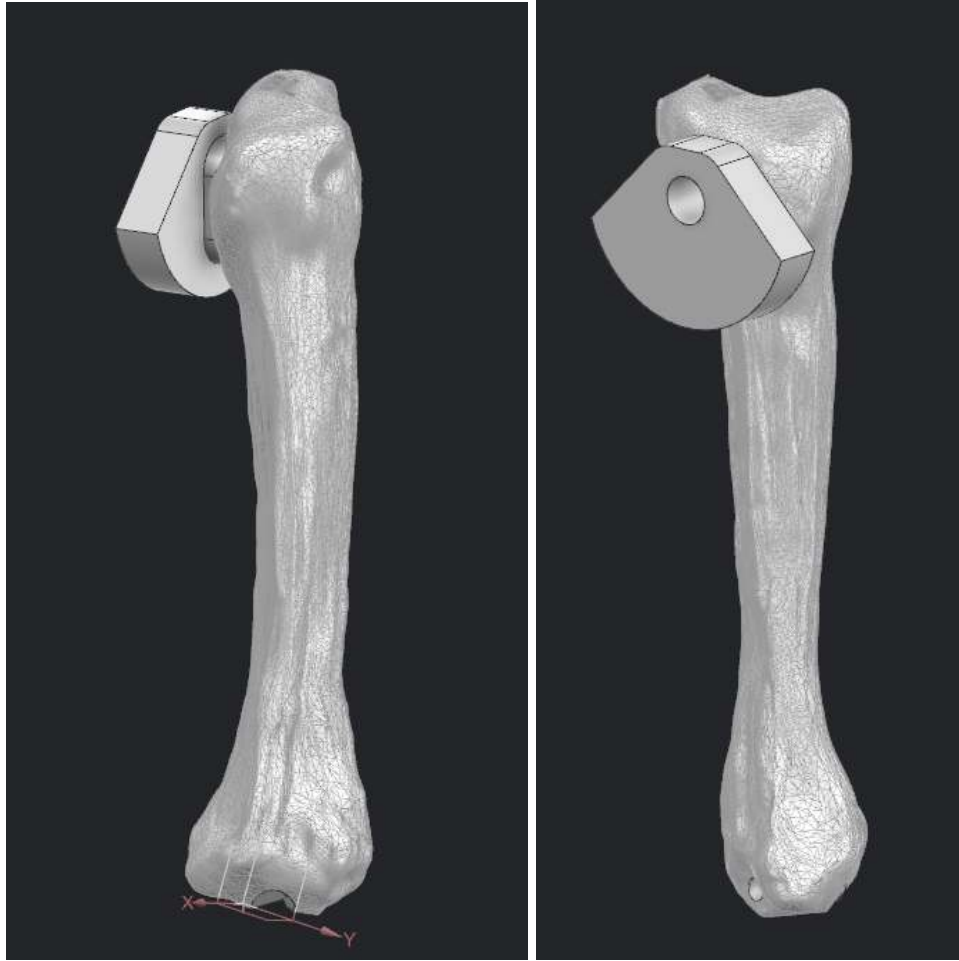
reduce slop with a large effective gear reduction however special attention will need to be placed on packaging.

| Hip Forward Drive Type    |                                                                  |                       |              |                |               |   |
|---------------------------|------------------------------------------------------------------|-----------------------|--------------|----------------|---------------|---|
|                           |                                                                  | Solution Alternatives |              |                |               |   |
| Key Criteria              | Importance Rating                                                | Benchmark Option      | Direct Drive | Gear Reduction | Capstan Drive |   |
|                           | <u>Concept Selection Legend</u><br>Better +<br>Same S<br>Worse - |                       |              |                |               |   |
|                           | Required Motor Torque                                            | 8                     |              | -              | +             | + |
|                           | Slop                                                             | 6                     |              | +              | -             | + |
|                           | Packaging                                                        | 8                     |              | +              | -             | - |
|                           | Noise Level                                                      | 2                     |              | S              | -             | + |
|                           | Joint Flexibility                                                | 4                     |              | +              | S             | S |
|                           | Tuneability                                                      | 4                     |              | -              | +             | + |
| Weighted Sum of Positives |                                                                  |                       | 18           | 12             | 20            |   |
| Weighted Sum of Negatives |                                                                  |                       | 12           | 16             | 8             |   |
| TOTALS                    |                                                                  |                       | 6            | -4             | 12            |   |

Figure 19: Hip Main Drive Matrix

Looking at figure 19 below, the initial model for a capstan drive is shown. The femur will contain the “Base plate” and a smaller drive wheel will be attached to a servo motor mounted to the hips of the dinosaur. This will allow for the motors to be set inwards to the dino in an attempt to hide them from main view. Another benefit to this drive type is that an effective gear ratio can be introduced, in this case the team will be able to achieve between a 2-4:1 gear reduction. This will lower the final speed of the joint, however it will greatly increase the effective torque output.

In the event that the final gear ratio found is not sufficient, it is possible to easily tune this ratio by revising the geometry of the base plate and wheel attached to the servo motor itself.



*Figure 20: Femur with base for capstan drive attached.*

Prior to the change in design requirements, The team had initially planned on having a 2nd degree of freedom in the hips. In order to achieve this, a similar capstan drive would have been selected, however the base plate and femur would not have been directly attached to one another. And would have included an additional hinge with a linkage to provide control of “splay” in the legs.

Figure 21 and 22 depicts a concept for a base plate for forward and aft movement for each leg. A main disc with a central hole for an axle shaft, with 2 tabs for the leg to both mount and pivot is provided. Another consideration may be to use a hollow axle that can have cables or wires passed through the center.

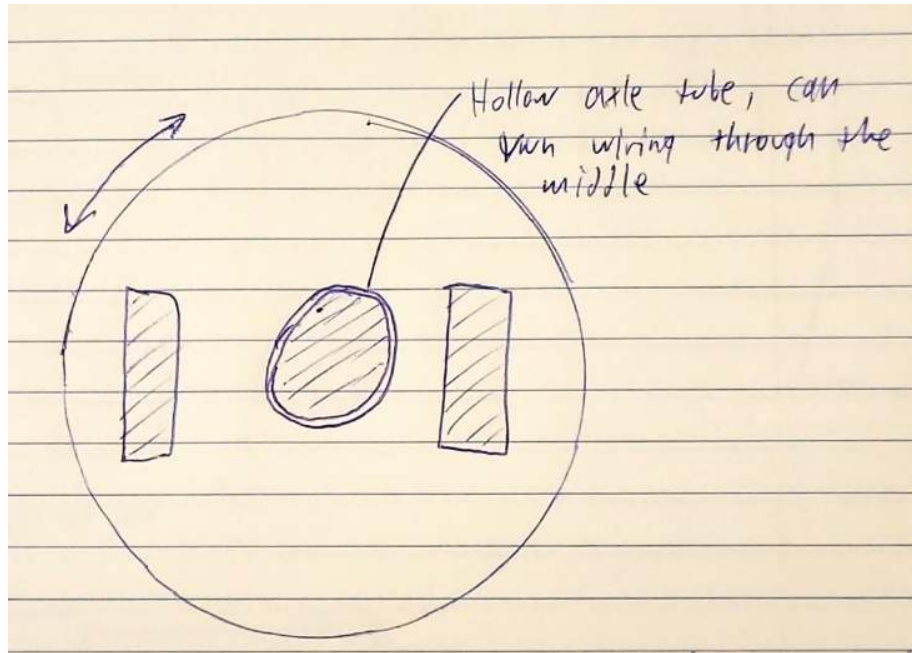


Figure 21: Hip Joint Base

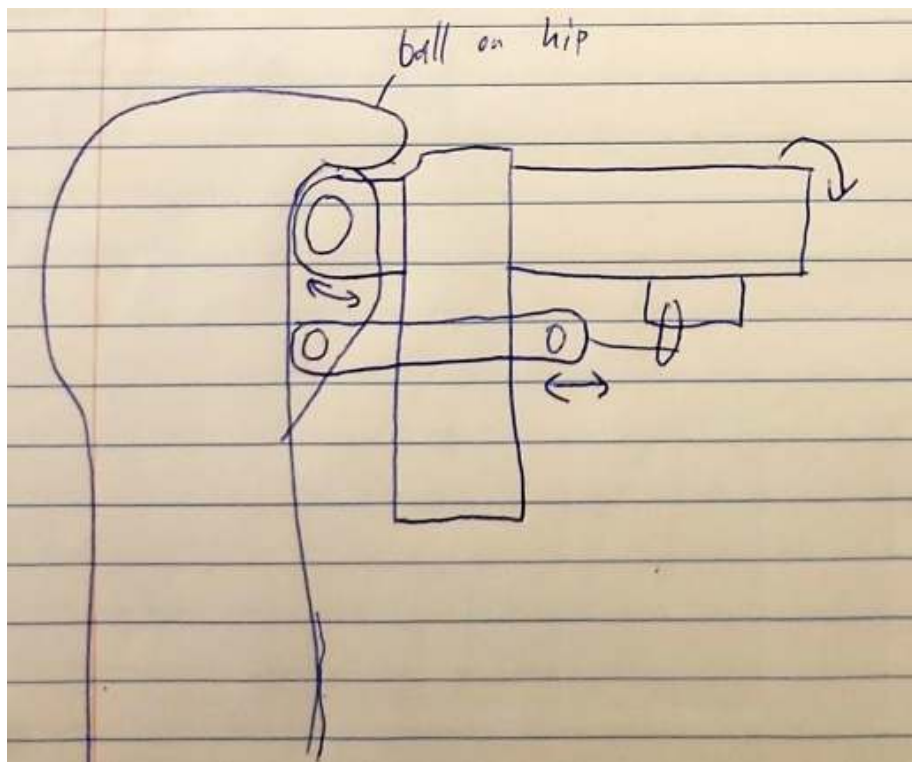


Figure 22: Hip Joint Side to Side Mechanism, rev A



## Knee Joints:

Physical knee joints will likely end up being simplified when compared to the hip, however packaging will become a very important topic. The knee will only need to have a single degree of freedom, so a simple hinge joint can be used, and hidden using the existing skeletal structure [Figure 22]. The challenge of the knee joint comes with the fact that it still needs to be an active joint, being precisely driven by an electric motor. And the fact that one of the primary goals of this project is to mimic realism and lifelike movement. This boils down to the fact that there is not much room to integrate a motor directly onto the knee itself, without it protruding and breaking visual accuracy. The ultimate goal will be to separate the joint itself from its drive motor, and have the joint effectively be controlled remotely by cables [Figure 23].

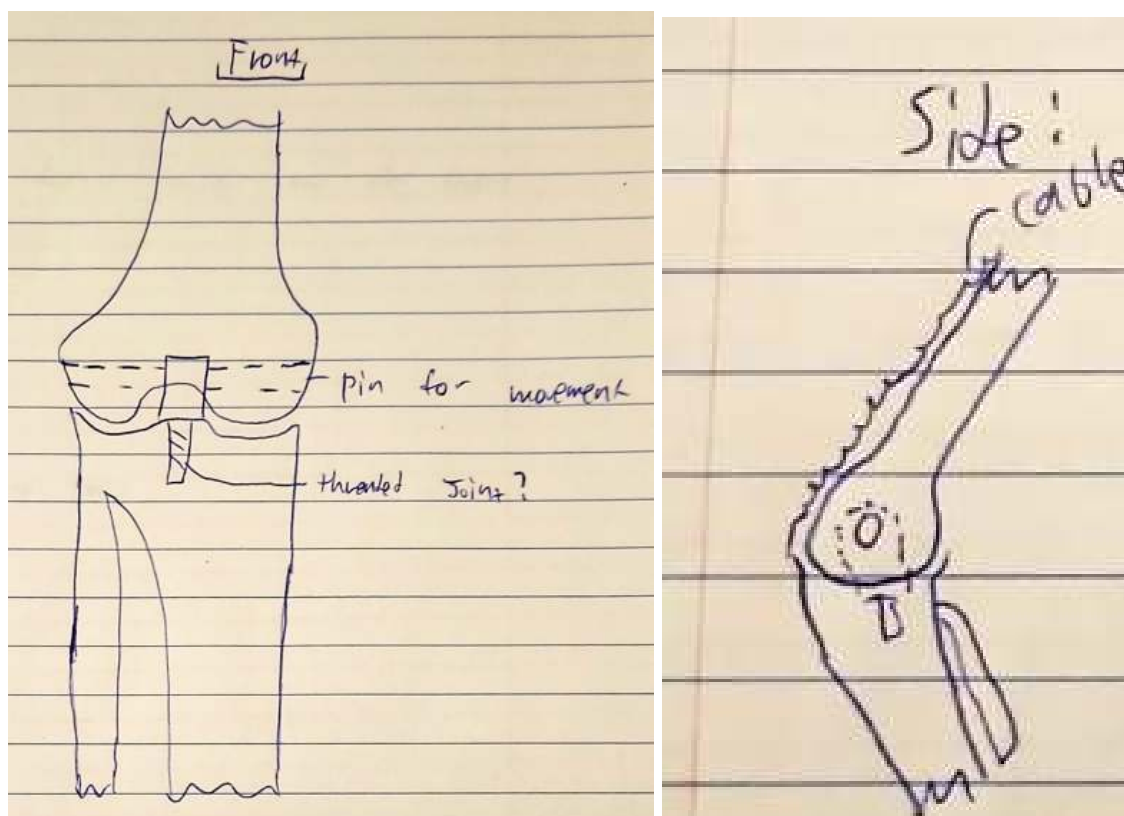
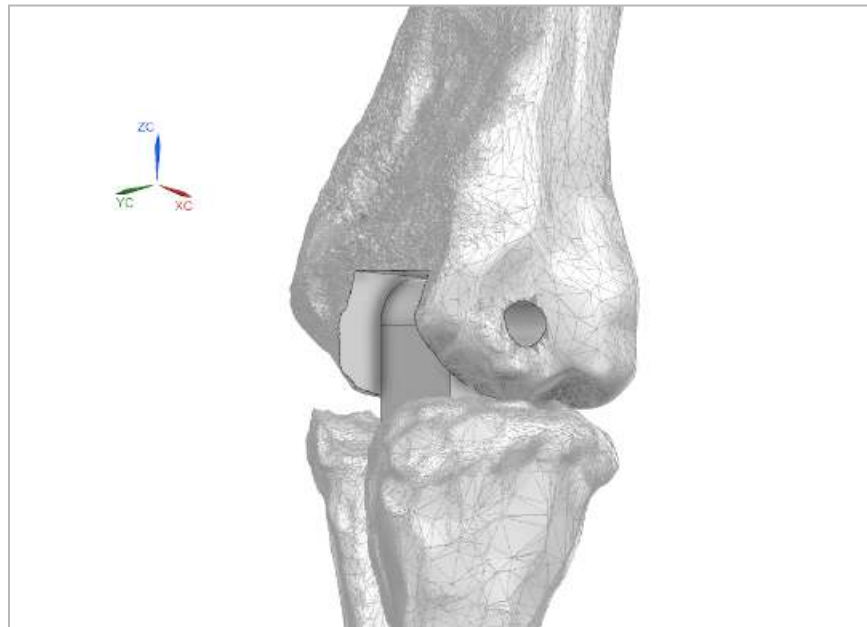


Figure 23: Knee base joint

On the topic of motors The team may be able to utilize linear actuators here, which would allow for the actuator to essentially act as a muscle with cables for tendons. However, their relatively slow speeds will likely not allow the team to meet the final walking speed goal set by the functional requirements, so this will likely take the form of a standard servo placed towards the interior of the dinosaur. However, it is curious to take inspiration from bio-mechanics through how muscles and tendons would have been set up if it were a living creature.

With the control motor being separated from the physical joint does present a few challenges, the more severe of which being that The team need a way to route the cables through another moving part, while attempting to minimise added slop or movement when the hips move separate to the knee. It may also be possible to counteract some of this on the software side of things. however an effort should be made to minimise this to simplify the finished software.



*Figure 24: Base model for Knee Hinge:*

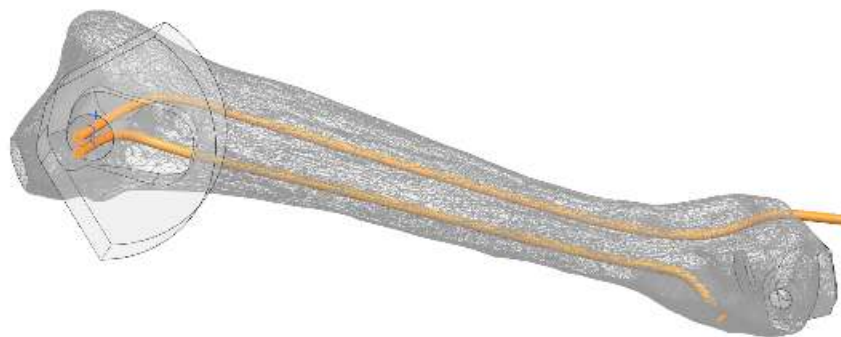
To begin the knee design, an initial hinge is needed to provide structural support and a rotation point. When looking at the knees of living creatures, they are not fixed at a pin, but

provide a sliding motion. This can be mimicked by moving the rotation point of the hinge upwards into the femur, here the top joint on the tibia will appear to slide over the femur instead of rotating at a hard point.



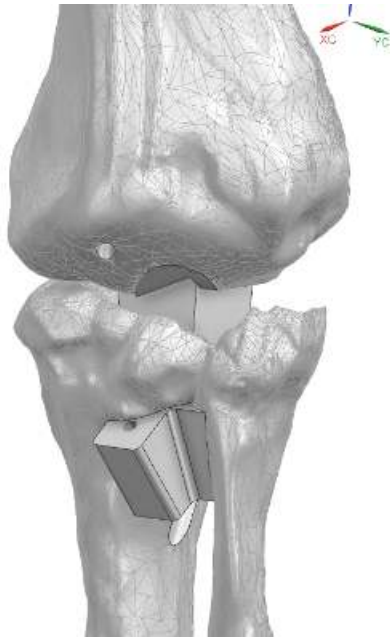
*Figure 25: Knee Range of Motion:*

Onto the topic of range of motion, in order to mimic biological movement, an overall angle goal is to be set at equal to or greater than 90 degrees. In the initial design, a range of motion has been set at 110 degrees from locked out, to fully bent back as seen in *figure 25*. The sliding effect of tuning the hinge point for the knee as previously discussed can also be seen.



*Figure 26: Internal cable routing tubes in the femur.*

To actuate the knee, it is greatly beneficial to relocate the drive motors away from the main view to keep a high level of visual accuracy. To accomplish this, 3D printing can be utilized to integrate cable guides/tubes into the femur itself in *figure 26*. These will exit in the middle of the drive axle on the hip joint, and exit through a hollow tube to then be connected to a servo motor. These cables will be attached to mounting blocks on the tibia (*figure 27*) and can be pulled from either direction to make it move.



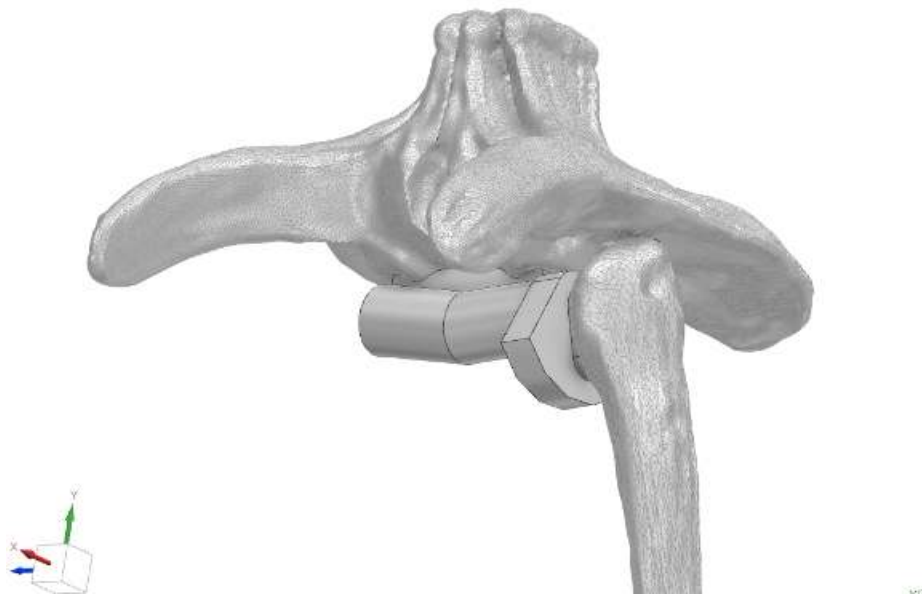
*Figure 27: Cable mounting blocks on the tibia.*

### **Ankle Joints:**

Looking at the ankles of the dinosaur. A stegosaurus has relatively stubby legs when compared to other dinosaurs and is similar to those present on an elephant. In this case, ankle movement will not play a critical element in the ability for the dinosaur to be able to walk. The team may be able to integrate an elastic or flexure joint such as the one used in [*Figure 11*]. however it will serve a greater benefit to simplify the overall leg design.

## Hips:

After the leg joints have been selected, they need to be mounted so they can rotate and support the full weight of the final prototype. To accomplish this, bearings can be mounted into some integrated supports for the hips. A nearly identical design will be used for the shoulder so only the rear legs will be shown in *figure 28* below. Servo motor mounts will also be mounted to this point and this is where the capstan drive wheel will be connected to the hips themselves. For the knees, a takeoff point can also be placed into the central part of this mount where the cables will exit, and connect to additional motors that will provide movement to the knees.



*Figure 28: Hips with a single leg attached at a pivot point.*

## Tail:

The tail of a Stegosaurus makes up nearly 50 percent of its entire body length. And although it will not directly impact the walking ability of the robot. It will be required to move for both balance and emotion. Per the current design requirements, the tail will need at minimum 8 separate vertebrae sections that are able to articulate a total of 45 degrees side to side across the length of the tail. In order to meet this requirement, the tail must first be effectively straightened

from the base model/ scan which was accomplished in Blender. The joints for each vertebrae can be fused to an extent, as the tail includes approximately 40 unique vertebrae so each individual one will not need to move.

| Tail Joint Types             |                   |                       |            |                 |                    |
|------------------------------|-------------------|-----------------------|------------|-----------------|--------------------|
| Key Criteria                 | Importance Rating | Solution Alternatives |            |                 |                    |
|                              |                   | Benchmark Option      | Pin Joints | Flexible Joints | Full Flexible Tail |
| Ease of manufacturing        | 2                 |                       | -          | +               | +                  |
| Flexibility                  | 8                 |                       | S          | S               | +                  |
| Realism                      | 6                 |                       | -          | +               | +                  |
| Controllability/ Constraints | 6                 |                       | +          | S               | -                  |
| Ease of Repair               | 4                 |                       | +          | -               | -                  |
| Sum of Positives             |                   |                       | 2          | 2               | 3                  |
| Sum of Negatives             |                   |                       | 2          | 1               | 2                  |
| Sum of Sames                 |                   |                       | 1          | 2               | 0                  |
| Weighted Sum of Positives    |                   |                       | 10         | 8               | 16                 |
| Weighted Sum of Negatives    |                   |                       | 8          | 4               | 10                 |
| TOTALS                       |                   |                       | 2          | 4               | 6                  |

Figure 29: Tail Joint Pugh Matrix

Looking into joint types there are a few options as always, first will be to add a mechanical hinge with a single pin similar to the knee. Another method may be to add flexure joints in between each segment, and make these joints from a flexible material such as TPU. However the final idea of making the entire tail a flexible unit can simplify the design, and make its movement appear more fluid.

An actuation method will also need to be designed, an interesting option will be through using a linked tendon. This allows for a pair of cables to act as a tendon, and will result in a curved movement path across the length of the joints or in our case vertebrae in the tail. An initial design is below [figure 30].

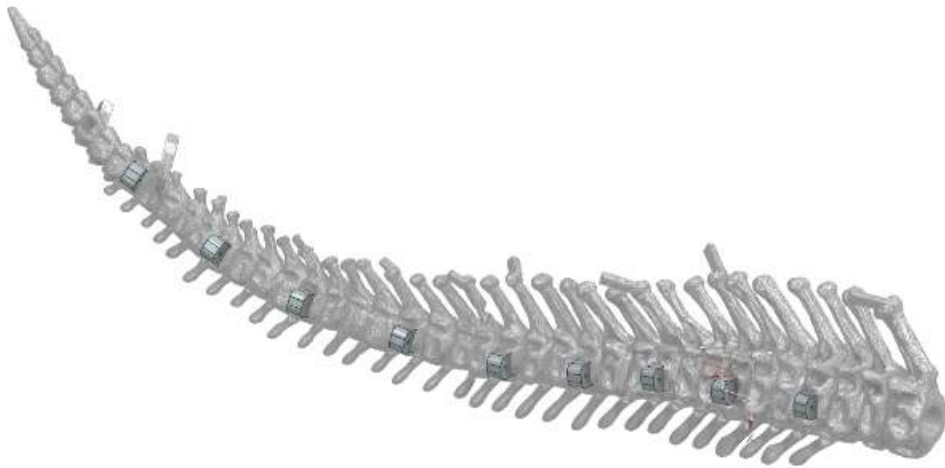


Figure 30: Tail with pull blocks for cables.

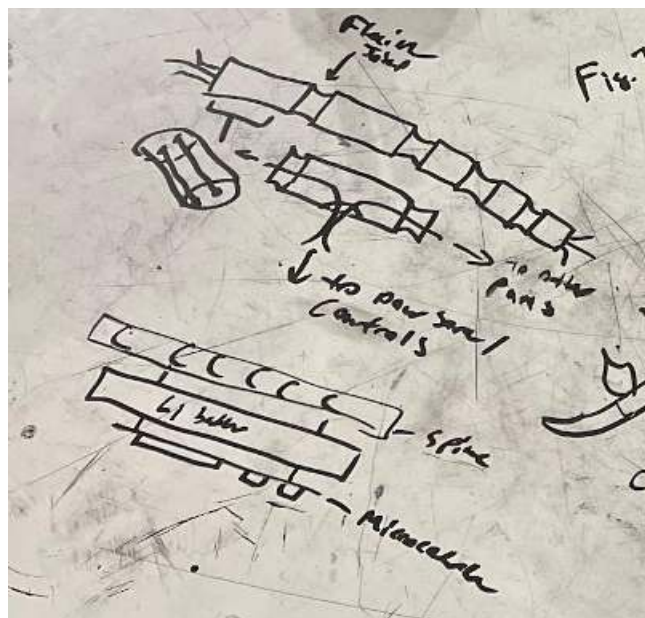


Figure 31: Example of Flexure Joints

## Neck:

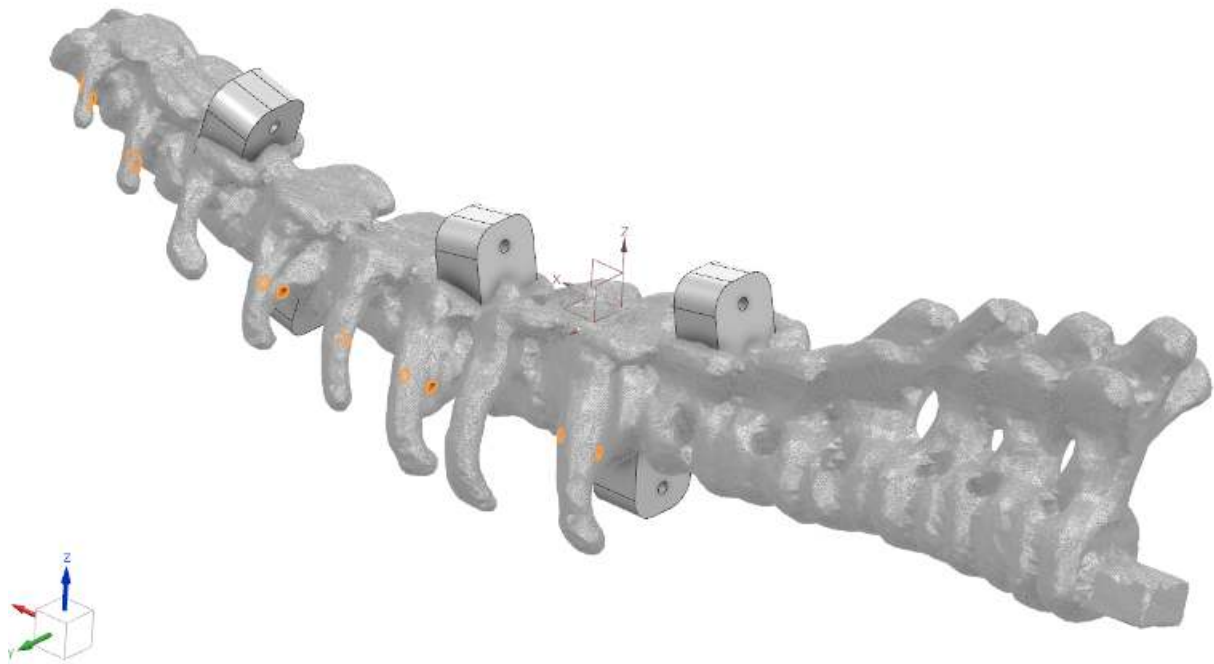
Similar to the tail, the neck will contain multiple vertebrae, in this case it will require a minimum of 3 moving sections. however it will need to be able to support the neck, move in 2 axes (up/down and left/right). Along with routing wires up to the lights in the eyes, linkage for the mouth, and being able to support the weight of the skull. The same joint types will be available for use, however the weighting and selection is different per figure 30. It can be seen that a base flexure joint is preferred due to its higher static capacity, as there are individual flexible sections unlike an entirely flexible neck. And will be easier to control in 2 axes as a pin style joint would require 2 pins in order to meet the flexibility goals.

| Neck Joint Types             |                   |                       |            |                 |                    |
|------------------------------|-------------------|-----------------------|------------|-----------------|--------------------|
| Key Criteria                 | Importance Rating | Solution Alternatives |            |                 |                    |
|                              |                   | Benchmark Option      | Pin Joints | Flexible Joints | Full Flexible Tail |
| Ease of manufacturing        | 2                 |                       | -          | +               | +                  |
| Flexibility                  | 8                 |                       | -          | S               | +                  |
| Realism                      | 6                 |                       | -          | +               | +                  |
| Controllability/ Constraints | 6                 |                       | -          | S               | -                  |
| Ease of Repair               | 4                 |                       | +          | -               | -                  |
| Static Weight Capacity       | 6                 |                       | +          | -               | -                  |
| Cable Packaging              | 8                 |                       | +          | +               | -                  |
| Sum of Positives             |                   |                       | 3          | 3               | 3                  |
| Sum of Negatives             |                   |                       | 4          | 2               | 4                  |
| Sum of Sames                 |                   |                       | 0          | 2               | 0                  |
| Weighted Sum of Positives    |                   |                       | 18         | 16              | 16                 |
| Weighted Sum of Negatives    |                   |                       | 22         | 10              | 24                 |
| TOTALS                       |                   |                       | -4         | 6               | -8                 |

Figure 32: Neck Joint Pugh Matrix



A preliminary model of the neck is shown below in *figure 33*, per the pugh matrix this will end up being a fully flexible section that will utilize tendon style drives. Another observation can be made in the side cable routing. The initial scan was first modified by sculpting the mesh in both blender, and then MeshMaker in order to add a solid structure in between the forward vertebrae structure and the vertebrae to mimic bone structure. And although it is not biologically accurate, it provides a clever way to minimize protrusions from the exterior boundaries of the neck itself.

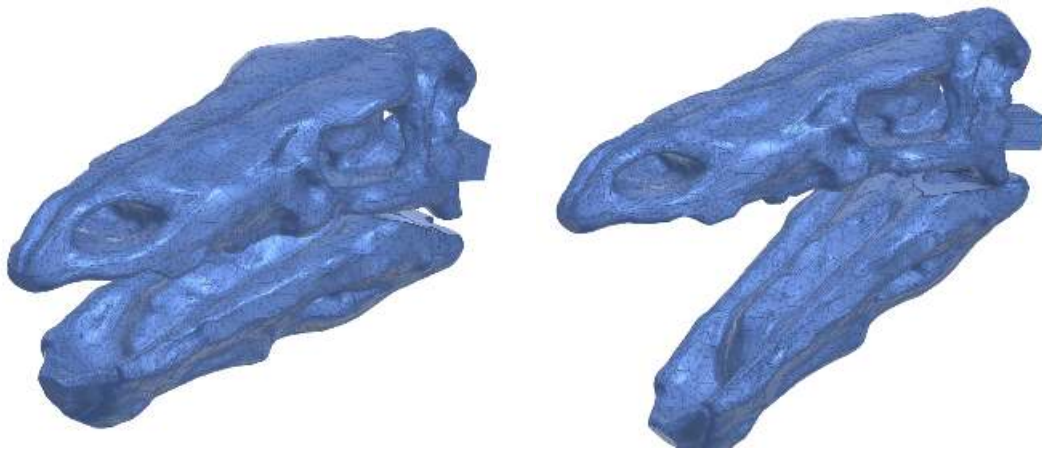


*Figure 33: Base Neck Design*

### **Skull/ Mouth:**

The skull and mouth will need to provide non-structural movement, and will not be critical to the final operation of the robot. however it will be necessary to provide an emotional element to the dinosaur. A simple hinged design will be sufficient to produce the necessary 20

degrees of movement outlined in the functional requirements. Movement will be provided from a micro linear servo that can fit within the boundary of the skull itself. Torque requirements will be minimal as it will only need to provide aesthetic movement and will not be interfacing with a solid surface.



*Figure 34: Skull with mouth closed vs open.*

To drive movement for the mouth 2 main options can be considered, that is in a servo and a linear actuator. In short, a rotational servo will be able to provide a higher speed, however it may become challenging when it comes time for packaging. Another option is to include a linear actuator, then run a linkage or cable down the neck into the main section of the robot. Where it can then be moved by a linear actuator.

## **Controls:**

In regards to the programming section of the project, the following section defines the generalized plan and functional requirements of the control code. It will go in depth on how software will interact with the physical parts of the robot and execute physical movements.

### **Code Structure Overview:**

Once powered on, the selected microcontroller will run a startup routine. It will initialize all hardware components, set initial parameters, and enter State 1 (Idle State). The initialization should confirm all systems are operational before proceeding. For example it will confirm that the remote control is connected and all motors are responsive. If any issues arise during start up, the user will be prompted with an audio cue, alerting that something has gone wrong. If no issues occur and the start up is successful, a different audio cue will play showing the user that it is safe to proceed with standard operation of the robot.

Once time to power off, the robot needs to shut down safely thus it will have a shutdown procedure to ensure all hardware components are powered down and data is saved to non-volatile memory. The shutdown procedure should also ensure that all motors are set to a safe position before fully powering down the entire robot.

Another system feature the code should account for is an emergency stop function. The emergency stop should immediately hold all motor activity and put the system into a safe state when triggered by a specific input. The input can be a button press, a remote command, or an automatically detected condition. One condition could be the gyroscope being a specific position which would indicate the robot has fallen over. Other possible triggers for the emergency stop to

activate could be a critical error being detected, overheating, motor stall, or a loss of communication with the remote control system.

For maintenance and troubleshooting purposes, there should be a dedicated section of the code to handle data logging. The system should record all variables, possible errors, connectivity issues, joint positions, response times, controller inputs, current power level, and the temperature of the system. All the data collected should be recorded and logged on an SD card or flash drive. This allows for an easier identification of when parts need to be replaced or when parts are not working.

Designated code must be included to handle and manage all incoming signals from the controller to execute the robots physical commands. Many microcontrollers and micro computers already come with built-in libraries and additionally there are plenty of publicly available libraries for different types of remote control systems that can be used to handle controller

inputs. The code needs to continuously monitor and process inputs from the remote control, such as joystick movements, button presses, and trigger presses to determine desired pre-coded actions and states.

In terms of audio management, the code is expected to play various sound effects or audio cues at specific events or states from start up. The sounds are used to signal when there has been an error or

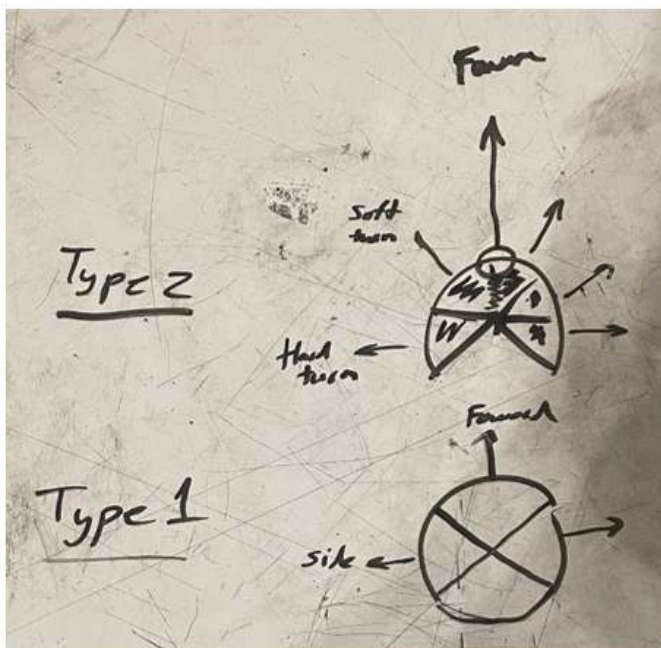


Figure 35: Joystick Inputs.

a successful startup. The main way the users will be able to determine if something is wrong or if all is good to proceed will be via audio cues. The audio will also be used for various animations to bring this robot to life.

In regard to power management, the code is expected to cue the user through some kind of visual or audio cue to indicate that the battery is low and a recharge is needed; This may be through an audio chime or a LED light indicator.

Included in the code, the robot is expected to have four different modes and to seamlessly transition between the states based on user input; these modes are currently defined as States 1-4.

### **State 1,2,3,4 Descriptions:**

State 1, otherwise known as the idle state, refers to when the robot's motors are set to a pre-coded position and are now waiting for further user input. This state is mainly for start-up purposes and as a safe transition state to other states.

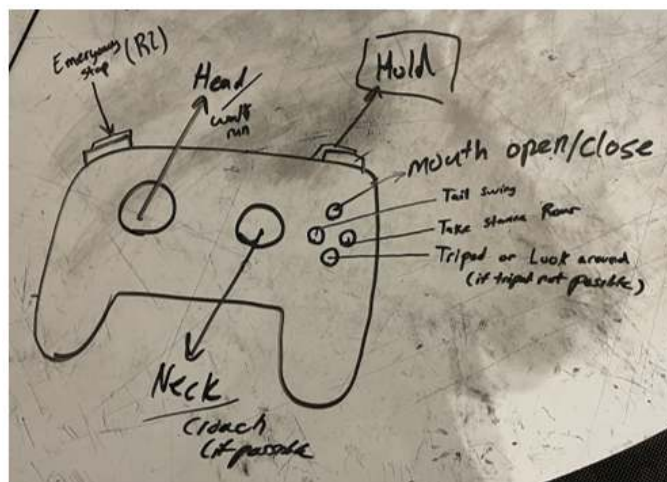


Figure 36: Theoretical key mapping of inputs.

State 2 refers to when the robot is in motion. This state is described as when the motors move according to registered inputs and the gait study in a predefined cycle. The stop and start of state 2 is all through user input, and shall remain in this state until a stop input is received.

The controller joystick and buttons can modify behavior, speed, and direction. If a stop command is received the robot is expected to transition from state 2 to state 1 before proceeding to any other states.

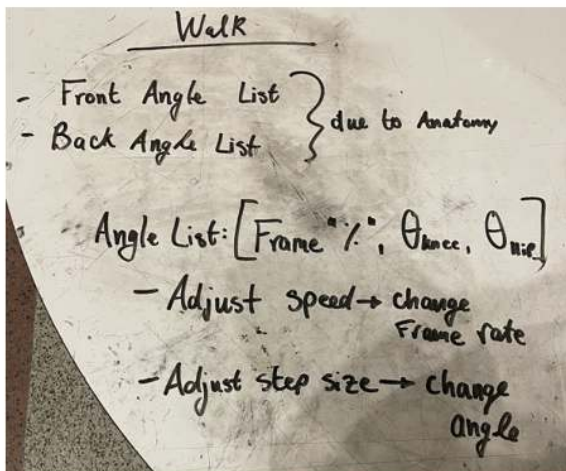


Figure 37: Concept of how the data from gait study will be used

State 3 is defined as an animation state. The motors are expected to move according to a predetermined animation indicated by an input and executed to completion before returning back to the idle state. Through the user's input they may disrupt and stop the animation halfway through and return to idle state. The robot is projected to have

three animations: a tail swing, a roar animation, and a third emotive animation which has yet to be determined. The tail swing animation is expected to have the robot look around and set up and swing its tail as if attacking. The roar animation is expected to have the robot dinosaur take a stance and open its mouth when an audio effect should sound.

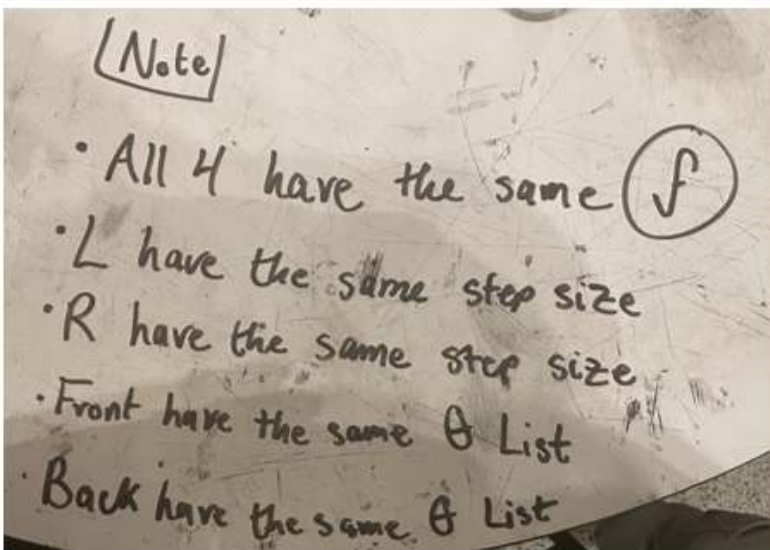


Figure 38: Concept of the data received from gait study

During the fourth and final state, State 4, the head, neck and tail will be active and will be controlled directly. The head, neck, and tail motors should move according to user input. The robot is expected to be able to move its head and neck left, right, up, and down. Likewise

the neck and tail should be able to move left and right. The jaw should also be able to open and close. The joystick and controller buttons are expected to be able to modify motor behavior, such

as speed and direction, to carry out these physical motions. Once state four has been initiated by the user, a second input must be signaled to exit this state and return back to the idle state.

### **Electrical System:**

The electrical system poses a unique problem due to the power supply needed to power all components within a highly constrained space. This requires a high density battery that can be easily accessed, recharged, or replaced to support repeated operation. The battery should be able to provide enough power to all subsystems for a long duration or for repeated use. The power must be able to supply two different voltages and current requirements. The control computer, microcontroller boards, and the audio modules operate at approximately 3.3-5 V with relatively low current demands. In contrast, the motor components require between 6-8.7 Volts and can draw higher current under stall conditions. The power system needs to be able to accommodate this, the following section outlines the power subsystem in detail, including the componentment, electrical considerations and rationale behind design choices to address constraints.

### **Battery**

Battery selection requires careful consideration of several factors, but first appreciate the class of battery for the system but be identified. The battery can fall into three categories: Primary non rechargeable batteries, secondary rechargeable batteries, and specialized industrial batteries. Common examples of primary non-rechargeable batteries are, alkaline batteries, zinc carbon batteries, silver oxide and zinc air batteries. Secondary batteries are rechargeable so these can include lithium ion batters, lithium polymer, nickel metal hydroxide, nickel cadmium, lead acid, and lithium iron phosphate batteries. Batteries like sodium ion and solid state batteries are classified as special industrial batteries, commonly defined by their advanced chemistries for high performance, long life cycle or unique conditions.



| Pugh Matrix                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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| <div><u>Concept Selection Legend</u><br/>Better +<br/>Same S<br/>Worse 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Figure 39: Battery Pugh Matrix

The selected battery needs to accommodate the following criteria: long life span, rechargeable, compact size, and it must be capable of supplying both the high-voltage, high current demands and the low voltage and low current demand without overloading or damaging the lower-voltage components of the robot.

The battery must be able to support an extended period of operation to meet the demands of museum use. If the battery were to wear down or need to be replaced, the selected battery must be easy to source from common suppliers. This will allow the museum to easily replace the



battery at a more economic cost. For example in a theoretical scenario if the robot were to have its power go out, the battery should be easily bought off of major online retailers like Amazon or any other in person option. This would significantly reduce downtime in the case where the battery needs to be replaced, as the battery will be easily accessible and might even allow for stocking spare if the selected battery for quick replacement.

Due to the different voltage and current demands, a single and multiple battery system was considered as well as a parallel battery system. The parallel battery arrangement would maintain the system required voltage but increase the current allowing for higher current accommodation that the motor system requires. Ultimately, a single battery system was selected, as suitable battery models were identified to satisfy the necessary high voltage high current requirement while remaining compact enough to fit within the tight space constraints of the animatronic .

To conclude, the key factors for battery selection were runtime, output voltage, volume, capacity, and overall weight. The main goal of the selected battery was to be easily usable while maximizing energy storage and minimizing size and mass. Having a higher power density was essential to achieve the desired longer operational periods for the anatomic. The team aims for the robot to have roughly 30 minutes of operation at maximum strain and an average operation time of 1 hour and 30 minutes to 2 hours under normal operation conditions.

Batteries commonly used in drones and RC vehicles were recognized as strong candidates for the robot. The two types of Lithium-polymer (Li-Po) type of batteries below offered compact sizing, rechargeability ease, and met several other requirements, making them well suited for our defined constraints.



*Figure 40: Lithium-polymer Battery*

## Charging Port

Because of maintenance and for the ease of the museum, a chargeable battery was determined to be the most practical choice. The rechargeable system eliminates the need for repeatedly opening or disassembling the animatronic to replace batteries. Instead the museum would operate on a plug and charge workflow, making it far more accessible. Due to this a charging port will be incorporated into the robot.

## Power Switch

To power on and off the robot, the team decided to integrate a power switch on the robot. Multiple switch types were considered including toggle switches, slide switches, and rocker switches as potential candidates [58]. Because the system is going to have a battery with a voltage output ranging from 7.4 to 12 Volts and may experience high current draw from the motors, the switch must be able to withstand these conditions without melting or welding shut.



*Figure 41: Toggle Switch*

The design will include a manual on and off switch in so that the system is not constantly running power when the battery is plugged. The selected switch was identified as a toggle switch, due to its ability to handle high-level current and its reliability to withstand the currents when the motors are stalled. The motors are expected to require at max stall is roughly 32 Amps of current which would strain the system but the toggle switch would be able to handle high current without any major concern. It is also important to note that the probability that all motors will ever achieve max stall conditions is low and is only taken into consideration as a precaution.

The animatic switch will be located in an easily accessible area, which will allow the users to toggle the power system on and off without worry of injury. The switch is only intended to be used before the robot is in operation. It should not be used while the robot is in motion. It should be powered off once all action has halted and it is in its idle state ready to be powered off to conserve the battery [58]

### **Low battery indicator LED light**

An LED light will be incorporated to provide a visual state of the battery's power level. When the battery is fully charged it the LED will indicate to the operator by displaying a designated color. When the battery is the LED will change to an alternate color to alert the user that it is time to charge the robot. This visual feedback allows the user to check the battery status without accessing any internal components.

### **Fuse**

For safety, a fuse will be integrated in the robot's power system to prevent any accidents or damage to the robot. It will serve as a safe guard to prevent any electrical failure or overheating.



*Figure 42: AMP Fuse*

### **Power BusBar**



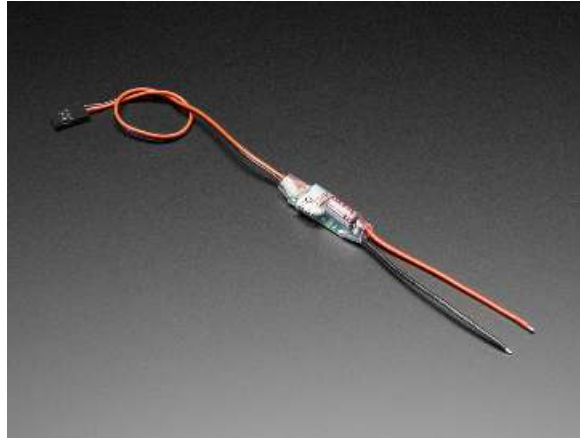
*Figure 43: BusBar*

Since a single battery system was selected to power two systems of varying requirements, a power bus is required to divert power to buck converters. This will allow us to convert voltages and current to the needs of the components of the system. Keeping the motors and MCU systems power separate allows us to prevent any possible backflow or damage to the control systems from the high voltage.

### **Buck converter less than 5V low amp**

The power system design includes a low-voltage low-current buck converter, in order to meet the needs of the lower voltage components without overloading the system. The buck

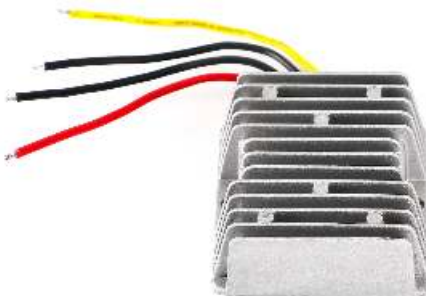
converter will allow us to convert the high voltage, medium to high current battery to lower voltage and lower current for the control system. This will be used to power the microcontroller and the peripherals included in the microcontroller.



*Figure 44: Low Amp Buck Converter*

### **Buck converter 6V high amp**

Likewise, a higher voltage high current buck converter will be used to help power the motor system. The motors require a higher operating voltage and current at varying levels. Using a buck converter ensures that the motors receive stable and sufficient power while isolating it from the lower voltage control system.



*Figure 45: High Amp Buck Converter*

## **Control System:**

Within the control system design, the team had to determine whether to develop a custom electronics solution or adopt an off-the-shelf control system. Both approaches had their own drawbacks and advantages but the final determinant was the museum's needs as the end user.

The museum requires a system that is easy to maintain, service and replace if a part fails. The museum staff needs to be able to learn how to assemble, disassemble and trouble shoot the control hardware, therefore it must be feasible to learn and understand within a short period of time. Because of this, an off the shelf would be more practical as a custom solution would be harder to learn, fix, and replace. Although a custom solution would allow for greater space efficiency, commercial microcontrollers would be readily available, more economic and familiar to a wide user base. Although the commercial control systems may have parts that will go unused in our project because they are part of the general purpose design, the benefits of maintenance and replaceability outweighs the cons.

## **Micro Controller/Micro Computer**

To begin the control system design, we had to decide whether a custom made PCB or MCU tailored to our robots needs or an off the shelf option was the more viable while considering user needs and the animatronic's constraints. Using a custom PCB or MCU would theoretically allow us to eliminate any parts or functions that would go unused in a commercially ready PCB/MCU and it would allow us to meet our packaging requirements and use the small amount of space we have efficiently.. In contrast, off the shelf options like Arduino boards, Raspberry Pi models, TI microcontrollers, and ESP32 variants come ready to use and are widely

supported. This was the first designation that set the foundation for the rest of the control architecture.

The team decided on an off the shelf variant as the benefits outweighed the drawbacks, the main determinants being the ease of maintenance, replaceability, and it being the more economical choice to be readily available. A custom board while would be more space efficient would only be hard to maintain, hard to replace, and may be more expensive due to it being custom built. After making the decision, a handful of options were considered with the top three finalists being Arduino Raspberry Pi and ESP32- S3. Below is a table of generalized comparisons between specific boards that were brought into consideration and the criteria considered in each. Throughout the decision process we continuously prioritized simplicity. A MCU/PCB that is too complex risks the system being too hard to understand and learn. A simple MCU/PCU may also minimize failure points, so when different subsystems become interconnected a more complex control system risks cascading issues. So to avoid this, selecting a reliable and straight forward off the shelf model was our main priority.

| Feature               | Arduino | Raspberry Pi                 | ESP32-S3-WROOM-1U  |
|-----------------------|---------|------------------------------|--------------------|
| Real-time performance | ✓       | X                            | ✓                  |
| Wireless built-in     | X       | ✓                            | ✓                  |
| Power consumption     | Low     | High                         | Low                |
| Size                  | Medium  | Large                        | Very small         |
| Motor control timing  | Weak    | Unreliable                   | Excellent          |
| Maintenance           | Easy    | High                         | Very easy          |
| Cost                  | Low     | Medium                       | Low                |
| Multitasking ability  | Weak    | Strong but non-deterministic | Strong + real-time |

Figure 46: Micro Controller Comparison

Before selecting the final microcontroller, the following array of various different controllers were individually reviewed and considered before ultimately falling short: Arduino Uno Rev 3, Arduino Mega 2560, Raspberry Pi 5, Raspberry Pi Zero 2 W, TI MSP-EXP430FR5994, TI MSP-EXP430FR231, STM32 Nucleo F446RE. The Arduino Uno Rev 3 is too limited in processing power and memory for our robot. The robot requires built-in wifi and bluetooth which the Uno Rev3 lacks. The Mega 2560 struggles with precise real-time motor coordination relying on a slow 16 MHz single core MCU making it fall short even if it offers more I/O. The Raspberry Pi options, such as the Pi 5 and Pi Zero 2 W are more like full Linux computers, not microcontrollers. These introduce limitations like non-deterministic timing, longer booting sequences, and vulnerability to SD card corruption. The TI MSP430 family, more specifically the TI MSP-EXP430FR5994 and TI MSP-EXP430FR231 are both constrained with low power and low memory. These models are not suitable for real time tasks, multitasking and data logging. Finally the STM32 Nucleo F44RE and other similar STM32 boards give us more power and flexibility giving an overall stronger performance but it is also a more complex toolchain and more technically intensive to work with. These models go against the goal of simplicity.

After more research we came across the ESP-32-S3-WROOM1U which was ultimately selected as our microcontroller. The ESP32-S3 is the correct size for our robot, since the stegosaurus has very limited internal space. It also has a lower power usage, so the efficiency helps extend the run time and avoids any power strain. Since the robot must control all the motors, sensor, communicate wirelessly and log data all at once it needs a microcontroller that can handle all this, the ESP32 can handle it reliably due to its high performance dual core processor. It has stable real time motor control with no OS interruptions as well having built in



blue tooth, an external antenna, Wifi capabilities, it is durable and stable, and has enough GPIOs, high-resolution PWM, and fast communication buses. The ESP32-S3-WROOM-1U hit all our target points but is still simple enough to not over complicate our robot, allowing us to still meet our user end needs.

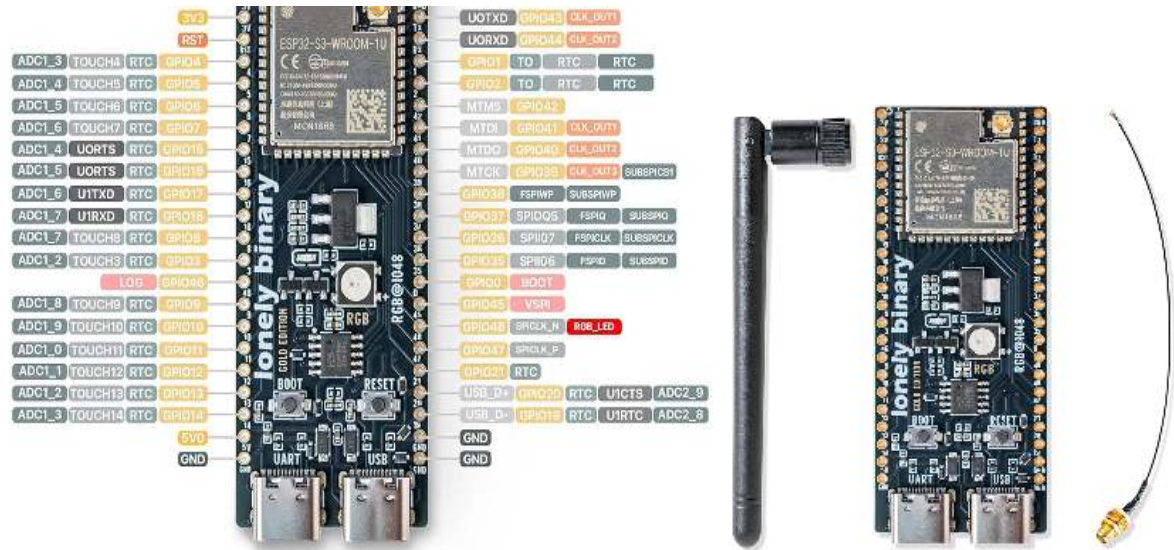


Figure 47: ESP32-S3-WROOM-1U

## User Input Interface

For our robotic stegosaurus, we considered several communication and control methods, keeping in mind that our expected operating distance is around **20 to 30 meters**. We considered methods like Infrared, Radio control, RF modules, Wi-Fi, and Bluetooth. At the end we selected bluetooth as our communication/ control method.

Infrared was not chosen due to its limited range, requiring the robot to be in the line-of-sight. This makes it impractical in a museum setting. Radio Control's bandwidth is so limited that it may not support the advanced sensor and motor data in the animatronic. While RF modules are flexible and have a more extended range, it would require extra hardware and

custom coding which would defeat the purpose of prioritizing simplicity. The second to last option we looked at before settling on bluetooth was Wi-Fi. Wifi gives us high data throughput and would allow the robot to be connected to phone or computers, the issue being it consumes more power and can cause latency issues for motor control. Finally our selected method was bluetooth due its low power, it falling into the 20-30 meter requirement, simple pairing, and it can be easily connected with commercially available controllers and smartphones.

### **Controller Options:**

In regards to the controller selection, we reviewed and researched multiple different types of controllers including PlayStation/Xbox controllers, Handheld RC/ Drone controllers, a mobile device or computer, and FPV flight controllers. Playstation and Xbox controllers are widely commercially available and ergonomic, making them comfortable to use. Libraries for the ESP32 already exist making integration of the controller simple and straightforward. Drone controllers/Handheld RC were ruled out due to them requiring extra adaption for our robot since they are typically designed for flying drones. While a computer or mobile drive covers more flexible control it is practical for museum works and adds set up complexity. A FPV flight controller is not suitable for a land based robot due to its general design purpose being for drones, so it would add complexity and need adaptation for our robot. Finally, after taking into account user needs and accessibility and taking into account our robot design we selected a Play Station controller.

Selecting a controller mainly took into account our selected communication method and feasibility as well as the overall complexity. Because Bluetooth is our communication method of choice, we were left with an Iphone/computer and a PlayStation. Choosing an Iphone would make the process for connecting more complex due to it probably needing a specialized app

which would have to be custom programmed. A computer is also just less efficient due to it not being easily transportable and posing a similar problem as the iPhone. The PlayStation controller The Playstation controller was chosen as it is something most people are familiar with so learning to use it will not be difficult for museum operators. Between the Xbox controller and the PlayStation, the PlayStation provided available libraries to help the set compared to the Xbox controller which does not offer the libraries nor the active support for use in microcontrollers.



*Figure 48: PS5 Controller*

## **Audio**

The audio plays a couple key roles within the project, we plan to use the audio as user feedback and safety cues. The audio will provide feedback to the operator when starting, stopping or encountering errors throughout the operation of the robot. We are also considering possibly having audio cues to signify battery level status conditions with regards to certain parts and connectivity with the user interface controller. The audio also plays a very key role in terms of entertainment for museum guests and making the project come to life. This enhances the visitor engagement. We could play roar sound effects, possibly footsteps sound effects as the dinosaur walks, maybe a tail swing sound effect. The overall goal is to enhance the engagement

and bring the dinosaur to life as much as possible. The audio effects should enhance physical motions. The audio needs to be synchronized with the motion, this adds complexity to the dinosaur but will make a big difference when done right.

The audio Hardware we selected is DFPlayer Mini MP3 Player (FIT0502). This Hardware works natively with ESP32 platforms as well as many other MCU platforms using a simple serial command to control. It takes MP3 and wave files directly from a microSD card making it very simple to use and easy to code. It also has a very low power requirement making it suitable for battery powered robotics without affecting the run time in a negative way. As we reiterate time and time again, size is a very big factor we need to consider due to our limited space and this component is very small and compact. It does not take up much space in the stegosaurus, allowing it for ease of use and easy to fit in.



*Figure 49: DFPlayer Mini*

In addition to the audio processor we also have a stereo enclosed speaker to actually play the audio out loud. We plan to package the speaker into the dinosaur itself. The speaker has a 3-Watt output which is loud enough to be heard within a museum environment while also maintaining a clear and crisp audio. The stereo feature allows for directional audio which is also a consideration that we wanted to have the option for. The speakers are also very durable and

economical as it only costs about \$5. This makes it the perfect speaker due to its low cost, easy replaceability while maintaining good quality.



*Figure 50: Speaker*

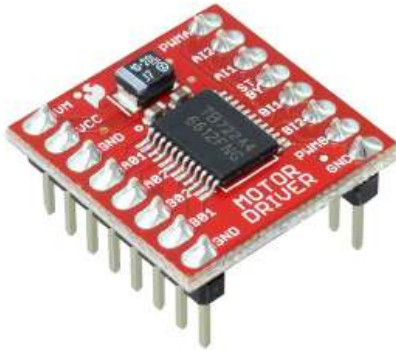
While the audio in the robot does promote engagement which is a user end request, the main purpose is ut being used for safety and operational feedback. The DFPlayer Mini and the 3W stereo enclosed speaker will give a compact and easy to use system that is compatible with the ESP32. This allows us to meet our goal of simplicity while maintaining reliability and low maintenance.

## **Motor Controllers**

In regards to motor drivers, we are incorporating two isolated drivers one for the servos and one for the linear motor. This is done to guarantee stable and isolated control of each subsystem.

One motor driver being used is the TB6612FNG Motor Driver; this driver will control the linear motor and the actuator. This drives the moth and fin movements of the robot. Because the linear motor and actuator requires 6V and higher current, which the ESP32 cannot supply directly, the TB6612FNg will help send lower voltage 3.3V signals to power the high current motors safely. This allows us to keep the logic and power domains electrically isolated. Keeping

them isolated allows for a smooth and safe operation of the linear motor and actuator while not damaging the eSP32 all while keeping a precise operation.



*Figure 51: TB6612FNG Motor Driver (EVAL Board)*

The second board being used is the PCA9685 16-channel PWM servo Driver Board. This board is used to control 16 servos that will animate the limbs, tails and ether body parts. The board provides smooth and coordinated movements via the 16 independent PWM channels with 12-bit resolution. The selected board uses I<sup>2</sup>C to communicate with the ESP32, this minimizes complexity via wiring and preserves ESP32 pins. Srvos operate at 6V and can draw higher current than the ESP32 can supply, therefore the PCA9685 acts as a safety buffer, converting 3.3V logic signals. This simplifies power distribution across all the servos and projects the ESP32 from overcurrent and voltage mismatches.

## PRODUCT SPECIFICATIONS

- Power supply: 5V (Pi connector) or 6V~12V (VIN terminal)
- Servo voltage: 5V
- Logic voltage: 3.3V
- Driver: PCA9685
- Control interface: I2C
- Mounting hole size: 3.0mm



Figure 52: PCA9685 16-Channel PWM Servo Driver Board

Using the TB6612FNG and the PCA9685 for the linear motor and serves respectively , precise, safe, and reliable control of the robot is achieved. Both also effectively protect the ESP32 to avoid any hazards and allow for smooth movement.

## Motors

For motor applications such as robotic systems that demand stable power output, precise control, and coordinated multi-axis motion, motors like servo, stepper, BLDC, and linear actuators are commonly used. These are commonly used in industrial application so they must satisfy the following criteria: energy efficiency, maintenance and long operational life span, high precision and control, high and low performance in regards to torque and speed output, rapid response, low noise and flexibility. For our project low noise is not a priority. [57]

For applications like our stegosaurus, we reviewed various types of motors and found that the most common and suitable motor types are servo motors, stepper motors, Brushless DC motors (BLDC motors) and linear motors/linear actuators. Servo Motors are able to provide precise position and speed control which would make the ideal for joint and articulated movement applications. They have a quick response and showcase relative motion. Because of

this they are commonly used on robotics, 3D printers, and CNC applications. Stepper motors allow for excellent precise incremental motion and repeatability but it lacks the ability to execute dynamic and lifelike motions. This makes stepper motors less ideal for a project like ours where we want dynamic motion. BLDC motors excel in efficiency, smoothness, and speed. They are more specialized towards continuous rotational movement rather than position specific tasks. Linear motors/actuators provide linear motion, which can be used for lifting, pushing, or extending components. This would be ideal for the mouth opening/closing of our robot. Alternate motors like AC, pneumatic, piezoelectric, magnetic field motors were deemed less practical due to their size, complexity or power requirements.

After identifying the most ideal motors for our robotic stegasaurs we assigned motor types different parts based on control, size, placement, and reliability. The main motors chosen are Servo motors and linear actuators. The servo motors were chosen based on the ease of control due to the simplicity of interfacing with the ESP32 via PWM signals. This makes the programming and motion control straightforward. Another reason was the precision and responsiveness a servo motor provides. This is critical for the limbs, tail, and neck movements of the dinosaur. The Servo also has a compact size and low maintenance fitting within the limited internal space and meeting the museum's low maintenance needs. Servo motors also handle movements like shoulder rotation, knee articulation, and tail motion without requiring complex control circuits. Linear actuators were to be used for the mouth and fin movements. A small linear actuator will be used in the mouth to attempt at mimicking accurate openings and closing jaw movements. The fin actuator will be embedded into the spine to allow for fin movement providing smooth oscillation and cam-like movements without taking up space.



The overall selection rationale was combining servos for its multi axis movements with linear actuaries for specialized linear motion to give the robot as much realism and flexibility as possible. The servos motors and linear actuators are compact, reliable, and low maintenance allowing us to meet museum needs while executing quality movement. These selected motors also allow us to avoid an overcomplicated design. The selection of motors allows for good energy efficiency, reliability, precision and torque making them the ideal motors for the project.



*Figure 53: Micro Actuator  
KT10-0903H*



*Figure 54: Micro Digital Linear Servo Actuator Servo*



Figure 55: DS3230MG High Torque 30KG Digital Servo Motor



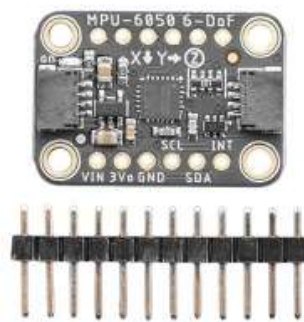
Figure 56: DS3230MG High Torque 30KG Digital Servo Motor



*Figure 57: Micro Servo*

## Peripherals

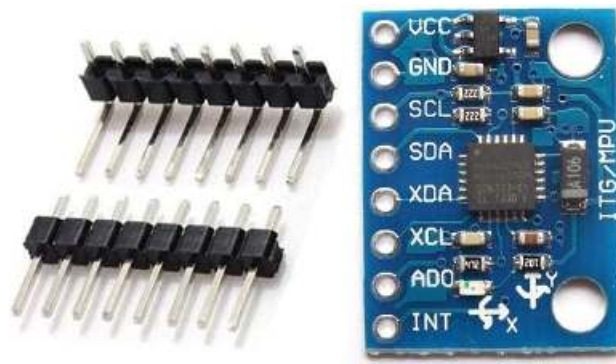
A gyroscope - STEMMA QT MPU-6050 (Accelerometer & Gyroscope / 6-Axis IMU) is needed to prevent the robot from tipping over. This gyroscope will allow an audio cue to play when fallen to alert staff. If the robot falls over then all inputs from the controller will no longer be taken to prevent itself from being damaged. We are also considering adding a gyroscope and having the system correct itself from falling. This feature would take a large amount of effort, but will be implemented if possible.



*Figure 58: STEMMA QT MPU-6050 (Accelerometer & Gyroscope / 6-Axis IMU)*

The “IMU - KEAcvise 6-Pack GY-521 MPU6050 Sensor Modules” The IMU - KEAcvise 6-Pack GY-521 MPU6050 Sensor Modules are used to monitor joint motion in order

to validate that the data being collected is consistent with expected ranges. By tracking accelerations and angular velocities across multiple joints, the sensors help ensure that the motion patterns are congruent with physiological norms rather than artifacts caused by noise, drift, or misalignment. Using several IMUs in tandem also allows for cross-validation between joints, improving the reliability and accuracy of the dataset. This monitoring process supports error detection and calibration, making it easier to identify anomalies such as unrealistic angles or sudden spikes in movement. Ultimately, the system provides quality assurance by confirming that the captured data is both accurate and meaningful, enabling more precise biomechanical analysis and ensuring confidence in the results.



*Figure 59: KEAcvise 6-Pack GY-521 MPU6050 Sensor Modules*

We will have a temperature sensor - MCP9808 High-Accuracy I2C Temperature Sensor. This is a safety feature to monitor the temperature of the robot and if the temperature reaches a critical level to prevent damage the robot will alert the operator via sound cue. These sensors will alert the user if the temperatures around the motors or battery exceeds a certain temperature, making sure nothing breaks down or overheats, or even explodes.



*Figure 60: MCP9808 High-Accuracy I2C Temperature Sensor*

The GPS - GNSS Module (SeeedStudio XIAO GNSS Add-on) allows us to keep track of distance and is meant to prevent the operator from driving the robot outside the distance of operation, again an audio cue will play to alert that the operator is getting close to being outside the operation zone. Having this system implemented in the robot is critical just in case any individual drives the robot too far and is at risk of disconnecting the robot from the controller, which would cause connection issues, and may cause malfunctioning issues.



*Figure 61: GNSS Module (SeeedStudio XIAO GNSS Add-on)*

All the peripherals are small, low-power, I<sup>2</sup>C compatible, and have robust ESP32/Arduino libraries, making development and maintenance .

### **Packaging:**

Fitting all the components in the robot can be tricky. All the electrical components should fit inside the ribcage. A housing will be designed for each component that will fit under the ribcage. As of now, wires will run through the bones. The bones will not be hollow to maintain

structural integrity. The housing that will hold all the electrical components will be designed to specifically hold certain components. This is so no electrical component is loose or rattling inside the housing to eliminate the risk of electrical component damage.

As of now the plan is to have a hollow cylindrical cutout for the wires to run through the bones up to the housing where most of the electrical components will go. Each leg will have 3 motors, and we will have wires running up through the leg bones, and up the chassis. Wires will also be about an inch long between each hinge so that no wire is stretched over the limit to reduce risk of electrical component damage.

# Preliminary Engineering Analysis

## Modeling & CAD Software Considerations

Although partially seen in concept generation, it is necessary to discuss the software selections and methods used to model the final design. Some considerations include availability, prior experience, and tools that are available within each respective software package.

For primary modeling, The team will be utilizing Siemens NX as this is what the past experience best aligns with. Additionally, NX is typically better at handling large scale assemblies with many parts. And although it does not contain as many analysis features as other programs, it provides more security in the assembly and modeling process so that models cannot “break” as easily.

Siemens NX has a very powerful simulation software and can run big mesh bodies in a reasonably short time. For the finished bone structure, we can run a simulation with different cases (different joint angles). One fast way to run a FEA simulation is just a simple 3D simulation with a load case and constrain the feet. There is another longer and more accurate way to run simulation through Siemens NX. First you want to export the assembly as a parasolid. Then you make a “fem” file and a “sim” file. Then in the part file you treat each bone as a “Midsurface by face pairs” then 2D “stitch” (in the fem) each bone together. Another alternative for stitching is 1D connection. 1D connection stitches pieces together but not physically. On the other hand, using the “stitch” tool physically bonds the bones together but might not work as well in complex geometries such as bones with many surfaces. 1D connection would be the better option. After stitching, you add mesh to the bones, and can select mesh size. Sometimes you will get a Jacobian error; one way to have less Jacobian errors is to make the mesh more fine. Even if there are some Jacobian errors (less than 10) you can still run the simulation and get

accurate results. After adding the mesh, you assign a material and a thickness. After assigning the material and thickness, you can now go in the “sim” file and add your constraints and loads. We can have different load cases as stated above. For example, we can have a load case where a child would kick the robot from the side, we could assume a force of 50 pounds. We would have different forces in different areas to see which bone needs to be adjusted. We can get an idea of how we might print the bones, what infill percent to choose, what infill pattern, and how to orient the bone on the build plate. We can also have different cases where the bones are at different angles in relation to other bones.

SolidWorks will also be used in conjunction with NX, this is able to provide more advanced mesh capabilities which is crucial when converting a 3D scan over to a solid and usable body which can be modified. Another advantage to solidworks is through its analysis package that is available to students. And can be used to run basic FEA to determine if the final designs will be able to hold up to both the durability requirements, along with basic strength calculations.

Another crucial determination to make is in the base 3D scan The team will be using to construct the dinosaur from. For this case there are 2 choices. In the prior milestone, it was discussed to use the scan from the Denver Museum of Natural History, however in the talks with Matt Dempsey, he informed the team that the Stegosaurus in Denver is effectively “squished”, as it was crushed under heavy rock when it died then was slowly warped over millions of years. The other option is to use Sophie from the British Museum of Natural History. This Dinosaur is better preserved, and a scaled, 3D printable model is already available on Makerworld [46].

An initial print of Sophie was made and it was found that the overall length was around the 24in mark. Although this does meet the minimum requirement The team has set. It will be



beneficial to stretch overall length to 36in in order to have more room for final packaging. So a scaling factor of 150% will be placed on all bones before they are modified. This shall place the final length to within 36in. however it is possible to adjust from there by up to 20%.

## Gait Studies

A gait study is a biomechanical analysis of how a living creature moves. Gait studies focus on a few kinematic and dynamic points of analysis. The first point of analysis is ground reaction force, which is the force of a limb hitting the ground. The second major point of analysis is joint angles, which is the calculation of one joint with respect to the bone members. Both points are measured throughout the stride of the animal. The gait can also be analyzed in terms of the angular velocity of each joint. The stride is divided into frames, where each frame can be assigned a ground reaction, force, and a list of angles with respect to multiple joints.

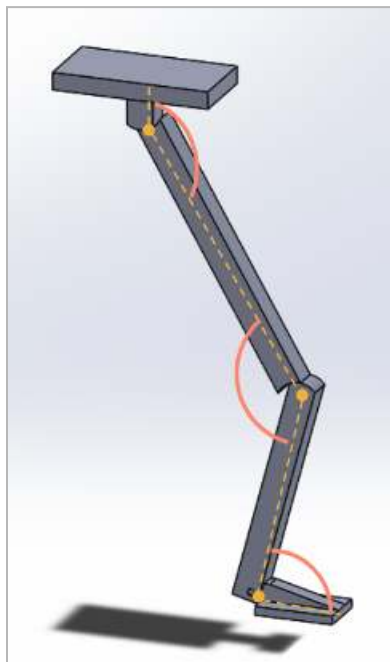
| Pugh Matrix                                                                                                               |                       |              |                  |                     |
|---------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------|------------------|---------------------|
|                                                                                                                           | Solution Alternatives |              |                  |                     |
|                                                                                                                           | Importance Rating     | Joint Angles | Angular Velocity | Torque Calculations |
| <div> <b>Concept Selection Legend</b><br/>           Better +<br/>           Same S<br/>           Worse -         </div> |                       |              |                  |                     |
| Key Criteria                                                                                                              |                       |              |                  |                     |
| Data Volume                                                                                                               | 5                     | -            | +                | +                   |
| Data Specificity                                                                                                          | 10                    | +            | -                | S                   |
| Speed Adjustments                                                                                                         | 8                     | +            | -                | -                   |
| Sum of Positives                                                                                                          |                       | 2            | 1                | 1                   |
| Sum of Negatives                                                                                                          |                       | 1            | 2                | 1                   |
| Sum of Sames                                                                                                              |                       | 0            | 0                | 1                   |
| Weighted Sum of Positives                                                                                                 |                       | 18           | 5                | 5                   |
| Weighted Sum of Negatives                                                                                                 |                       | 5            | 18               | 8                   |
| TOTALS                                                                                                                    |                       | 13           | -13              | -3                  |

Figure 62: Gait Study Method Concept Generation

The team considered these methods of compiling data for a gait study and have determined they will use joint angles to coordinate the control system and the joint actuators.

The use of angular velocity provides a more limited data package, as the animatronic will not require positional data for each point in its stride. However, this limited data will lack the specificity needed. The control system would not be able to [Figure 62:] account for the position of the limbs at each frame of motion, meaning if anything unexpected occurred, the animatronic would have no capacity to correct itself.

Similar problems arise when designing the gait using torque. Torque calculations would essentially provide the angular acceleration of the joints, which would similarly reduce the data package. But, without positional data, the animatronic would not be able to handle unexpected changes to its joint angles. Additionally, both the angular velocity and torque methods would struggle to handle changes to speed.



*Figure 63: Hind Limb Simplified Model with Notate Joint Angles*

The team ultimately decided that the use of joint angles will lead to the greatest success. A data point would be required for each frame of motion; however, the control system would have positional data for the entire stride. Further, because the data is purely positional, the control system can more easily adjust the speed of the animatronic by increasing the frequency of the frames.

Another difficulty of this project is the creature in question is extinct, leaving no surviving examples to calculate a gait study. In order to move forward, the team has [Figure 63] analyzed the anatomy of the dinosaur and made comparisons to living creatures with similar anatomy.

The stegosaurus has a notably different skeletal structure from any reptile, mammal, or avian. The four limbs are dramatically shorter than the rear limbs, which results in hips that are substantially taller than the shoulders. Additionally, the stegosaurus has facing back knees, but backwards facing front knees. No other living creature has this anatomy. The closest resemblance the team has found is a human walking on all fours. This, paired with the hip to shoulder displacement has led the group to utilize human gait studies as the initial point of data collection. The benefit of using human data is that much more of it exists than any other creature.

| Pugh Matrix                                                                         |   |                       |                     |            |          |                 |             |
|-------------------------------------------------------------------------------------|---|-----------------------|---------------------|------------|----------|-----------------|-------------|
|                                                                                     |   | Solution Alternatives |                     |            |          |                 |             |
|                                                                                     |   | Importance Rating     | Stegosaurus Anatomy | Human Gait | Cow Gait | Rhinoceros Gait | Lizard Gait |
| <u>Concept Selection Legend</u><br>Equivalent +<br>Similar S<br>Far from Accurate - |   |                       |                     |            |          |                 |             |
| Key Criteria                                                                        |   |                       |                     |            |          |                 |             |
| Hip Angle w/ Respect to Anatomy                                                     | 7 |                       | S                   | -          | +        | -               |             |
| Knee Angle w/ Respect to Anatomy                                                    | 7 |                       | -                   | S          | +        | -               |             |
| Ankle Angle w/ Respect to Anatomy                                                   | 2 |                       | -                   | S          | +        | -               |             |
| Counterbalance                                                                      | 5 |                       | -                   | S          | -        | S               |             |
| Body Mass/Ground Reaction Forces                                                    | 2 |                       | -                   | S          | +        | -               |             |
| Hip-to-Shoulder Displacement                                                        | 6 |                       | +                   | -          | -        | -               |             |
| Femur Ratio                                                                         | 7 |                       | +                   | -          | -        | -               |             |
| Tibia Ratio                                                                         | 7 |                       | S                   | -          | -        | -               |             |
| Foot Ratio                                                                          | 7 |                       | S                   | -          | +        | -               |             |
| Fore-leg Anatomy                                                                    | 5 |                       | +                   | S          | S        | -               |             |
| Hind-leg Anatomy                                                                    | 5 |                       | +                   | S          | S        | -               |             |
| Shoulder Range of Motion                                                            | 7 |                       | -                   | -          | -        | -               |             |
| Hip Range of Motion                                                                 | 7 |                       | -                   | -          | -        | -               |             |
| Hip/Shoulder Degrees of Freedom                                                     | 4 |                       | -                   | S          | S        | -               |             |
| Sum of Positives                                                                    |   |                       | 0                   | 0          | 5        | 0               |             |
| Sum of Negatives                                                                    |   |                       | 7                   | 7          | 6        | 13              |             |
| Sum of Sames                                                                        |   |                       | 7                   | 7          | 3        | 1               |             |
| Weighted Sum of Positives                                                           |   |                       | 23                  | 0          | 25       | 0               |             |
| Weighted Sum of Negatives                                                           |   |                       | 34                  | 48         | 39       | 73              |             |
| TOTALS                                                                              |   |                       | -11                 | -48        | -14      | -73             |             |

Figure 64: Various Animal Gait Comparisons

However, human data points will not be sufficient to animate the stegosaurus gait. While it may guide the joint angles of the hind legs, it will not be able to provide any information on the synchronicity of the four limbs. Further, scarring remaining on many stegosaurus bones paired with the anatomy of modern day reptiles seems to indicate that the stegosaurus tail had some range of motion. It can be predicted that the tail acted as a counterbalance while the

dinosaur walked. Additionally, the structure of the shoulders would have prevented the stegosaurus from any dramatic movement that is common in the majority of modern animal gaits.



*Figure 65: Skeletal remains of a stegosaurus compared to a modern Rhinoceros*

The stegosaurus's anatomy was compared to four different living creatures, and their gaits were analyzed. The goal was to see how much data from these available creatures could be extrapolated to the stegosaurus. Working backwards, using the gaits of a lizard provides nearly no useful information. The only benefit to a lizard gait is that it has a tail analogous to a stegosaurus. However, lizards have their four limbs out horizontally from the body, as opposed to vertical, putting their humerus and femur parallel to their spine. Analyzing the stegosaurus bone structure, this would have been impossible as the density of the femur and humerus suggest that there was axial loading on these bones.

A rhinoceros gait has benefit in that the rhinoceros has a comparable body mass to a stegosaurus. Additionally, the structure of the rhinoceros foot and ankle is similar. However, the rhinoceros hips and shoulders create an axis parallel to the ground, while the stegosaurus had a sharp incline from the shoulders to the hips. Further, the ratio of the rhinoceros femur to tibia is significantly smaller than that of the stegosaurus.

The gait of a cow also lacks similarity to the stegosaurus. It has benefit in the body mass and counterbalancing effects of its gate, but lacks in most bone ratios.

The most comparable gait to the stegosaurus would be a human, walking on all fours. The benefit of this gait is the orientation of the front limbs and hind limbs. Further, the slope from the hip to the shoulder is comparable to the stegosaurus. The human does lack similar body mass and ground reaction forces, however, because the gait study will focus on joint angles, body mass is not as important. All four of these alternative animal gaits have strong hip and shoulder ranges of motion, as well as stronger degrees of freedom in the hip and shoulder joints. Stegosaurus would have very limited range of motion, particularly in the shoulders, and though designed as a ball and socket joint for the hip and shoulder, the bone structure surrounding the pelvis and shoulder blades, limited the degrees of freedom in the hip and shoulder movement.

To complete the gait study, the team will need to assemble a list of tabulated data, including the angles of the joints at specific frames of motion. Then, the team will need to coordinate the motion of the four limbs and tail with respect to each other. The control system can then read this data, and relay the information to the joint actuators, because the data will be based off of frames of motion, the control system can increase the frequency of the frames to increase the speed.

A joint study must be conducted for a walking gait as well as a running gate. Walking gaits never contain a frame of motion without points of contact. By contrast, a running gate has a moment of “airtime“. Due to the mass and control systems of the animatronic, it would be prudent not to design the run gate with a frame of no contact, however, the team can create a gait that is visually similar.

## Materials

The team intends to 3D print the majority of the structure of the stegosaurus. The figure below compares the properties of many materials [54].

|                                  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|----------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                                  | <a href="#">Learn More</a>                                                        | <a href="#">Learn More</a>                                                        | <a href="#">Learn More</a>                                                        | <a href="#">Learn More</a>                                                        | <a href="#">Learn More</a>                                                        | <a href="#">Learn More</a>                                                        | <a href="#">Learn More</a>                                                        | <a href="#">Learn More</a>                                                        | <a href="#">Learn More</a>                                                          | <a href="#">Learn More</a>                                                          | <a href="#">Learn More</a>                                                          | <a href="#">Learn More</a>                                                          | <a href="#">Learn More</a>                                                          |
| <b>Compare selected</b>          | <input type="checkbox"/>                                                          | <input type="checkbox"/>                                                          | <input type="checkbox"/>                                                          | <input type="checkbox"/>                                                          | <input type="checkbox"/>                                                          | <input type="checkbox"/>                                                          | <input type="checkbox"/>                                                          | <input type="checkbox"/>                                                          | <input type="checkbox"/>                                                            | <input type="checkbox"/>                                                            | <input type="checkbox"/>                                                            | <input type="checkbox"/>                                                            | <input type="checkbox"/>                                                            |
| Ultimate Strength                | 47MPa                                                                             | 35-45MPa                                                                          | 55 MPa                                                                            | 50MPa                                                                             | 55 MPa                                                                            | 40-55MPa                                                                          | 45-47 MPa                                                                         | 50 MPa                                                                            | 25MPa                                                                               | 52 MPa                                                                              | 25-30MPa                                                                            | 40 MPa                                                                              | 50MPa                                                                               |
| Stiffness                        | 2.1 GPa                                                                           | 1.7 GPa                                                                           | 3.5 GPa                                                                           | 10 GPa                                                                            | 3 GPa                                                                             | 2 GPa                                                                             | 12 GPa                                                                            | 2 GPa                                                                             | 2.1 GPa                                                                             | 2.1 GPa                                                                             | 10 GPa                                                                              | 6 GPa                                                                               | 2.1 GPa                                                                             |
| Durability                       | 3 GPa                                                                             | 3 GPa                                                                             | 4 GPa                                                                             | 7 GPa                                                                             | 8 GPa                                                                             | 10 GPa                                                                            | 3 GPa                                                                             | 10 GPa                                                                            | 10 GPa                                                                              | 9 GPa                                                                               | 4 GPa                                                                               | 3 GPa                                                                               | 7 GPa                                                                               |
| Maximum Service Temperature      | 98 °C                                                                             | 60-70 °C                                                                          | 52 °C                                                                             | 100 °C                                                                            | 73 °C                                                                             | 80-95 °C                                                                          | 52 °C                                                                             | 95 °C                                                                             | 121 °C                                                                              | 180 °C                                                                              | 52 °C                                                                               | 52 °C                                                                               | 75 °C                                                                               |
| Coefficient of Thermal Expansion | 90 ppm/°C                                                                         | 150 ppm/°C                                                                        | 60 ppm/°C                                                                         | 80 ppm/°C                                                                         | 60 ppm/°C                                                                         | 95 ppm/°C                                                                         | 57 ppm/°C                                                                         | 80 ppm/°C                                                                         | 60 ppm/°C                                                                           | 150 ppm/°C                                                                          | 11.7 ppm/°C                                                                         | 30 ppm/°C                                                                           | 85 ppm/°C                                                                           |
| Density                          | 1.04 g/cm <sup>3</sup>                                                            | 1.19-1.23 g/cm <sup>3</sup>                                                       | 1.24 g/cm <sup>3</sup>                                                            | 1.03-1.04 g/cm <sup>3</sup>                                                       | 1.23 g/cm <sup>3</sup>                                                            | 1.06-1.14 g/cm <sup>3</sup>                                                       | 1.3 g/cm <sup>3</sup>                                                             | 1.07 g/cm <sup>3</sup>                                                            | 1.2 g/cm <sup>3</sup>                                                               | 0.9 g/cm <sup>3</sup>                                                               | 2.4 g/cm <sup>3</sup>                                                               | 1.15-1.25 g/cm <sup>3</sup>                                                         | 1.23 g/cm <sup>3</sup>                                                              |
| Weldability                      | 10-140                                                                            | 130-150                                                                           | 110-140                                                                           | 125-132                                                                           | 120-160                                                                           | 125-165                                                                           | 130-160                                                                           | 135-140                                                                           | 140-175                                                                             | 160-1120                                                                            | 120-1120                                                                            | 125-150                                                                             | 140-1110                                                                            |
| Printability                     | 4 GPa                                                                             | 2 GPa                                                                             | 6 GPa                                                                             | 2 GPa                                                                             | 8 GPa                                                                             | 8 GPa                                                                             | 6 GPa                                                                             | 2 GPa                                                                             | 2 GPa                                                                               | 4 GPa                                                                               | 2 GPa                                                                               | 6 GPa                                                                               | 2 GPa                                                                               |
| Strain Temperature               | 220-250 °C                                                                        | 225-265 °C                                                                        | 190-220 °C                                                                        | 200-240 °C                                                                        | 230-250 °C                                                                        | 220-270 °C                                                                        | 200-230 °C                                                                        | 235-255 °C                                                                        | 260-310 °C                                                                          | 220-250 °C                                                                          | 190-220 °C                                                                          | 190-220 °C                                                                          | 185-200 °C                                                                          |
| Use Temperature                  | 95-110 °C                                                                         | 45-60 °C                                                                          | 45-60 °C                                                                          | 100-115 °C                                                                        | 75-90 °C                                                                          | 70-90 °C                                                                          | 45-60 °C                                                                          | 80-110 °C                                                                         | 80-120 °C                                                                           | 85-100 °C                                                                           | 45-60 °C                                                                            | 45-60 °C                                                                            | 45-60 °C                                                                            |
| Heated Bed                       | Required                                                                          | Optional                                                                          | Optional                                                                          | Required                                                                          | Required                                                                          | Required                                                                          | Optional                                                                          | Required                                                                          | Required                                                                            | Required                                                                            | Optional                                                                            | Optional                                                                            | Required                                                                            |
| Recommended Build Surface        | Kapton Tape, ABS Slurry                                                           | PT, Painter's Tape                                                                | Painter's Tape, Glue Stick, Glass Plate, PET                                      | Glass Plate, Glue Stick, Kapton Tape                                              | Glue Stick, Painter's Tape                                                        | Glue Stick, PET                                                                   | Painter's Tape, Glue Stick, Glass Plate, PET                                      | Glue Stick, PET                                                                   | PET, Commercial Adhesive, Glue Stick                                                | Packing Tape, Polysporylon Sheet                                                    | Painter's Tape, Glue Stick, PET                                                     | Painter's Tape, Glue Stick, PET                                                     | PT, Painter's Tape                                                                  |
| Other Hardware Recommended       | Heated Bed, Enclosure Recommended                                                 | Part Cooling Fan                                                                  | Part Cooling Fan                                                                  | Heated Bed, Enclosure Recommended                                                 | Heated Bed, Part Cooling Fan                                                      | Heated Bed, Enclosure Recommended, May Require All Metal Hotend                   | Part Cooling Fan                                                                  | Heated Bed                                                                        | Heated Bed, Enclosure Recommended, All Metal Hotend                                 | Heated Bed, Enclosure Recommended, Part Cooling Fan                                 | Wear Resistant or Stainless Steel Nozzle, Part Cooling Fan                          | Part Cooling Fan                                                                    | Heated Bed, Part Cooling Fan                                                        |

|                         |     |          |     |      |      |       |                    |     |               |              |                |               |     |
|-------------------------|-----|----------|-----|------|------|-------|--------------------|-----|---------------|--------------|----------------|---------------|-----|
| Heatable                | —   | ✓        | —   | —    | —    | ✓     | —                  | —   | —             | ✓            | —              | —             | ✓   |
| UL94V-0                 | —   | ✓        | —   | —    | —    | —     | —                  | —   | —             | —            | —              | —             | —   |
| Impact Resistant        | ✓   | —        | —   | ✓    | —    | ✓     | —                  | ✓   | ✓             | —            | —              | —             | —   |
| Soft                    | —   | ✓        | —   | —    | —    | —     | —                  | —   | —             | ✓            | —              | —             | ✓   |
| Corrosion Resistant     | —   | —        | —   | —    | —    | —     | ✓                  | —   | —             | —            | ✓              | ✓             | —   |
| UV Resistant            | —   | —        | —   | —    | —    | —     | —                  | ✓   | —             | —            | —              | —             | —   |
| Water Resistant         | —   | —        | —   | —    | ✓    | —     | —                  | —   | —             | ✓            | —              | —             | —   |
| Shrinkable              | —   | —        | —   | ✓    | —    | —     | —                  | —   | —             | —            | —              | —             | ✓   |
| Heat Resistant          | ✓   | —        | —   | ✓    | —    | ✓     | —                  | ✓   | ✓             | ✓            | —              | —             | —   |
| Chemically Resistant    | —   | —        | —   | —    | ✓    | —     | —                  | —   | —             | —            | —              | —             | —   |
| Fully Recyclable        | —   | ✓        | —   | —    | ✓    | ✓     | —                  | —   | ✓             | ✓            | —              | —             | ✓   |
| Heated bed not required | —   | ✓        | ✓   | —    | —    | —     | ✓                  | —   | —             | —            | ✓              | ✓             | —   |
|                         | ABS | Hesbolic | PLA | HIPS | PETG | Nylon | Carbon Fiber Hitec | ASA | Polycarbonate | Polysporylon | Metal Filament | Wood Filament | PVA |

Figure 66: Materials and property comparison [54].

Materials under consideration include: PLA, PETG, ABS, ASA, NYLON, TPU, and Carbon fiber reinforced (nylon, onyx,). When it comes to the bulk of the material, ASA will be

used because of its availability and properties. ASA is strong and has deflection properties similar to ABS, and it can handle heat better than PLA and ABS. When prototyping, it is beneficial to see deflection; the team will use whatever material available and save ASA for the final product.

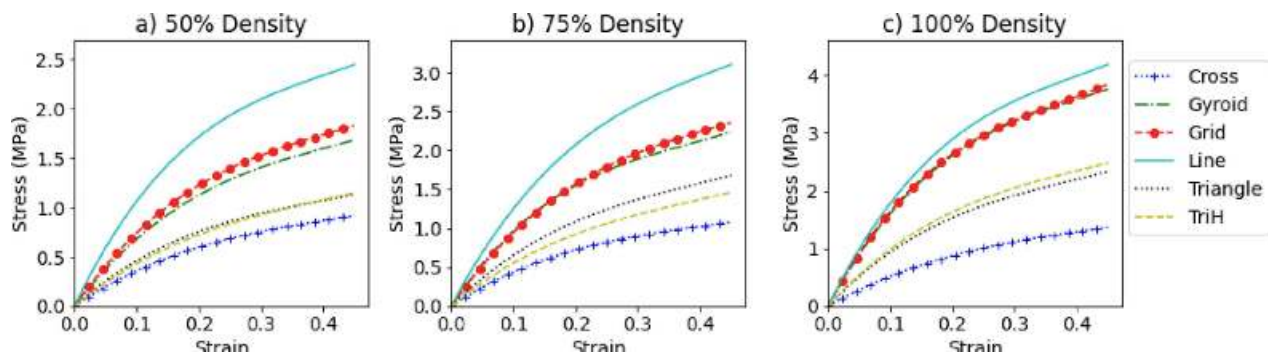


Figure 67: Infill Patterns at Different Densities [55].

When it comes to infill patterns, hex, or honeycomb, has the highest strength in all directions [55]. If a part is experiencing load in one direction, line and grid are the strongest option. Gyroid is also another alternative to hex/honeycomb. In regards to infill density, anything above 40% is adding more weight than strength, and it is not worth it to spend more money to get a slightly stronger bone [55]. Taking into consideration the motors strength, having low density bones should be considered when calculating torque required when deciding on which motors will be chosen. Further, having a lower infill density is better for materials like ASA because it will bend before breaking, so it acts like a shock absorber if any load is applied in a rapid manner. All these materials do not have a problem with clogging when printing. TPU may be used for damping and vibration. TPU can be challenging to print, but the team should be able to overcome this.



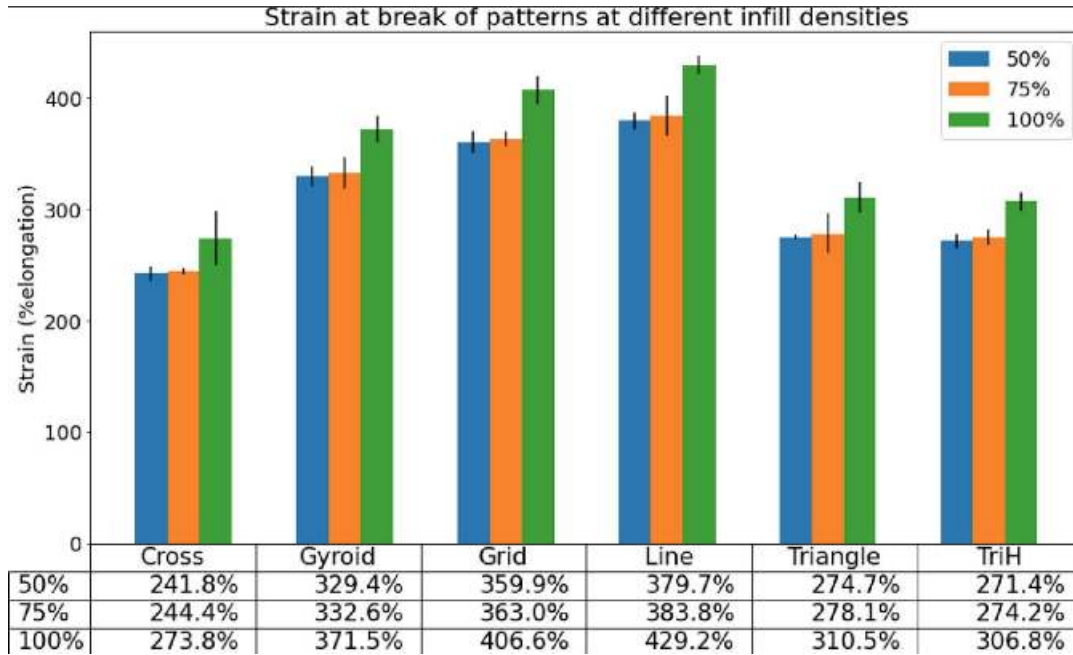


Figure 68: Compares different infill patterns with density and shows the strain % elongation [55].

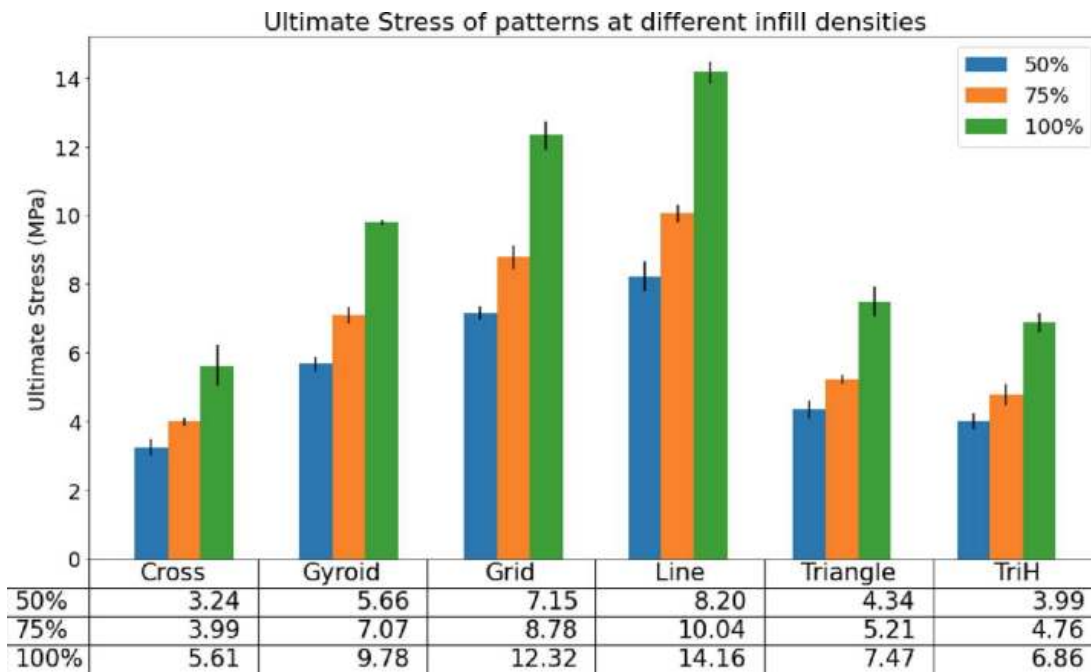


Figure 69: Compares different infill patterns with density and shows the ultimate stress in MPa [55].

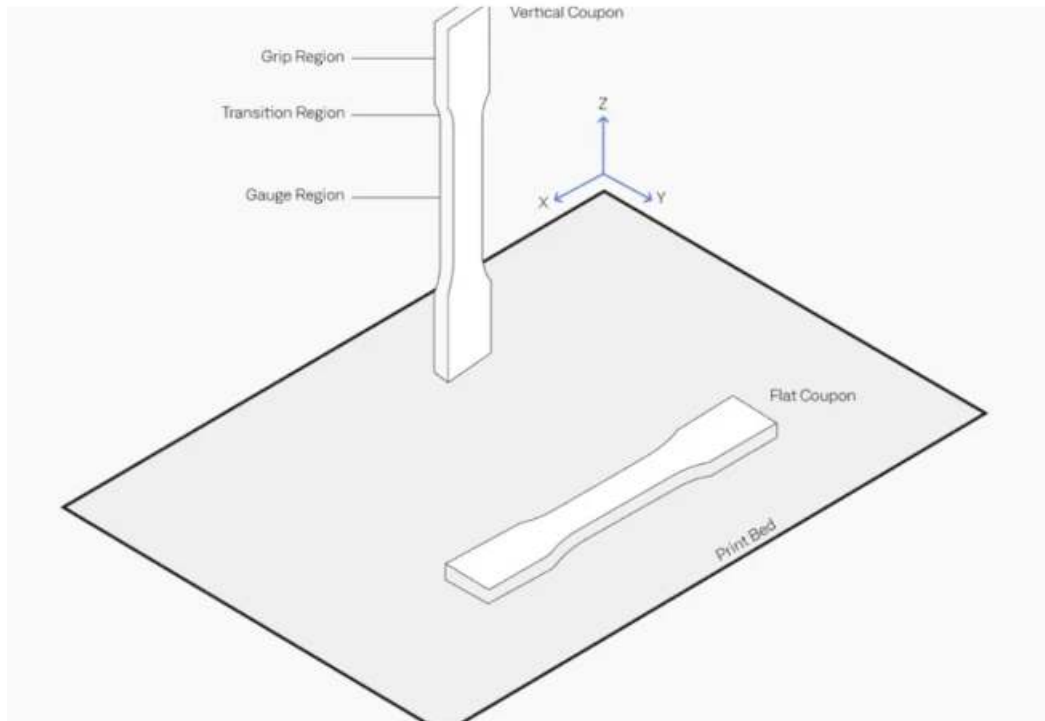


Figure 70: Importance in printing orientation [56].

Printing orientation is extremely important when 3D printing; simply changing the orientation can make a big difference when it comes to the success of a 3D printed object [56]. For example, in figure 70, when testing the print in tension (pulling the grip regions), the vertical coupon will easily snap as the layer lines are laid horizontally which create many small notches which set up the part for failure. On the other hand, the flat coupon (printed horizontally) will take much more tensional force as the layer lines run along the part.

Another thing to take into consideration is print quality. Errors in prints such as “u” and “v” notches can cause a 3D printed part to fail when undergoing tension forces. Having a notch as seen in figure 6 in a 3D printed part puts the structural integrity in danger [57]. This is because notches create a place where cracking begins. This is why quality is important when it comes to these 3D printed bones.

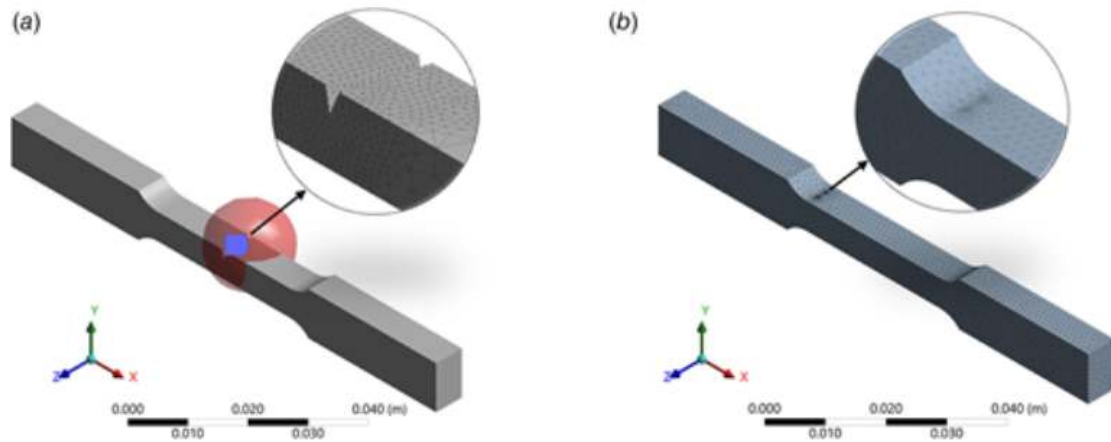


Figure 71: [57] Importance of print quality

## Preliminary Cost Estimation

Appendix I represents the basic cost estimates for the final components of the dinosaur, a 20% contingency is also included in the event that spare parts are needed, or a shortened lead time is required to meet the deadline as outlined in our Gantt chart. Although final costs will not be known until our final component selections have been made. It gives us a rough idea of what to expect in the project. These values will most certainly change in the future, but we anticipate the final ballpark cost of the first prototype to be around \$1000 based on the figure below.

Included are estimated for all the primary components and materials needed for the robot. Such as motors, actuators, onboard electronics, batteries, and basic material cost values.

## Gantt Chart and Major Deliverables

Appendix G contains the full project GANTT chart. This project runs from August 2025 through April 2026, resulting in an eight month schedule. Five months are remaining in this timeframe.

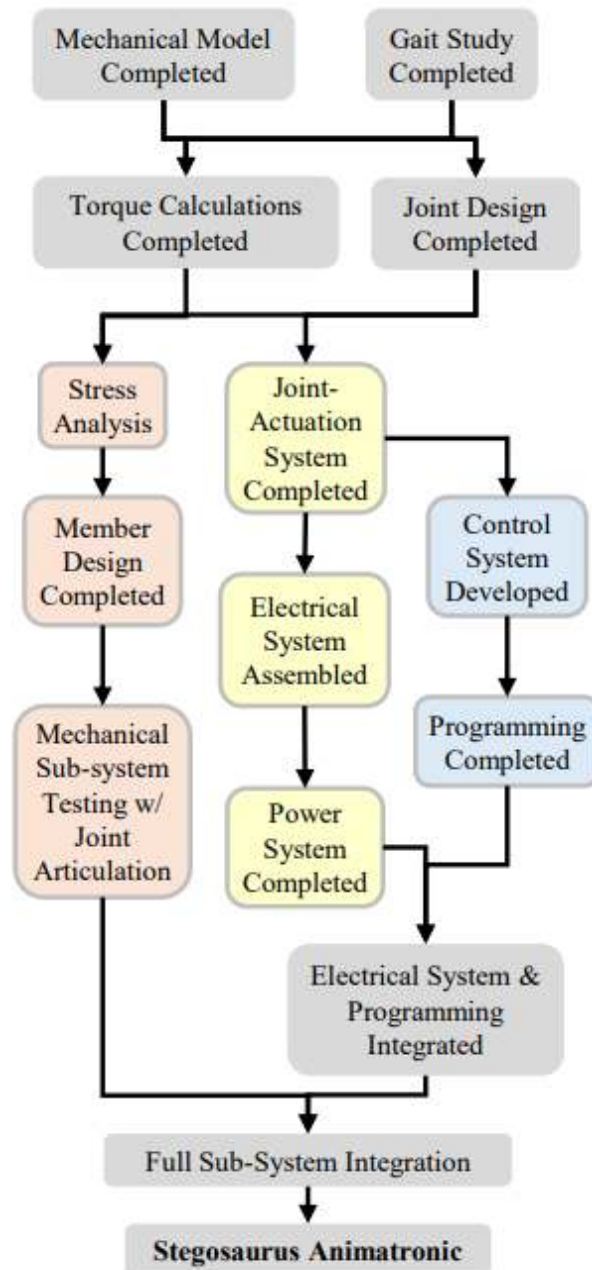


Figure 72: Deliverables Flow Chart

Initial deliverables include converting a 3-D scan of an actual stegosaurus skeleton into a workable model. Another major deliverable includes the gait study. The completion of the gait study and mechanical model will allow the team to finalize joint design and complete the necessary torque calculations for each joint. From this point, the project will follow three major tracks: the mechanical system, the electrical system, and the control system.

The electrical system begins with developing the joint articulation system. Once the team is sure how to move the joints and what torque will be required, it can then determine the electrical system to connect all the joints. From there, the power system can [Figure 72:] be determined.

The mechanical system will utilize the joints design and torque calculations to begin preliminary stress analysis on the joints and members. This will allow the team to appropriately select materials. Then, utilizing the joint articulation system from the electrical path, mechanical subsystems, such as one leg, can be tested using the joint articulation system.

The programming aspect requires the joint articulation system to be determined. From there, the control system can be developed, and the programming completed, before they are integrated with the electrical system. The last major deliverable includes integrating all subsystems to develop the full stegosaurus animatronic. Each stage of this process requires subsystem testing to ensure success.

As per the requirements of the Senior Design course, the completion of this project includes presentation material. The team will compile data that has been assembled from throughout the project and present it in a showcase format.

# **System Concept Evaluation Plans**

## **Gait Study**

For the gait study, once a list of joint angles per frame is compiled, this data needs to be attached to the mechanical model to see if the data will generate a smooth gait on the model. From this point, iterations may be made to the list of joint angles.

## **Torque Calculations**

Once the mechanical model and gait study are completed, torque calculations can be conducted. The mechanical model will indicate where the insertion points of the “tendons” will be placed. Knowing the angle per frame, as well as the insertion point will allow the team to calculate the torque on each joint. Torque calculations will also provide data for an effective gear ratio from each joint to calculate the desired motor torque.

## **Finite Element Analysis**

The team will then conduct stress analysis on the model components. The torque calculations will allow the team to determine the stresses that each member will experience. It is important to understand the axial and shear stresses that will occur. This will determine the method of printing for the members.

The structure will require testing of material, infill densities, infill patterns, etc. The primary risks in the joints of the robot will be their ability to meet the final torque and speed requirements. After running FEA, a reasonable safety factor for each bone can be determined depending on the stresses each bone will undergo. The team can also reinforce each bone with supporting material to reduce risk. In the FEA, different load cases will be considered to account

for different ways this robot can experience external forces, such as from a falling object or a kick from a child. Although the electrical components are extremely sensitive, the structural body will be protective to defend the components from unexpected impact.

$$\sigma = \frac{P}{A_0} \quad \epsilon = \frac{l - l_0}{l_0} \quad \sigma = E\epsilon \quad \tau_{\max} = \frac{Tr}{J}$$

*Figure 73: Material Stress Formulas [68]*

## **Electrical System**

Before full assembly, the electrical system will be tested by subsystems. This will protect the components so that if a test fails, the entire system is not destroyed. Additionally a factor of safety needs to be determined with each motor so that they are properly sized and issues such as overheating and excessive wear can be mitigated.

For motor safety, the team will need to make sure the motors can handle exceptionally high torques to take into account any external forces that will be applied from testing and public use. Safety factors will need to be doubled or tripled when it comes to calculating torque required to have the robot properly move under certain loads. The animatronic will also have an indicator when it comes to battery life, as having the battery fall below a certain percentage can cause it to burn or explode.

## **Programming**

The team intends on using multiple methods to test the code of the robot. First, each function will be tested individually. Functions include the gait and walking, as well as the animations of the head and tail movement. After each of the coded functions are tested, the total code will be assembled and tested to ensure that they work in conjunction with one another. In

addition, there will be a data logging feature which will store all the data and variables of the motors, microcontrollers, power system, temperature, and any other data collected by sensors in an individual file with a time stamp. This will allow the team to know when errors occur, which improves diagnostic work.

## **Integration**

In order to integrate all subsystems, the team will do tests on smaller components. For example, before assembling the entire animatronic, the team will do a proof of concept on one leg. The joint actuation system will be tested with the mechanical model and the control system to ensure all our coordination. This will allow the team to make edits to the design based on their findings.



## Failure Modes and Effects Analysis

In terms of our project, there are a lot of moving parts and variables to consider. A full table is given in Appendix H. There are six major areas to consider for this project: balance, structure, power, joint articulation, control systems, safety, and environment.

The robot must be able to balance both statically and dynamically. Any case of the robot falling over or not being able to handle its environment to a certain degree constitutes a design failure. Its main structure also must be able to handle the weight and reaction forces of the robot as it moves.

The whole robot must be powered by the same power system, and all constituent components cannot cause the power system to become overburdened. There is a real risk when using a rechargeable Li-po battery that charging speeds, discharging speeds, and possible punctures can have an adverse reaction on the health of the battery and possibly start a fire. Special care must be taken in this regard when it comes to consumer safety.

Our robot is also heavily focused on having realistic movement, which depends heavily upon the accuracy and reliability of the joints and articulation systems. If those systems are inaccurate or in their operation damage the robot or something in their environment, this poses a problem. The control systems of the robot are the center of anything and everything actually working. Operators of the robot must have full control over the robot at all times and the robot's detection systems must be calibrated properly and send proper signals to the robot's microcontroller. Errors or defects in this system could cause problems with balance, action, navigation, and realism. The robot *must* shut off if communication with the controller is lost at any time.

Lastly and most importantly is the safety and operational environment of the robot. At all times, the safety and well-being of both the operators of the robot and the general public is of paramount importance. For proper operation of the robot, the intended operating environment must be respected, as extreme heat, cold, and wet weather or surrounding conditions (i.e. a pool, puddle, lake, or any body of water) must be strictly avoided for proper operation of the robot. The robot is not designed to be struck or fall in any significant way, doing so may damage the robot permanently.

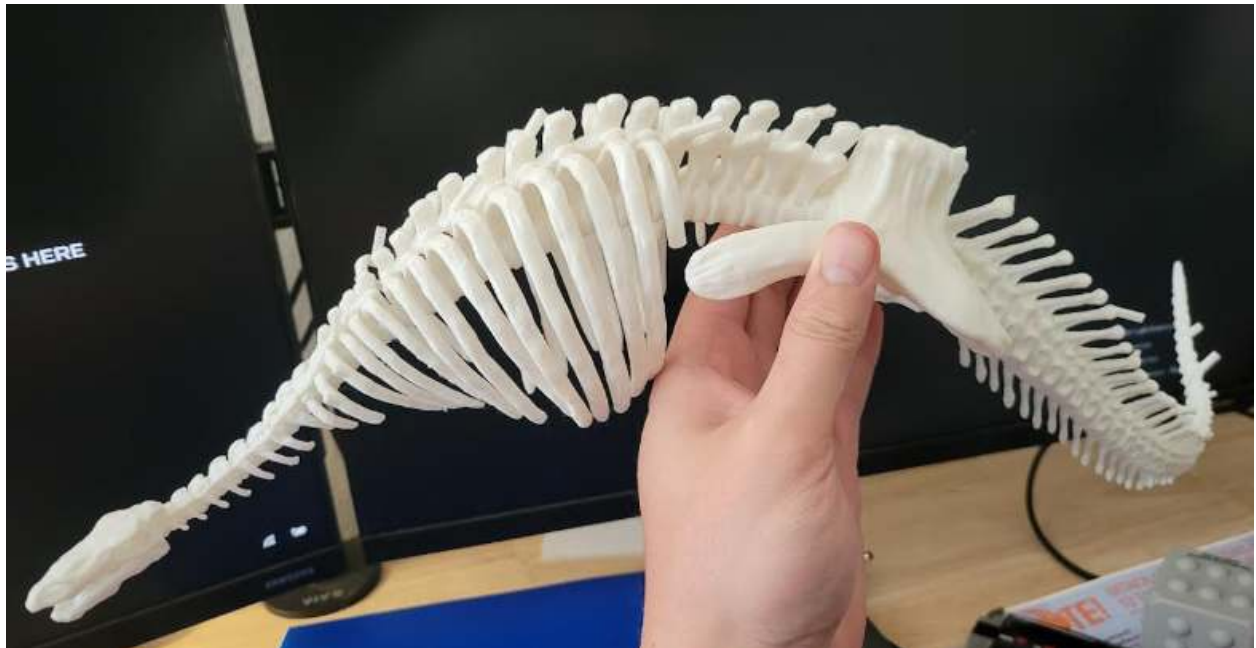
In general, the robot is intended to be operated in an air-conditioned, indoor space, by a trained operator. It is not meant to carry loads of any kind, nor to be used as a toy and handle things like falls or hits. It is not waterproof, and should have monitored interactions with the general public.

## Significant Accomplishments and Future Work

### Physical Prototyping:

Although prototyping is not expected until later in Senior Design 2. It is a useful tool we can use in order to gain a better grasp of physical sizing and overall packaging requirements that will be needed further down the line.

Initially a base 3D print was made of the skeleton from Makerworld [46] and can be seen in figure something. Although a very small portion of this initial design will remain unmodified, it was critical to get an idea of the physical proportions of the dinosaur.



*Figure 74: Initial Stegosaurus Print*

From this point, basic measurements have been taken off of this model in order to get a rough idea of the amount of space that is available in the inner section of the body for packaging. Also a scale factor was found from this initial print to be 150% in order to attain the desired 3ft length from head to tail as discussed in concept generation.

On the topic of concept generation, multiple revisions of the leg joint designs have been made, these were used to make changes to the model and concepts themselves to make them more efficient, however there is still much work to be done so each joint can operate smoothly.



*Figure 75: Final conceptual print of the legs with knee straight and bent.*

Smaller sections of the flexible elements have also been made to gain a greater understanding of the physical flexibility and the effect that varying infill densities will have on flexibility. There is a tradeoff between flexibility and structural integrity that needs to be analyzed as we get further towards our final design.



*Figure 76: Initial Tail Section from TPU*

### **Future Joint/ Structural Work**

As touched on above an extensive effort will need to be made in order to properly size the motors and main electrical components of the robot. Along with that a more extensive material analysis will need to be conducted on other hardware, such as the cables, bearings, and other hardware that will need to be selected to complete final assembly of the robot. Such examples can be an analysis on cable types, the current leg uses UHMWPE cord which is chosen for its superior elastic resistance and overall low frictional coefficient. However more options may need to be looked into as they may be a better fit for each specific application.

On the physical structure of the robot, although we will likely not be running into issues or significant consequences for material failure. There needs to be a plan implemented to run strength calculations on both the main structure of the robot, along with the individual joints. As doing so can easily prevent reliability issues in the future especially considering the final use case may include facing a public audience at a museum.

### **Contact With Museum**

The team got in contact with Lawrence Hall of Science UC Berkeley, and the museum offered to help us as we move forward with the project. The museum is also interested in having our robot in their indoor facility where visitors would interact with the robot. The team is in contact with paleontologists.

### **Future Contact With Museum**

The museum will reach out to our team occasionally to monitor our progress with the robot. Our team is in frequent contact with paleontologists and with the museum to get feedback on our work, and possible specifications they may want. They will also want a meeting in the beginning of the Spring semester to see how far the project has come.

## Conclusions and Recommendations

This project was developed by Dr. Kurt Stresau, who wanted to see his daughter's stuffed dinosaur toy come to life. It is also inspired by the film, *The Night at the Museum*, where a T-Rex skeleton begins walking around. While the team has no specific stakeholder, the project is being developed with a museum application in mind. Key requirements of the project include an anatomically accurate skeleton model of a stegosaurus, and the stegosaurus model must have a realistic walk.

In order for the animatronic to walk, a gait study must be performed. The team has determined that the bulk of the gait study will be based off of a human gait study. The team will compile data from human joint angles and apply them to the stegosaurus model. From there, adjustments must be made in order to coordinate the four limbs and the tail, as well as adjust for discrepancies that arise from differences in anatomy between humans and the stegosaurus.

In order for a mechanical model to walk, adjustments to the joints have been made. The team will then use a series of servos to actuate these joints. Only 2 degrees of freedom will be required for the legs. Hips will be attached on hollow shafts, allowing for knee actuation cables to pass through. Knees have a hidden joint and are tuned so the joint appears to slide instead of rotate. The joints will be cable driven by a remotely mounted servo motor and have a single flexible section. They will be driven by a single servo using a tendon style joint to have a smooth curve. The neck design is similar to the tail, but will move in two axes. To produce side to side movement, cables are hidden into the existing bone structure using a modified mesh. Back plates and spines will help to disguise the cable pull blocks. For the skull, a simple hidden hinge will allow a range of 20 degrees of movement, and will be driven by a hidden micro servo inside of

the skull.

For the code overview, there will be four states. State 1 is the idle state, where the motors are set to position and wait for command to transition to other states on input, and will be the default state on startup. State 2 is the active walk state, where the motors move according to registered inputs. During State 2, other inputs from the joystick and buttons can modify motor behavior (speed, direction, etc). State 3 is the active animation state, where the motors move according to predetermined animation initiated by an input and executed to completion, then return to state 1. During State 4, other inputs from the joystick and buttons will cycle the animatronic through predetermined animations.

When it comes to the bulk of our material, ASA will be used because of its availability and properties. ASA is strong and has deflection properties similar to ABS, and it can handle heat better than PLA and ABS due to its thermal properties. Hex, or honeycomb, is the strongest infill pattern when it comes to strength in all directions. If a part or bone is experiencing load in one direction, line and grid are the strongest option. Gyroid is also another alternative to hex. When it comes to infill density, anything above 40% is adding more weight than strength, and there is a tradeoff. The team has also taken into consideration motor strength, as having lighter bones should be considered when calculating torque. Printing orientation is extremely important when it comes to the stresses a print will experience. Errors in prints such as “u” and “v” notches can cause a 3D printed part to fail when undergoing tensile and stress forces. The initial design has been analyzed for point of failure throughout all the subsystems.

In general, the team has accomplished the majority of the preparatory work necessary for a successful project. The goals for the following semester will be to generate prototypes of the



animatronic and use testing and analysis to eliminate faults. The goal of this team is to utilize fundamental engineering principles to bring ROAR to life.

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# Appendices

## I. Appendix A: Engineering Design Competence Evaluation

| Project Title: <u>Bring ROAR to Life Team 1</u> |                                   | Semester: <u>Fall 2025</u> |                                                  |                                                 |                                |                                              |                       |
|-------------------------------------------------|-----------------------------------|----------------------------|--------------------------------------------------|-------------------------------------------------|--------------------------------|----------------------------------------------|-----------------------|
| MECHANICAL                                      | Critical /<br>Main<br>contributor | Strong<br>contributor      | Necessary<br>but not a<br>primary<br>contributor | Necessary<br>but only a<br>minor<br>contributor | Only a<br>passing<br>reference | Not<br>Included in<br>this Design<br>Project | Report<br>Section Pg. |
| Thermal-Fluid<br>Energy Systems                 |                                   |                            |                                                  | X                                               |                                |                                              | 74                    |
| Machines &<br>Mechanical Systems                | X                                 |                            |                                                  |                                                 |                                |                                              | 34-49                 |
| Controls &<br>Mechatronics                      | X                                 |                            |                                                  |                                                 |                                |                                              | 50-78                 |
| Materials Selection                             |                                   | X                          |                                                  |                                                 |                                |                                              | 85-89                 |
| Modeling &<br>Measurement<br>Systems            |                                   | X                          |                                                  |                                                 |                                |                                              | 34-49,<br>77-78       |
| Manufacturing                                   | X                                 |                            |                                                  |                                                 |                                |                                              | 77-78<br>85-89        |

## II. Appendix B: Functional Project Requirements

| FUNCTIONAL REQUIREMENTS |                                                                           |                                                                                                                                                  |                                                                                                  |
|-------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Req. #                  | REQUIREMENT                                                               | TARGET                                                                                                                                           | VALIDATION METHOD                                                                                |
| <b>S</b>                | <b>STRUCTURAL REQUIREMENTS</b>                                            |                                                                                                                                                  |                                                                                                  |
| F.S1                    | The animatronic shall balance statically.                                 | The animatronic must be able to remain upright on its 4 limbs on a smooth surface within a range of +/- 10° in any cardinal direction.           | Test: Stand animatronic on a surface, changing its degree of incline in the cardinal directions. |
| F.S2                    | The animatronic shall balance while walking.                              | Barring any external disturbances, animatronic must retain balance when cycling through all its gaits.                                           | Test: The animatronic will cycle through all gaits on a smooth, flat surface.                    |
| F.S3                    | Joints and structure should support the weight and movement of the robot. | The animatronic shall not weigh more than 50 pounds, including all systems involved. The animatronic shall be able to support its entire weight. | Analysis: G force calculations.<br>FEA.                                                          |
| <b>M</b>                | <b>MATERIAL REQUIREMENTS</b>                                              |                                                                                                                                                  |                                                                                                  |
| F.M1                    | Shall be durable.                                                         | Robot shall be able to withstand falls, small drops, and regular handling.                                                                       | Testing.                                                                                         |
| F.M2                    | Shall use additive manufacturing.                                         | 3D printing shall be considered for rapid prototyping and custom parts.                                                                          | Material Analysis and Selection.                                                                 |
| F.M3                    | Shall be economical                                                       | Material shall be economical to produce and use standard parts when possible.                                                                    | Analysis on material costs.                                                                      |
| F.M4                    | Shall be reliable.                                                        | Material can consistently withstand the loads and forces applied during operation.                                                               | Analysis.                                                                                        |
| F.M5                    | Shall be easily repairable.                                               | Shall utilize sub-assemblies that are easily replaceable.                                                                                        | Analysis.                                                                                        |
| <b>P</b>                | <b>Power REQUIREMENTS</b>                                                 |                                                                                                                                                  |                                                                                                  |

|             |                                                                                    |                                                                                                                |                    |
|-------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------|
| F.P1        | Power source shall have a long run time.                                           | The robot runs for at least 30 mins before signaling low battery.                                              | Test.              |
| F.P2        | Power systems shall comply with all relevant FCC and Industry safety standards.    | Refer to relevant standards.                                                                                   | Analysis, testing. |
| F.P4        | Power distribution infrastructure will not interfere with mechanical articulation. | Joints will be able to have full range of motion without being impeded or damaging the Power system.           | Test.              |
| F.P5        | The power system shall be light weight.                                            | The Power system and battery combined must weigh less than 5 lbs.                                              | Analysis.          |
| F.P6        | Robot will visibly and audibly communicate diagnostic data.                        | When the robot reaches 20% battery left, the robot will lay down and its eyes will flash red.                  | Test.              |
| F.P7        | Robot shall be rechargeable.                                                       | There will be a USB C port on the back of the robot.                                                           | Visual.            |
| F.P8        | Software will monitor Power source health and fuel consumption.                    | When the robot reaches below 5% Power, it will automatically shut off.                                         | Test.              |
| F.P9        | The Power system shall use an internal Power source.                               | the whole robot will function off the same single battery, subsystems will not have independent Power systems. | Visual.            |
| F.P10       | The Power system shall support all current requirements.                           | The main Power source shall be able to handle the max current loads from all internal electronics.             | Test.              |
| <b>A</b>    | <b>ARTICULATION REQUIREMENTS</b>                                                   |                                                                                                                |                    |
| F.A1        | Shall have a high resolution.                                                      | An individual joint should be able to stop and hold within 2 degrees of the desired angle.                     | Testing.           |
| <b>F.A2</b> | Shall be flexible.                                                                 | Each leg shall have a minimum of 2 degrees of freedom.                                                         | Analysis.          |

|          |                                                  |                                                                                                                                         |                   |
|----------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| F.A3     | Shall emulate realistic movement.                | The general walking gate and structure will appear to move in a biological way.                                                         | Analysis on Gate. |
| F.A4     | Shall have adequate range of motion.             | Each joint and limb will move XX degrees to allow for full range of motion.                                                             | Analysis.         |
| F.A5     | Shall be able to ROAR.                           | The stegosaurus mouth will open at a 20° angle with respect to the top jaw with a tolerance of +/- 2°.                                  | Test.             |
| F.A6     | Shall be able to turn to avoid obstacles.        | The animatronic shall be able to turn at a 1:1 body length to turn radius ratio.                                                        | Test.             |
| F.A7     | Shall be able to move its tail.                  | Can wag tail at minimum of 45 degrees from the center axis.                                                                             | Test.             |
| F.A8     | Shall be compact.                                | Wiring and circuitry shall be hidden in the main body of the robot.                                                                     | Analysis.         |
| <b>C</b> | <b>CONTROL SYSTEM REQUIREMENTS</b>               |                                                                                                                                         |                   |
| F.C1     | Shall have a control system.                     | Shall be controlled by a single person.                                                                                                 | Test.             |
| F.C2     | Shall respond to user input.                     | Control system shall have a low latency and update rate, with a command latency <= 100ms (Input to Response).                           | Test.             |
| F.C3     | Shall provide and store continuous diagnostics.  | Stores Joint positions, IMU, Errors, Response Time, Controller Inputs, Power, and Temperature.                                          | Demonstration.    |
| F.C4     | Shall have pre-recorded movement.                | The interaction shall fully execute within 10 to 20 seconds then return to its idle state. The animation can be canceled mid execution. | Test.             |
| F.C5     | User shall be able to control individual motion. | The controller will provide of head and neck motors in the yaw and pitch directions and hold the position within Plus of minus 2        | Test.             |

|          |                                                   |                                                                                                                           |                |
|----------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------|
|          |                                                   | degrees of the commanded orientation.                                                                                     |                |
| F.C6     | Shall make noise/sound.                           | ROAR volume shall be at minimum 70 decibels.                                                                              | Test.          |
| F.C7     | ZMovement and walking shall be easily controlled. | The controller shall control direction of gait and speed.                                                                 | Test.          |
| F.C8     | Shall have an operators instruction manual.       | Easy to operate and learn control mapping, with an included instruction manual.                                           | Visual.        |
| F.C9     | Shall have a balance feedback system.             | Stop animation and return to idle if animation is interrupted or loses balance.                                           | Test.          |
| F.C10    | Shall be connected wirelessly.                    | Shall have at least a 50ft range.                                                                                         | Test.          |
| <b>E</b> | <b>SAFETY AND ENVIRONMENT REQUIREMENTS</b>        |                                                                                                                           |                |
| F.E1     | Shall operate in dry conditions.                  | Robot shall not be submerged in water or exposed to rain.                                                                 | Test.          |
| F.E2     | Shall operate in standard atmospheric conditions. | Standard atmospheric conditions are defined as a temperature of 15°C (59°F) and a pressure of 101,325 pascals (14.7 psi). | Test.          |
| F.E3     | Shall have an integrated shut down system.        | Program will be able to sense if the robot falls over and turns off.                                                      | Test.          |
| F.E4     | Shall have a mechanical shut down system.         | Will be able to turn off in less than 2 sec if a switch is pressed.                                                       | Test.          |
| F.E5     | Shall shut down if remote contact is lost.        | Shall shut down within 5 seconds of loss of contact.                                                                      | Demonstration. |
| F.E6     | Shall have low emissions.                         | Shall not damage its surrounding environment or objects.                                                                  | Analysis.      |

### III. Appendix C: Component Project Requirements

| COMPONENT REQUIREMENTS |                |                                |              |                                                                                                                                        |                   |       |       |
|------------------------|----------------|--------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------|-------|
| Req. #                 | System         | Component                      | Stretch Goal | Requirement                                                                                                                            | Validation Method | Units | Range |
| C.C1                   | Control System | On/Off switch                  |              | Shall be easily accessible                                                                                                             | Inspection        | N/A   | N/A   |
| C.C2                   | Control System | On/Off switch                  |              | Shall immediately cut all Power to the robot.                                                                                          | Inspection        | N/A   | N/A   |
| C.C3                   | Control System | Remote Transmitter/ Controller |              | Shall have a minimum range of [50] ft.                                                                                                 | Testing           | Ft    | >= 50 |
| C.C4                   | Control System | Remote Transmitter/ Controller |              | Shall stop all movement if the robot loses range or remote contact.                                                                    | Testing           | N/A   | N/A   |
| C.C5                   | Control System | Controller                     |              | Will contain a computer that is able to be programmed allowing for precise control of joints and motion.                               | Testing           | N/A   | N/A   |
| C.C6                   | Control System | Motor Controller               |              | Will interface between the primary computer and the motors, combining positioning requests and handling motor electrical requirements. | Testing           | N/A   | N/A   |
| C.C7                   | Control System | IMU                            | X            | Will be able to keep track of inertia when the robot moves.                                                                            | Testing           | N/A   | N/A   |
| C.C8                   | Control System | Camera                         | X            | Will be able to take in visual data which will affect how the robot                                                                    | Testing           | N/A   | N/A   |

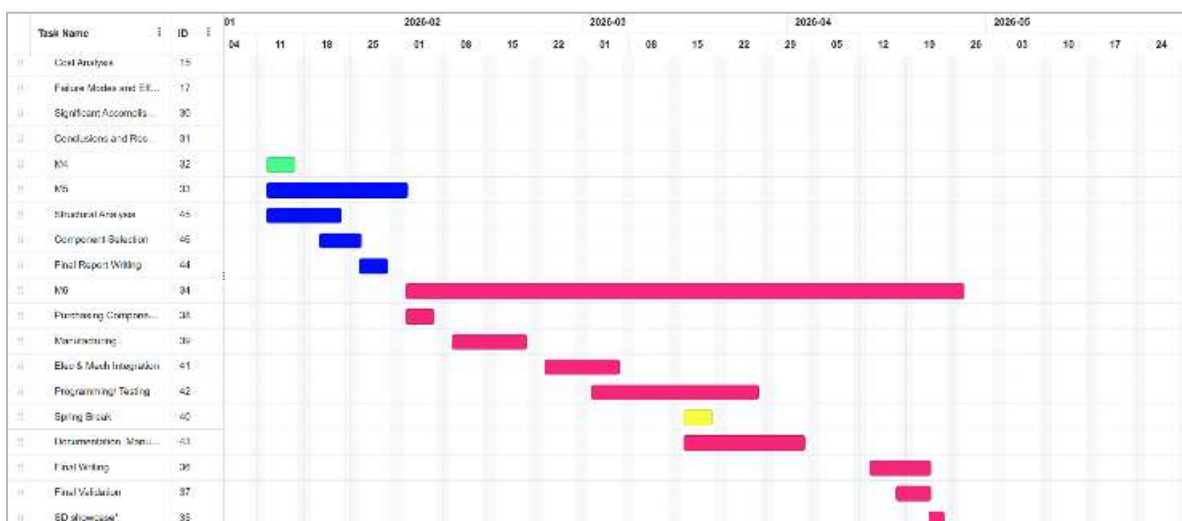
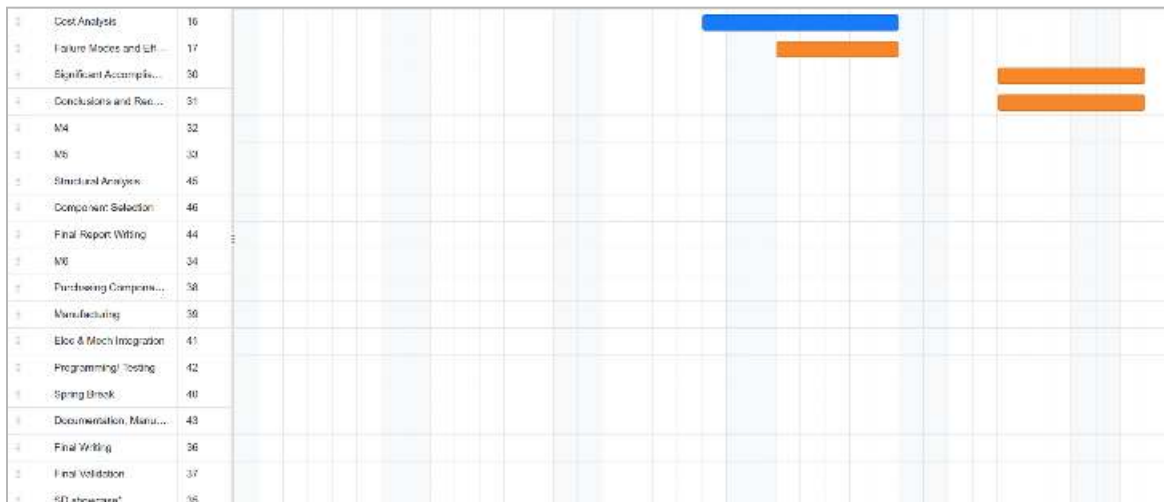
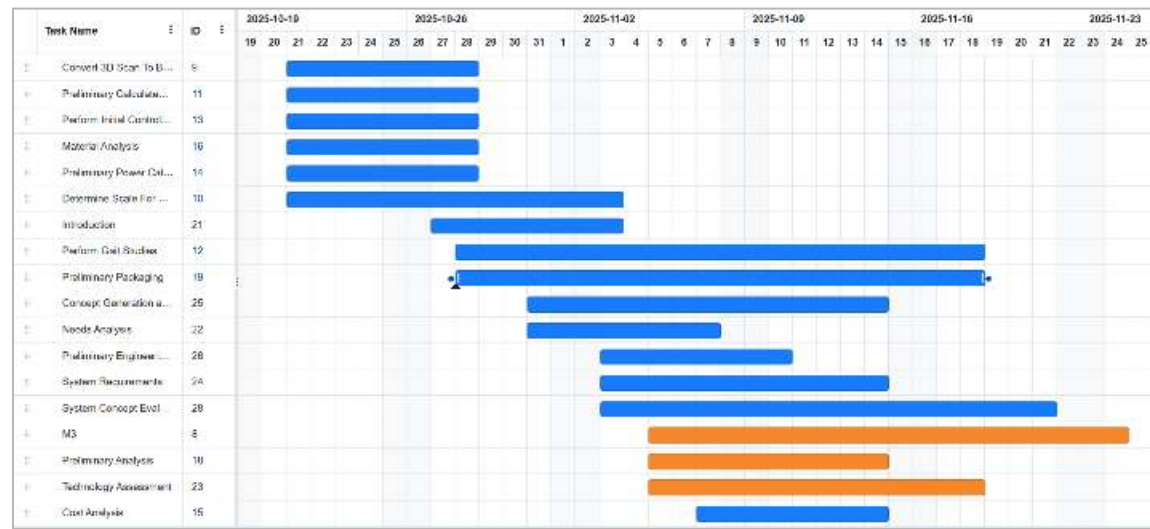


|       |                   |                  |   |                                                                                  |          |         |       |
|-------|-------------------|------------------|---|----------------------------------------------------------------------------------|----------|---------|-------|
|       |                   |                  |   | moves.                                                                           |          |         |       |
| C.C9  | Control System    | Gyroscope        | X | Determine whether the position of the robot is upright or if it has fallen over. | Testing  | N/A     | N/A   |
| C.C10 | Control System    | GPS              | X | Determine where the robot is located in space.                                   | Testing  | N/A     | N/A   |
| C.C11 | Control System    | I/O              |   | Input/Output connections to support everything and there are enough connections. | Analysis | N/A     | N/A   |
| C.E   | Electrical System |                  |   |                                                                                  |          |         |       |
| C.E1  | Electrical System | Battery Pack     |   | Shall have a run time of at least [30] min.                                      | Test     | min     | >=30  |
| C.E2  | Electrical System | Battery Pack     |   | Shall charge in under [180] min.                                                 | Analysis | min     | <=180 |
| C.E3  | Electrical System | Drive Motors     |   | Shall have adequate torque in order to move each limb.                           | Analysis | N/M     | N/A   |
| C.E4  | Electrical System | Drive Motors     |   | Shall have adequate speed in order to replicate a realistic gate.                | Analysis | Rad/Sec | N/A   |
| C.E5  | Electrical System | Animation Motors |   | Shall have [XX] torque in order to move non-structural elements.                 | Analysis | N/M     | N/A   |
| C.E6  | Electrical System | Speaker          |   | Will be able to communicate audio at a volume of at least [XX] decibels.         | Testing  | dB      | >=70  |
| C.E7  | Electrical System | Eyes             |   | Will be able to display at least 2 color.                                        | Analysis | Count   | >=2   |
| C.S   | MAIN              |                  |   |                                                                                  |          |         |       |

|       |                |                        |  |                                                                                                               |            |         |       |
|-------|----------------|------------------------|--|---------------------------------------------------------------------------------------------------------------|------------|---------|-------|
|       | STRUCTURE      |                        |  |                                                                                                               |            |         |       |
| C.S1  | Main Structure | Power Module           |  | Shall not exceed the physical dimensions of the robot.                                                        | Analysis   | N/A     | N/A   |
| C.S2  | Main Structure | Power Module           |  | Shall be able to store and protect all main electrical components with the exception of motors.               | inspection | N/A     | N/A   |
| C.S3  | Main Structure | Power Module           |  | Shall be impact resistant                                                                                     | Inspection | N/A     | N/A   |
| C.S4  | Main Structure | Hardware/<br>Fasteners |  | Shall use off the shelf, metric Nuts & Bolts unless a specific part requires otherwise.                       | Inspection | N/A     | N/A   |
| C.S5  | Main Structure | Hardware/<br>Fasteners |  | Shall use off the shelf standard size Bearings, Pins, & other hardware.                                       | Inspection | N/A     | N/A   |
| C.S6  | Main Structure | Bones                  |  | Main bone elements will and geometry will be within [XX] percent of measurements taken from an existing scan. | Analysis   | Precent | <= 20 |
| C.S7  | Main Structure | Back Bone              |  | The main structure, spine & rib cage, shall be able to support [XX] weight.                                   | Analysis   | LBS     | TBD   |
| C.S11 | Main Structure | Fins                   |  | Shall have fins along its spine, as the fossil record shows.                                                  | Analysis   | Count   | 11    |
| C.S12 | Main Structure | Tail Spikes            |  | Shall have tail spikes in accordance with the fossil.                                                         | Analysis   | Count   | 4     |
| C.S13 | Main Structure | Feet                   |  | Interface material shall provide adequate friction to be able to have proper traction.                        | Analysis   | N/A     | N/A   |

|      |        |            |  |                                                                          |          |         |           |
|------|--------|------------|--|--------------------------------------------------------------------------|----------|---------|-----------|
| C.J  | JOINTS |            |  |                                                                          |          |         |           |
| C.J1 | Joints | Leg Joints |  | Each leg will have at least 2 degrees of freedom.                        | Analysis | N/A     | N/A       |
| C.J2 | Joints | Leg Joints |  | Each joint shall be able to carry the weight of the robot.               | Analysis | FOS     | $\geq 2$  |
| C.J3 | Joints | Leg Joints |  | Each joint shall be able move [XX] degrees in order to emulate movement. | Analysis | Degrees | TBD       |
| C.J4 | Joints | Neck       |  | The neck will have at minimum [3] moving parts or vertebrae.             | Analysis | Count   | $\geq 3$  |
| C.J5 | Joints | Tail       |  | The tail will have at minimum [8] moving parts or vertebrae.             | Analysis | Count   | $\geq 8$  |
| C.J6 | Joints | Mouth      |  | The mouth shall be able to open at least [20] degrees.                   | Analysis | Degrees | $\geq 20$ |

## IV. Appendix D: GANTT Chart



## V. Appendix E: Failure Modes and Effects Analysis

| Structure                            |                                                                                  |                                                                         |          |                                                                |             |                                                       |           |     |
|--------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------|----------|----------------------------------------------------------------|-------------|-------------------------------------------------------|-----------|-----|
| Function                             | Failure Mode                                                                     | Effect                                                                  | Severity | Causes                                                         | Probability | Controls                                              | Detection | RPN |
| Balance statically                   | Loses balance/doesn't balance                                                    | Robot falls over                                                        | 3        | weight imbalance, external strike, uneven ground               | 3           | Internal gyroscope                                    | 1         | 9   |
| Balance dynamically                  | Fails to balance during motion                                                   | Robot falls during motion                                               | 3        | incorrect data input, gyroscope malfunction, joint malfunction | 7           | Microcomputer system check                            | 3         | 63  |
| Materials                            |                                                                                  |                                                                         |          |                                                                |             |                                                       |           |     |
| Function                             | Failure Mode                                                                     | Effect                                                                  | Severity | Causes                                                         | Probability | Controls                                              | Detection | RPN |
| Joints and structures support weight | Robot too heavy                                                                  | joint breaks, weight imbalance, energy inefficient, slow movement speed | 5        | internal parts too heavy, load placed on top of robot          | 3           | scale                                                 | 1         | 15  |
| Parts are durable                    | Part breaks from material fatigue                                                | Robot cannot walk, head or tail no longer move properly                 | 7        | overuse, external strain on parts                              | 3           | Factor of Safety                                      | 5         | 105 |
| Use of 3D printing                   | Part breaks along layer line fault                                               | legs/body stop functioning. Robot can no longer support weight          | 7        | Shear force along printing layer line                          | 3           | infill density, Factor of Safety                      | 2         | 42  |
| Economical                           | Parts/manufacturing over budget                                                  | decreases potential customer base                                       | 1        | design flaw                                                    | 1           | project budget, design choices                        | 3         | 3   |
| Economical                           | Product outside of consumer target range                                         | decreases potential customer base                                       | 2        | design flaw                                                    | 1           | project budget, design choices                        | 3         | 6   |
| Reliable                             | Part or parts of robot consistently break and/or require replacement             | Maintenance cost increase, consumer interest decrease                   | 5        | Part weakness, shear forces, environmental damage              | 5           | infill density, Factor of Safety, structural supports | 3         | 75  |
| Repairable                           | Robot maintenance excessively time consuming                                     | Maintenance cost increase, consumer interest decrease                   | 3        | design flaws                                                   | 2           | design choices                                        | 3         | 18  |
| Reparable/reliable                   | Robot maintenance excessively expensive                                          | Maintenance cost increase, consumer interest decrease                   | 5        | design flaws, part choices                                     | 2           | material/design choices, testing                      | 2         | 20  |
| Parts are durable                    | Parts experience severe deflection without breaking                              | Robot cannot walk, head or tail no longer move properly                 | 7        | overuse, external strain on parts                              | 5           | Factor of Safety                                      | 5         | 175 |
| Replaceable                          | Robot maintenance excessively complex or consistently requires expert assistance | Maintenance cost increase, consumer interest decrease                   | 7        | part choice, design flaws                                      | 2           | documentation, training, design choices               | 2         | 28  |
| Power                                |                                                                                  |                                                                         |          |                                                                |             |                                                       |           |     |
| Function                             | Failure Mode                                                                     | Effect                                                                  | Severity | Causes                                                         | Probability | Controls                                              | Detection | RPN |
| Runtime                              | Robot has low runtime                                                            | Use in exhibits is limited                                              | 5        | battery capacity, excess weight, motor inefficiencies          | 6           | battery choice, testing                               | 3         | 90  |
| Standards                            | Robot fails to meet a relevant FCC/Engineering standard                          | Lawsuit :(                                                              | 9        | design flaw, law oversight                                     | 1           | research                                              | 5         | 45  |
| Power distribution infrastructure    | Wires break                                                                      | Part or whole robot stops working until problem is fixed                | 9        | Material fatigue, part interaction                             | 3           | design choices, testing                               | 4         | 108 |

| Power distribution infrastructure                    | Wires short                                                      | Potential fire hazard, risk of electrocution                                      | 9        | wire breakage, water damage                                                            | 3           | design choices, testing              | 3         | 81  |
|------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------|-------------|--------------------------------------|-----------|-----|
| Overall function                                     | Critical infrastructure not integrated properly                  | robot fails to work correctly                                                     | 7        | human error, broken/damaged parts                                                      | 3           | testing, visual verification         | 5         | 105 |
| Weight                                               | Power system overburdens robot/causes excessive strain on joints | Increased power usage, possible joint breakage                                    | 5        | decreased runtime, battery strain                                                      | 2           | design choices, testing              | 2         | 20  |
| Visible and audible communication of diagnostic data | Sound levels too loud                                            | annoying to consumer                                                              | 2        | voltage to speaker too high, code problem                                              | 3           | testing                              | 1         | 6   |
| Visible and audible communication of diagnostic data | Sound levels too soft                                            | unusable to consumer, audio cue misinterpreted or missed                          | 6        | voltage to speaker too low, code problem                                               | 3           | testing                              | 1         | 18  |
| Interference with mechanical articulation            | something interferes with joint movement                         | Part or whole robot stops working, potential wire short                           | 7        | foreign objects, wires, internal defects                                               | 4           | shielding, design choices            | 4         | 112 |
| Rechargeable                                         | Overcharging                                                     | potential battery explosion                                                       | 9        | code bug, microcontroller defect, voltage regulator defect                             | 1           | testing, voltage regulator           | 5         | 45  |
| Rechargeable                                         | Undercharging                                                    | potential battery explosion, decreased runtime                                    | 8        | code bug, microcontroller defect, voltage regulator defect                             | 1           | testing, voltage regulator           | 5         | 40  |
| Safety hazard                                        | Battery puncture                                                 | Fire :(                                                                           | 10       | external damage to battery                                                             | 1           | shielding                            | 3         | 30  |
| Software monitored                                   | Software crashes                                                 | Robot stops working until problem is rectified                                    | 7        | code bug, microcontroller defect                                                       | 2           | testing                              | 6         | 84  |
| Software monitored                                   | Software receives bad sensor information                         | robot behaves erratically or not at all                                           | 6        | external interference with sensor, damage to sensor, microcontroller problem, code bug | 3           | testing, calibration                 | 5         | 90  |
| Internal power source                                | Exceeds weight requirement                                       | Increased power usage                                                             | 4        | human error                                                                            | 2           | scale                                | 2         | 16  |
| Design error                                         | Exceeds internal spatial dimensions                              | Less space for critical infrastructure, more vulnerable if not properly protected | 4        | human error                                                                            | 2           | product description                  | 3         | 24  |
| Improper safety hazard                               | Malfunction damages other interior components                    | robot stops working or causes further damage to itself or environment             | 8        | manufacturing defect, microcontroller problem, voltage regulator problem               | 2           | testing of parts                     | 1         | 16  |
| Power system supports all requirements               | Robot components exert electrical load above battery rating      | battery strained beyond recommended use, damage or possible explosion             | 9        | human error, design error                                                              | 2           | component amperage load testing      | 1         | 18  |
| Articulation                                         |                                                                  |                                                                                   |          |                                                                                        |             |                                      |           |     |
| Function                                             | Failure Mode                                                     | Effect                                                                            | Severity | Causes                                                                                 | Probability | Controls                             | Detection | RPN |
| High resolution movement                             | Irregular or erratic movement                                    | unpredictable or dangerous behavior                                               | 5        | code error, joint problem, motor problem                                               | 5           | testing, torque and ROM limitations  | 7         | 175 |
| Degrees of freedom                                   | Failure to move in a prescribed degree of freedom                | partial or complete failure in locomotion                                         | 9        | blocked joints, code error, mechanical failure, foreign object                         | 4           | testing, material and design choices | 7         | 252 |

|                    |                                                         |                                                                        |   |                                                                |   |                                 |   |    |
|--------------------|---------------------------------------------------------|------------------------------------------------------------------------|---|----------------------------------------------------------------|---|---------------------------------|---|----|
| Realistic movement | Jerky or robotic movements                              | consumer immersion lost, interest decreases, potential loss of balance | 4 | blocked joints, code error, mechanical failure, foreign object | 3 | visual verification, gait study | 7 | 84 |
| Range of motion    | Failure to move smoothly through entire range of motion | potential loss of balance/ability to move                              | 1 | blocked joints, code error, mechanical failure, foreign object | 3 | testing, gait study             | 3 | 9  |
| Range of motion    | Range of motion blocked by something                    | robot loses balance/ability to move                                    | 7 | foreign debris                                                 | 4 | visual verification, shielding  | 3 | 84 |
| Roar               | Fails to produce sound                                  | customer expectations not met                                          | 3 | speaker defect, code error, microcontroller problem            | 2 | test                            | 1 | 6  |
| Communication      | Produces sound without movement                         | customer expectations not met, no movement                             | 7 | speaker defect, code error, microcontroller problem            | 2 | test                            | 1 | 14 |
| Avoid obstacles    | Hits obstacle                                           | loss of balance, damage to robot or environment                        | 8 | operator failure, detection failure                            | 3 | test, training                  | 1 | 24 |
| Tail movement      | Tail only partially moves                               | fails to meet standards                                                | 3 | foreign object, material fatigue, broken part                  | 3 | test                            | 2 | 18 |
| Jaw movement       | Jaw doesn't move or too much load                       | fails to meet standards                                                | 2 | material experiences stresses past the yield point             | 3 | tests, FEA                      | 1 | 0  |
| Tail movement      | No tail movement at all                                 | fails to meet standards                                                | 3 | foreign object, material fatigue, broken part                  | 2 | test                            | 2 | 12 |
| Compact            | Robot dimensions exceed requirements                    | problems with shipping and handling, excessive weight                  | 5 | design failure                                                 | 2 | size limitation                 | 1 | 10 |

| Control System                      |                                               |                                                           |          |                                                     |             |                                                 |           |     |
|-------------------------------------|-----------------------------------------------|-----------------------------------------------------------|----------|-----------------------------------------------------|-------------|-------------------------------------------------|-----------|-----|
| Function                            | Failure Mode                                  | Effect                                                    | Severity | Causes                                              | Probability | Controls                                        | Detection | RPN |
| Has a control system                | loses control system                          | impossible to control, feedback loop or complete inaction | 8        | damage to control system, control system defect     | 4           | shielding, integration                          | 3         | 96  |
| Responds to user input              | User input misinterpreted by CS               | erratic behavior                                          | 5        | control system defect                               | 2           | Testing, component choice                       | 3         | 30  |
| Responds to user input              | No response to user input/ blocked user input | erratic or no behavior                                    | 4        | RC signal blockage, code issue, controller defect   | 4           | testing, environment                            | 3         | 48  |
| Provides and stores diagnostic data | Reports inaccurate diagnostic data            | erratic or no behavior                                    | 4        | memory corruption, diagnostic tool damage/defect    | 3           | testing, code debugging                         | 2         | 24  |
| Provides and stores diagnostic data | Fails to store diagnostic data                | control system accuracy compromised                       | 3        | memory corruption, diagnostic tool damage/defect    | 3           | testing, debugging                              | 2         | 18  |
| Pre-recorded movement               | pre recorded movement does not play           | breaks immersion, fails design standards                  | 3        | memory issue, code issue, microcomputer defect      | 2           | testing, debugging, memory integrity validation | 2         | 12  |
| User-controlled individual motion   | User error, robot piloted to dangerous area   | robot breaks either structurally or electronically        | 8        | operator                                            | 5           | training, documentation                         | 3         | 120 |
| Makes noise/sound                   | audio files do not work                       | no sound, breaks immersion, fails design specifications   | 3        | code problem, file corruption                       | 2           | testing                                         | 5         | 30  |
| Makes no sound                      | audio devices broken or compromised           | no sound, breaks immersion, fails design specifications   | 3        | external damage, water damage, manufacturing defect | 4           | protection, testing                             | 2         | 24  |

|                               |                                                |                                                                                                  |   |                                                    |   |                                                                 |   |     |
|-------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------|---|----------------------------------------------------|---|-----------------------------------------------------------------|---|-----|
| Easy to control               | controls exceedingly complex                   | operator fails to properly operate robot                                                         | 8 | design flaw, human error                           | 4 | testing                                                         | 3 | 96  |
| Comes with instruction manual | instructions unclear                           | owner does not know how to use robot, potential unintended damage or failure to operate normally | 8 | design flaw, human error                           | 4 | Documentation, feedback                                         | 1 | 32  |
| Lost instruction manual       | Instruction manual lost, damaged, or destroyed | owner does not know how to use robot, potential unintended damage or failure to operate normally | 8 | shipping problem, user error                       | 3 | protection, documentation, redundancy (have more than 1 manual) | 1 | 24  |
| Balance feedback              | Inaccurate data                                | Robot loses/fails to balance dynamically                                                         | 6 | sensor defect, code error, microcontroller problem | 4 | testing, sensor calibration                                     | 5 | 120 |
| Wireless connection           | Robot walks too far away                       | loses connection                                                                                 | 3 | environment                                        | 5 | testing, automatic shutdown                                     | 4 | 60  |

| Safety and Environment             |                                          |                                                        |          |                                                                            |             |                                                           |           |     |
|------------------------------------|------------------------------------------|--------------------------------------------------------|----------|----------------------------------------------------------------------------|-------------|-----------------------------------------------------------|-----------|-----|
| Function                           | Failure Mode                             | Effect                                                 | Severity | Causes                                                                     | Probability | Controls                                                  | Detection | RPN |
| Standard atmospheric conditions    | Excessive heat                           | electronics/material damage                            | 8        | environment                                                                | 5           | operating environment and safety constraints              | 1         | 40  |
| Standard atmospheric conditions    | Excessive cold                           | material damage                                        | 5        | environment                                                                | 5           | operating environment and safety constraints              | 2         | 50  |
| Standard atmospheric conditions    | High humidity                            | electronics damage                                     | 8        | environment                                                                | 5           | operating environment and safety constraints              | 5         | 200 |
| Standard atmospheric conditions    | Condensation                             | electronics damage                                     | 8        | environment, rapid temperature change from hot to cold                     | 5           | operating environment and safety constraints              | 5         | 200 |
| Automatic shut down                | Signal failure                           | robot continues to operate outside of user definition  | 9        | code error, microcontroller defect                                         | 2           | test                                                      | 2         | 36  |
| Shut down on loss of communication | Signal failure                           | robot continues to operate outside of user definition  | 9        | code error, microcontroller defect                                         | 1           | test                                                      | 1         | 9   |
| Low emissions                      | Non-recyclable materials used            | Environmental impact from end of product life cycle    | 4        | material design choice                                                     | 1           | material design choices                                   | 1         | 4   |
| Human interaction                  | Digits/body parts pinched in joints      | medical injury, lawsuit                                | 10       | human error, insufficient safety protocol                                  | 6           | operating environment and safety constraints              | 1         | 60  |
| Human interaction                  | Jaw bite force causes soft tissue damage | medical injury, lawsuit                                | 10       | loss of control, human error (sticking hand into jaw), no detection system | 2           | safe distance from people, controlled by trained operator | 1         | 20  |
| Human interaction                  | Robot damaged or hit by human            | robot functions compromised, possible need for repair. | 7        | operator error, environment, other human error                             | 5           | operating environment and safety constraints              | 1         | 35  |
| Operator interaction               | Does not respond to operator input       | inactive or erratic behavior                           | 7        | control system error                                                       | 3           | testing feedback                                          | 1         | 21  |
| Operator interaction               | Operator improperly trained              | Damage to robot or environment                         | 6        | human error, improper training                                             | 5           | testing feedback                                          | 1         | 30  |
| Operator interaction               | Operator improperly trained              | robot fails to move                                    | 5        | human error, improper training                                             | 5           | testing feedback                                          | 1         | 25  |
| Operator interaction               | Operator improperly trained              | erratic behavior                                       | 5        | human error, improper training                                             | 5           | testing feedback                                          | 1         | 25  |



## VI. Appendix F: Cost Analysis

| Subsystem      | Component                                                                                                                        | Qty | Unit Cost | Total Cost | Vendor / Link                                                                                                                     |
|----------------|----------------------------------------------------------------------------------------------------------------------------------|-----|-----------|------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Control        | ESP32-S3-WROOM-U1                                                                                                                | 1   | \$20.00   | \$20.00    | Espressif – <a href="https://www.espressif.com/">https://www.espressif.com/</a> or Amazon                                         |
| Audio          | DFPlayer Mini MP3                                                                                                                | 1   | \$5.90    | \$5.90     | DFRobot – <a href="https://www.dfrobot.com">https://www.dfrobot.com</a>                                                           |
| Audio          | 3W 8Ω Speaker                                                                                                                    | 1   | \$3.50    | \$3.50     | Adafruit – <a href="https://www.adafruit.com">https://www.adafruit.com</a>                                                        |
| Mobility       | DS5180 High-Torque Servo                                                                                                         | 2   | \$34.99   | \$69.98    | Amazon                                                                                                                            |
| Mobility       | DS3230MG Servo                                                                                                                   | 6   | \$14.34   | \$86.04    | Amazon                                                                                                                            |
| Mobility       | Micro Linear Servo                                                                                                               | 5   | \$12.35   | \$61.75    | SparkFun – <a href="https://www.sparkfun.com">https://www.sparkfun.com</a>                                                        |
| Mobility       | Goteck 1.5g Micro Digital Linear Servo Actuator                                                                                  | 1   | \$11.58   | \$11.58    | Amazon                                                                                                                            |
| Mobility       | Linear Actuator - KT10-0903H                                                                                                     | 1   | \$16.59   | \$16.59    | Pololu – <a href="https://www.pololu.com">https://www.pololu.com</a>                                                              |
| Drivers        | PCA9685 16-Ch Servo Driver                                                                                                       | 1   | \$13.00   | \$13.00    | Amazon                                                                                                                            |
| Drivers        | TB6612FNG Motor Driver                                                                                                           | 1   | \$14.50   | \$14.50    | SparkFun – <a href="https://www.sparkfun.com">https://www.sparkfun.com</a>                                                        |
| Controller     | PS4/PS5 Controller                                                                                                               | 1   | \$54.76   | \$54.76    | Amazon                                                                                                                            |
| Power          | High Volt Buck Converter                                                                                                         | 1   | \$18.75   | \$18.75    | Ebay                                                                                                                              |
| Power          | Low Volt Buck Converter                                                                                                          | 1   | \$9.95    | \$9.95     | Adafruit – <a href="https://www.adafruit.com">https://www.adafruit.com</a>                                                        |
| Power          | 25 AMP Inline Blade Fuse Assembly                                                                                                | 1   | \$10.00   | \$10.00    | Great Plains Industries                                                                                                           |
| Power          | 50A 12V DC Heavy Duty Toggle Switch                                                                                              | 1   | \$7.13    | \$7.13     | Walmart                                                                                                                           |
| Power          | Busbar Power Distribution Block                                                                                                  | 1   | \$23.99   | \$23.99    | Daier Switches                                                                                                                    |
| Power          | 2S LiPo Battery                                                                                                                  | 1   | \$29.75   | \$29.75    | Amazon                                                                                                                            |
| Safety Sensors | MPU-6050 IMU                                                                                                                     | 1   | \$12.50   | \$12.50    | DigiKey                                                                                                                           |
| Safety Sensors | Additional GY-521 IMUs (Pack)                                                                                                    | 1   | \$15.00   | \$15.00    | Amazon                                                                                                                            |
| Safety Sensors | MCP9808 Temp Sensor                                                                                                              | 1   | \$4.95    | \$4.95     | Adafruit                                                                                                                          |
| Safety Sensors | GNSS GPS Module                                                                                                                  | 1   | \$20.00   | \$20.00    | SeeedStudio – <a href="https://www.seeedstudio.com">https://www.seeedstudio.com</a>                                               |
| Print Material | TPU Filament                                                                                                                     | 1   | \$19.00   | \$19.00    | Amazon                                                                                                                            |
| Print Material | Additional Filaments                                                                                                             | 4   | \$32.00   | \$128.00   | Amazon                                                                                                                            |
| Joint Cable    | UHMWPE Cord                                                                                                                      | 1   | \$13.00   | \$13.00    | Amazon                                                                                                                            |
| Hardware       | Cables (All internal wiring, Power distribution, Servo extensions, Sensor cables, Audio wiring, Cable management supplies, Pins) | 1   | \$95.00   | \$95.00    | N/A                                                                                                                               |
| Hardware       | Spacers                                                                                                                          | 1   | \$10.00   | \$10.00    | Amazon                                                                                                                            |
| Hardware       | Screws                                                                                                                           | 1   | \$5.00    | \$5.00     | Walmart - 1023PCS M3 Bolt and Nuts Washers Assortment Kit, Hex Flat Head Cap Machine Screws with Washers 4mm to 40mm Black - \$20 |
| Hardware       | Washers                                                                                                                          | 1   | \$5.00    | \$5.00     |                                                                                                                                   |
| Hardware       | Fasteners                                                                                                                        | 1   | \$5.00    | \$5.00     |                                                                                                                                   |
| Hardware       | Nuts                                                                                                                             | 1   | \$5.00    | \$5.00     |                                                                                                                                   |
| Misc           | Misc                                                                                                                             | 1   | \$35.00   | \$35.00    | N/A                                                                                                                               |

|                         |          |
|-------------------------|----------|
| <b>Subtotal:</b>        | \$829.62 |
| <b>20% Contingency:</b> | \$165.92 |
| <b>Total:</b>           | \$995.54 |

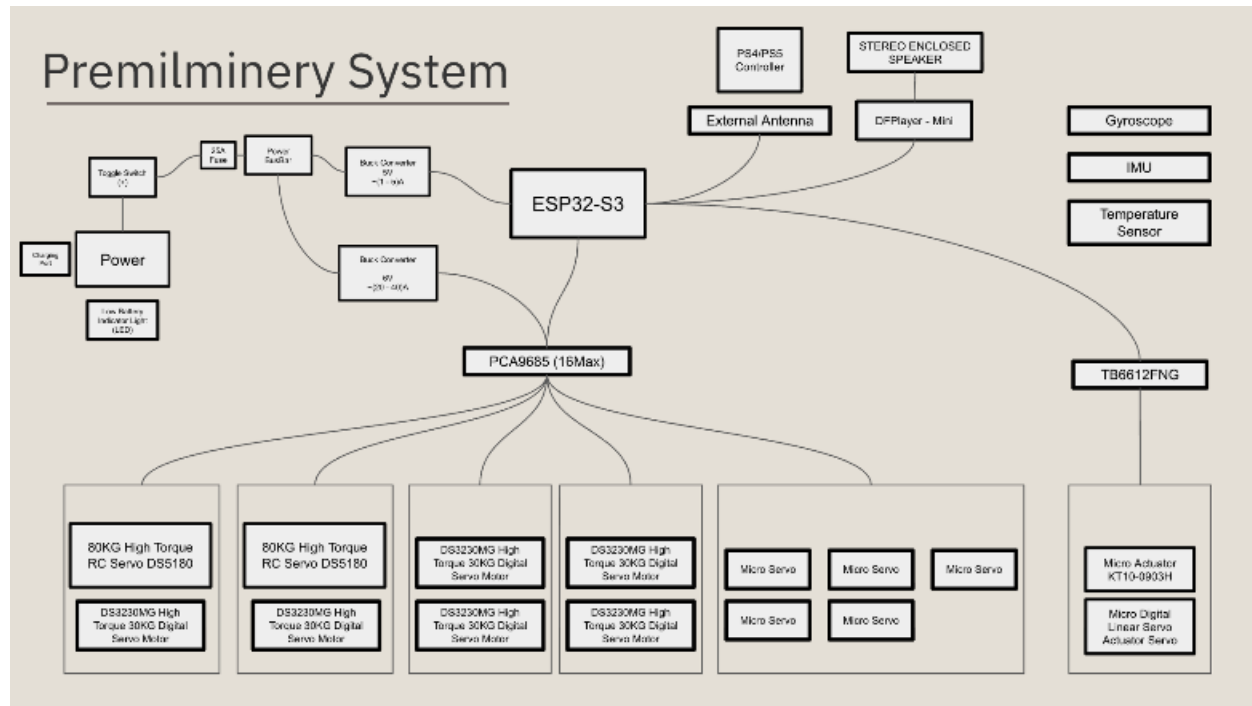
**VII. Appendix G: Programming Language Selection Table**

| <b>Factor</b>                                                      | <b>Description</b>                                                                                                        |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Execution speed and real-time capabilities</b>                  | Critical for time-sensitive control tasks where delays can compromise performance.                                        |
| <b>Hardware compatibility and low-level access</b>                 | Ability to interact directly with hardware components such as sensors, actuators, and controllers.                        |
| <b>Library and API support (e.g., for sensors, motors, vision)</b> | Availability of pre-built modules and frameworks to accelerate development in areas like sensing, motion, and perception. |
| <b>Community support and documentation</b>                         | Access to tutorials, forums, and technical references that reduce development time and troubleshooting difficulty.        |
| <b>Integration with existing robotics platforms</b>                | Ease of interoperability with frameworks like ROS and embedded systems, ensuring scalability and modularity.              |

## VIII. Appendix H: Techniques Used in Robotics

| Techniques In Robotics                                          |                                                                                                                                                                                                                                      |                                                                       |                    |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------|
| Technique                                                       | Purpose                                                                                                                                                                                                                              | Application Area                                                      | Source / Reference |
| <b>PID Controller</b>                                           | Cascaded PID for accurate foot-tip trajectory tracking on a quadruped leg                                                                                                                                                            | Quadruped leg servo control / trajectory tracking                     | [1]                |
| <b>LQR (Linear Quadratic Regulator)</b>                         | Optimal feedback (Riccati/LQR) integrated with quadruped planning/control to improve robust motion tracking                                                                                                                          | Dynamic quadruped locomotion (trotting/pacing)                        | [11]               |
| <b>Kalman Filter (KF)</b>                                       | State estimation under Gaussian noise                                                                                                                                                                                                | General robotics state estimation                                     | [12]               |
| <b>Extended Kalman Filter (EKF)</b>                             | Classical Gaussian state estimation (derives and analyzes the KF)                                                                                                                                                                    | General robotics state estimation                                     | [12]               |
| <b>Particle Filter</b>                                          | Sequential Monte Carlo methods; practical tutorial and use for localization                                                                                                                                                          | General estimation + (see below PF legged)                            | [13]               |
| <b>Model Predictive Control (MPC)</b>                           | Representation-Free MPC (RF-MPC): represents orientation with a rotation matrix, uses variation-based linearization and QP transcription; demonstrated real-time 250 Hz control and dynamic motions including a controlled backflip. | Quadruped robot dynamic locomotion and acrobatics (Panther platform). | [9]                |
| <b>A Algorithm*</b>                                             | Foundational text on graph/discrete planning incl. A* properties and use                                                                                                                                                             | General motion/discrete path planning                                 | [14]               |
| <b>RRT / RRT*</b>                                               | Sampling-based motion planning                                                                                                                                                                                                       | General motion planning                                               | [15]               |
| <b>Deep Neural Networks (DNNs)</b>                              | Perception, classification, control policies                                                                                                                                                                                         | Vision-based Navigation, Object Recognition                           | [18]               |
| <b>CNNs (Convolutional Neural Nets)</b>                         | Feature extraction from images                                                                                                                                                                                                       | Robotic Vision, SLAM                                                  | [17]               |
| <b>Reinforcement Learning (RL)</b>                              | Learning control policies via rewards                                                                                                                                                                                                | Autonomous Navigation, Manipulation                                   | [16]               |
| <b>Inverse Kinematics</b>                                       | Convert end-effector position to joint angles                                                                                                                                                                                        | Humanoid Robots, Manipulators                                         | [27]               |
| <b>Forward Kinematics</b>                                       | Compute end-effector position from joint states                                                                                                                                                                                      | Robot Motion Modeling                                                 | [27]               |
| <b>SLAM (Graph-based, EKF, etc.)</b>                            | Simultaneous localization and mapping                                                                                                                                                                                                | Mobile Robots, Autonomous Vehicles                                    | [19]               |
| <b>Bezier and B-Spline Curves (Parametric Animation Curves)</b> | Smooth trajectory generation through parametric curves                                                                                                                                                                               | Motion planning for arms, mobile robots                               | [20]               |
| <b>Principal Component Analysis (PCA)</b>                       | Dimensionality reduction and feature extraction                                                                                                                                                                                      | Sensor fusion, gesture recognition, object tracking                   | [21]               |
| <b>Trajectory Optimization (CHOMP)</b>                          | Gradient-based trajectory optimization                                                                                                                                                                                               | Motion planning for manipulators and legged robots                    | [22]               |
| <b>Trajectory Optimization (STOMP)</b>                          | Stochastic trajectory optimization with constraints                                                                                                                                                                                  | High-dimensional motion planning                                      | [23]               |
| <b>Central Pattern Generators (CPGs)</b>                        | Generate rhythmic signals for periodic movement (e.g., walking gaits)                                                                                                                                                                | Quadruped locomotion, bio-inspired robots                             | [24]               |
| <b>IMU Fusion (e.g., EKF)</b>                                   | Combine IMU data with other sensors (e.g., encoders, GPS) to estimate pose                                                                                                                                                           | Balance control, walking on uneven terrain                            | [25]               |
| <b>ROS Teleop / Joystick Input</b>                              | Manual control of robot motion using joystick or keyboard commands                                                                                                                                                                   | Manual operation, remote teleoperation                                | [6]                |
| <b>Direct Control Loops</b>                                     | Send low-level velocity or position commands directly to actuators                                                                                                                                                                   | Low-latency control, gait tuning                                      | [26]               |

## IX. Appendix I: Preliminary System



## X. Appendix J: Table of Necessary Motors

| Robot Part                     | Movement / Action     | Motor Type / Example                                           | Notes                                                         |
|--------------------------------|-----------------------|----------------------------------------------------------------|---------------------------------------------------------------|
| Mouth                          | Open / close          | 1 × Micro Linear Servo Actuator                                | Lightweight, precise linear motion                            |
| Neck                           | Up, down, left, right | 2 × Servo Motors                                               | Medium torque servos for multi-axis head control              |
| Fins                           | Cam movement / wave   | 1 × Linear Motor Servo                                         | Smooth oscillatory motion, Lightweight, precise linear motion |
| Tail                           | Up, down, left, right | 2 × Servo Motors                                               | Requires torque and precise multi-axis control                |
| Shoulders (2 joints)           | Y-axis rotation       | 2 × High-torque Digital Servo per joint (DSServo DS5180, 80kg) | Strong enough to handle limb weight                           |
| Knees and shoulders (2 joints) | Y-axis rotation       | 2 × DS3230MG Metal Gear Digital Servo per joint (30kg)         | High precision for leg articulation                           |
| Knees (4 joints)               | Y-axis rotation       | 4 × DS3230MG Metal Gear Digital Servo per joint (30kg)         | High precision for leg articulation                           |

|                   |                  |          |                                |
|-------------------|------------------|----------|--------------------------------|
| Ankles (4 joints) | Spring mechanism | No motor | Passive movement using springs |
|-------------------|------------------|----------|--------------------------------|